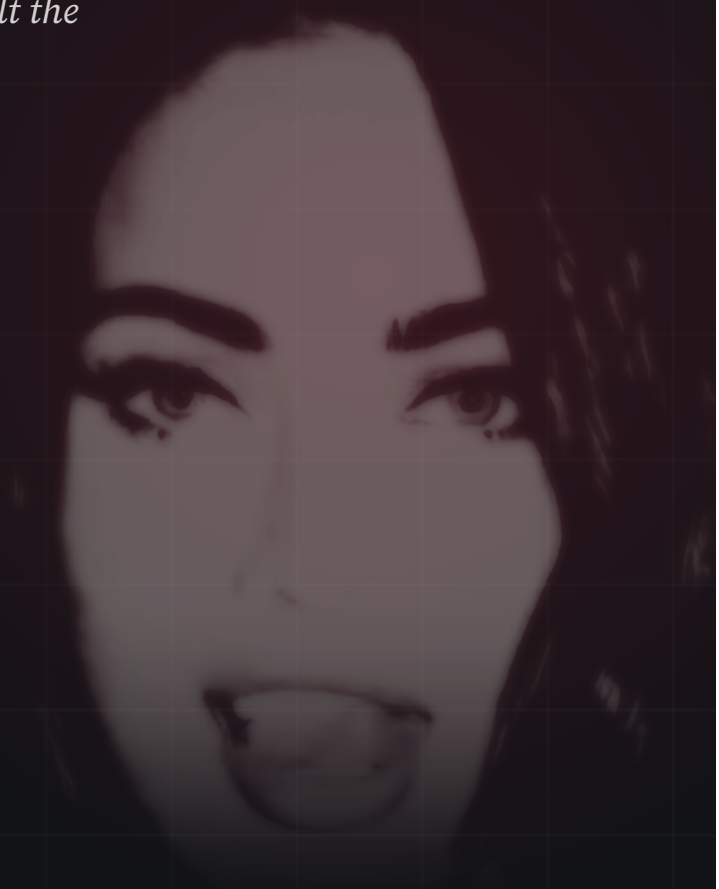
NOIR EDITION · DEFINITIVE

THE 9.0 MANUAL

Everything you do between now and the exam, in the order that works — and every result the paper can actually ask you for.



Viluxent (mkb6)
TMUA9. (jamiebaek.)

DEFINITIVE EDITION · TWO BOOKS IN ONE · METHOD, THEN CONTENT
SECTIONS 1–80
FORMAT AND CONTENT CHECKED AGAINST THE CURRENT UAT-UK SPECIFICATION

Contents

BOOK ONE — THE METHOD

What you do. Diagnose yourself, attack a problem you do not recognise, turn one question into a rule, get faster, and know what to practise and in what order.

Ninety seconds 4

PART I WHERE YOU ARE 6

1 What you are actually walking into 7
2 What a 9.0 actually costs 8
3 Where your hours go 9
4 Why past papers stop working 9
5 Find out where you actually are 10
6 Six things you probably believe that are not true 12

PART II THE ENGINE 14

7 Before you write anything 15
8 When you are stuck, what to actually do 16
9 "I have no idea where to start" 18
10 Struggling well, and knowing when you are not 20
11 How long to spend on a hard question 21
12 Reading the examiner's cues 22
13 Paper 2 is a different sport 24

PART III EXTRACTION 26

14 How to actually learn from a question 27
15 Why did you really get it wrong 29
16 Reading the wrong answers 32
17 The circuit card 33
18 Anki, done properly 34
19 When to redo a question 36
20 Lots of questions, or one question properly? 37

PART IV SPEED 38

21 Six kinds of slow 39
22 Using the answer choices 40
23 The noteboard nobody practises 41

PART V MATERIAL 43

24 What each resource is for 44
25 The order everybody agrees on 44
26 The six stages 45
27 The full order, resource by resource 46
28 Which community papers to bin 50
29 Things that feel productive and are not 50
30 Stop grading yourself 51

PART VI YOUR PLAN 53

31 Find yourself here 54
32 How long have you got? 56
33 Building your week 57
34 How to tell you are improving 58

PART VII THE EXAM 60

35 The last two weeks 61
36 Exam day 61
37 When it goes wrong in the room 63

PART VIII THE ROADMAP 65

38 Seven phases 66
39 One page to keep 67
The work that counts 68

Contents

BOOK TWO — THE CONTENT

What you know. Every result the paper can ask for, written as a trigger you will meet rather than a topic you were taught, priced by what it saves and tiered by what it costs.

PART IX THE LIBRARY 69

40 Why Book Two was rebuilt 70
41 Tiers, and how they were decided 72
42 The frame, and the card it becomes 73

PART X ALGEBRA 75

43 Polynomials: degree, roots and shape 76
44 Quadratics: the discriminant machine 80
45 Roots and coefficients, and related equations 82
46 Hidden polynomials, and the constraint the substitution carries 83
47 Exponentials and logarithms 85
48 Inequalities and casework 87
49 Ranges: what a function can and cannot reach 89

PART XI FUNCTIONS, GRAPHS AND CALCULUS 91

50 Graphs, symmetry and transformations 92
51 Symmetry inside a definite integral 94
52 Modulus, floor and ceiling 96
53 Trigonometry: counting, not computing 97
54 Areas: one identity, not eight formulae 99
55 Curvature, the trapezium rule, and honest estimates 102
56 Calculus checkpoints 103

PART XII GEOMETRY 106

57 Circles and configurations 107
58 Lines, distances and reflections 109
59 Triangle machinery, priced honestly 111
60 Awkward areas, and how to decompose them 113

PART XIII NUMBER, SEQUENCES AND COUNTING 116

61 Sequences and series: the traps that get set 117
62 Sigma, telescoping and the method of differences 119
63 Recursions: solve, iterate, or spot the loop 121
64 Number, divisibility and counting 122
65 Selections and arrangements 124
66 Diophantine problems 126

PART XIV PAPER 2, PROPERLY 128

67 The vocabulary, as a machine 129
68 Negation and quantifier order 132
69 Counterexamples, and the two jobs they must do 134
70 The flawed-step catalogue 135
71 Proof: choosing the method, ordering the argument 137
72 "Exactly one of these is true" 139

PART XV THE 9.0 LAYER 141

73 Ratio, invariance and scaling 142
74 The break-even rule 143
75 Invariance as an eliminator 144
76 Composite drills 145

PART XVI BEYOND, AND HOUSEKEEPING 147

77 Beyond the specification 148
78 If you are sitting the ESAT 148
79 What changed in this edition, and what was corrected 149
80 Where to find everything 151
One page to keep 153
Learn the trigger, price the move 155

Page numbers are the numbers printed at the foot of each page. A section written as §14 always means section 14 of this book.

OPENING

Ninety seconds

There is one moment that decides your score, and it is not the one you are worried about.

It is not the night before. It is not the topic you never quite nailed. It arrives somewhere around question eleven, when you read something you have never seen in your life and feel your stomach drop.

Everybody gets that moment. The person who walks out with a 4.0 gets it and the person who walks out with a 9.0 gets it. What separates them is entirely what happens in the ninety seconds afterwards.

One of them freezes, reads the question again, reads it a third time hoping it will change, and then starts writing something. Anything. Because writing feels better than sitting in the blank.

The other one has somewhere to go. Not a memorised method for that question, because no such thing exists. A procedure. A set of moves they have run so many times that their hand starts moving while they are still frightened.

That is the whole thing. That is what this book is for.

> *You are not trying to get good at the TMUA. You are trying to become someone who can be handed a problem they have never seen and reliably find a way in.*

Which is fortunate, because the second thing is the only one that is actually trainable. Nobody can teach you the questions on the next paper. Everybody can be taught what to do when they don't recognise one.

Here is what to understand before reading another page. The work that raises your score is almost never the work that feels like progress. Sitting a paper feels like progress and produces a number you can look at. Going back over a topic you already know feels like progress and produces a warm sense of coverage. Sitting with one question you cannot do for twenty-five minutes, and then writing down the single sentence you failed to think, produces nothing you can show anybody.

That last one is the one that works. Everyone at the top of this scale has made that trade. Most people who plateau never do, and it is not because they are lazy. It is because the alternative looks so much more like studying.

So expect the score to sit still for a while. The ceiling moves before the number does. Almost everyone who abandons this method abandons it about ten days before it would have worked.

One more thing, and then we start.

Don't burn yourself out. But don't waste this either. If the goal is a 9.0 then your ordinary Tuesday has to look like the Tuesday of somebody whose goal is a 9.0, because that is all a 9.0 ever is: an accumulation of unremarkable evenings.

There is somebody else out there right now working for the exact place you want. They are finishing another paper, correcting another mistake, going back over a question they got wrong last week, while everybody else is saying they will start tomorrow.

You do not have to be the smartest person in the room. You have to be the one who keeps showing up after it stops being exciting.

Stay disciplined. Stay consistent. Trust this for longer than feels reasonable.

See you all with those 9.0s.

Viluxent (mkb6) · TMUA9. (jamiebaek.)

HOW TO USE THIS BOOK

There are two books here and they answer different questions.

Book One, §1–39	*What you do.* How to diagnose yourself, how to attack a problem you do not recognise, how to turn one question into a rule, how to get faster, what to practise and in what order, and what to do in the room. Parts II and III are the engine. If you only ever re-read two things, re-read those.
Book Two, §40–80	*What you know.* Every result the test can actually ask for, written as a trigger you will meet rather than a topic you were taught, priced by what it saves, and tiered by whether the specification can ask for it at all.

Short on time. Go to §5 and find out where you actually are. Then §14 for how to learn from a question, then §27 for the order to do things in. If the exam is roughly a month away, add §32. If you are already above 7.0 and stuck, go to §31.4 and then Part XV.

Navigating it. This PDF is linked throughout. Every row of the contents, every section list on a part divider, and every §NN reference in the text jumps straight there, and the running head at the top of each page brings you back to the contents. Your reader's own bookmark panel carries the same structure.

Reading Book Two. Do not read it front to back. §42 tells you the four-minute routine for a single result. Everything labelled CORE belongs in your daily deck. Everything labelled EDGE belongs in your head but not in your reflexes. Everything labelled BEYOND is for a quiet month early on, not for the run-in.

01

PART I

WHERE YOU ARE

Before you choose what to do, work out what the exam is really asking of you, what a 9.0 actually costs, and which of three very different problems you have. Almost everybody skips this and then trains the wrong thing for four months.

1 What you are actually walking into
2 What a 9.0 actually costs
3 Where your hours go
4 Why past papers stop working
5 Find out where you actually are
6 Six things you probably believe that are not true

Read once. Come back to §5 every three weeks and measure yourself again.

1 What you are actually walking into

Two papers. Seventy-five minutes each, twenty multiple-choice questions each, sat one after the other with no guaranteed break. No calculator, no formula booklet, no dictionary. Paper 1 is Applications of Mathematical Knowledge. Paper 2 is Mathematical Reasoning. The raw marks from both are combined and turned into a single score from 1.0 to 9.0, given to one decimal place. Nothing is deducted for a wrong answer.

Four things in that description matter enormously, and most candidates act as though three of them are not true.

1.1 It is a computer exam, and you are probably not practising like it

Since 2024 you sit the TMUA on a screen at a Pearson VUE test centre. Questions come one at a time. There is a timer, a flag button, and a review screen showing what is answered, unanswered and flagged. You get an erasable noteboard and a pen. You cannot bring paper in and you cannot write on the question.

Now think about how you have been practising. Printed papers. Notes in the margin. Options crossed out right next to the question. Conditions circled in the text. On the day, none of that exists.

This is a genuine and completely avoidable loss of marks, and hardly anybody talks about it. §23 deals with it properly. The short version is that the elimination technique in §22 has to be trained on a board, because on screen there is nothing to cross out.

1.2 The famous 3 minutes 45 seconds is a trap

Twenty questions in seventy-five minutes averages three minutes forty-five per question. Every guide repeats it. It is true and almost useless, because nobody who finishes at the top spends 3:45 on twenty consecutive questions. A real 9.0 sitting looks more like this.

QUESTION BAND	HOW MANY	TIME EACH	WHAT IS HAPPENING
Instant	8 to 10	45 to 90 s	You recognise the structure, write two lines, move on.
Standard	6 to 8	2:30 to 3:30	You know the route within thirty seconds and execute it.
Genuinely hard	3 to 4	5 to 7 min	You have to invent the idea there and then.
Committed		**about 56 min**	Leaving about 19 minutes to sweep the paper again.

That changes what you are training for. You are not training to average 3:45. You are training so that ten questions cost you less than ninety seconds each, because that is what buys you six minutes on the four that actually deserve them. Everything in Part IV exists to move questions up that table.

1.3 Nothing is deducted, so a blank is simply a wasted mark

Five options, so a blind guess is worth 0.2 of a mark. Kill two options that cannot possibly be right and it is worth 0.33. Across five guessed questions that is the gap between one mark and nearly two. Some Paper 2 questions offer more than five options, which makes elimination worth more rather than less.

Never leave anything blank. Put an answer in *before* you flag a question, so that running out of time can never turn a flag into a zero.

1.4 The scale is equated, not marked

Since 2024, UAT-UK equates the papers statistically, adjusting for how hard the particular questions turned out to be, and publishes no raw-mark conversion table for current sittings. This is the most misunderstood fact about the exam and §30 is devoted to it. For now, hold onto one consequence: a score you calculate for yourself on a practice paper is not a score. It is a number you invented.

1.5 The fifth thing, which is not in any specification

Since the test moved on screen, questions have been reported as longer and more often set inside an applied or wordy context. That does not change what is examinable. It changes the first fifteen seconds of every question, because there is now a translation step before any mathematics starts. §7 is where that is trained, and §21 treats reading speed as its own separate skill, because for a lot of candidates it is now the binding constraint.

---

2 What a 9.0 actually costs

Be honest about the size of the target, because the size of the target decides the method.

Published data on recent sittings: about ten per cent of candidates score above 7.0. The most common score in the two most recent published sittings was 3.5. The 90th percentile sits around 6.5. Roughly one in a hundred reaches 9.0. Back when raw-mark tables were still published, in 2023, you needed somewhere around 38 out of 40 overall for a 9.0, and 19 out of 20 on a single paper converted to 9.0 on that paper's scale.

A 9.0 does not mean "very good". It means "lost almost nothing".

You are not training to be reliable on the questions you recognise. You are training to be reliable on the ones you have never seen, because at that level those are the only ones left. Sit with that for a second, because it settles every argument in this book. If a 9.0 needed 30 out of 40, the right strategy would be to get very fast and very accurate on familiar material and write off the hard tail. It doesn't. It needs the hard tail. And the hard tail is unfamiliar by construction, because the examiners will never reuse a question you have already met.

> **WHAT FOLLOWS FROM THAT**
>
> Preparation that makes familiar questions faster has a ceiling somewhere around 7.5. Getting past it needs preparation that changes what you can do with a problem you have never seen. Those are two different activities.

> Most people only ever do the first one, and cannot work out why the line has gone flat. If your score has stalled, you are almost certainly not lacking effort. You are running the wrong one of the two.

3 Where your hours go

Every hour you spend lands in one of two accounts.

3.1 The capability account

Work that makes you able to solve things you currently cannot solve at any speed. Pulling official papers apart. Attacking unfamiliar problems. Repairing a topic the log says is broken. Building and reviewing your circuits. Proving results you have only ever used.

3.2 The performance account

Work that turns capability you already have into marks under constraints. Timed papers. Decision drills. Elimination practice. Stamina. Getting used to the interface. The exam-day protocol.

Both are necessary. The mistake is almost always identical: people run about ninety per cent performance and ten per cent capability, because performance work feels like real preparation. It produces a number. It takes exactly seventy-five minutes. It goes in a timetable. Capability work is slow, unmeasurable and uncomfortable, and it is where the marks are.

Jamie Baek draws the sharpest version of this line, splitting skillset development from performance and arguing for roughly eighty per cent on the first. The analogy he uses is exact. No serious athlete improves by playing more matches, and nobody has ever fixed a jump shot in a game.

Where this book goes further: the right ratio is not eighty-twenty for everybody. It depends entirely on what your diagnostic says, which is §5.

4 Why past papers stop working

Do enough past papers and something real happens. You start to recognise shapes. Areas between a line and a curve. Three-term arithmetic and geometric conditions. The negation of a nested quantifier. Tangent and chord configurations. A geometric series hiding inside a diagram of nested squares.

This is not shallow and it is not cheating. It is exactly what makes ten questions a paper take under ninety seconds, and it is worth an enormous amount.

It is also, on its own, a ceiling.

WHAT RECOGNITION GIVES YOU	WHAT IT CAN NEVER GIVE YOU
Speed on structures you have met before.	Any traction at all on a structure you have not met.
Confidence, which stops you second-guessing.	The ability to *generate* a route instead of retrieving one.

WHAT RECOGNITION GIVES YOU	WHAT IT CAN NEVER GIVE YOU
A clean first pass, and therefore time for the hard tail.	Anything on the hard tail itself.
Accurate self-assessment on familiar material.	Accurate self-assessment on unfamiliar material, where it quietly flatters you.

That last row is the dangerous one. Somebody who has done fifteen papers and scores well on them has no way of knowing whether they are good at mathematics or good at those fifteen papers. From the inside the two feel identical. The only way to tell them apart is to sit in front of a problem whose structure you genuinely have never met, and watch what your brain does in the first four minutes.

The single most valuable thing you can train is what happens inside you when you read a question and have no idea. Most candidates have trained that feeling to produce panic. You are going to train it to produce a procedure. Part II is that procedure. It is why this book exists.

---

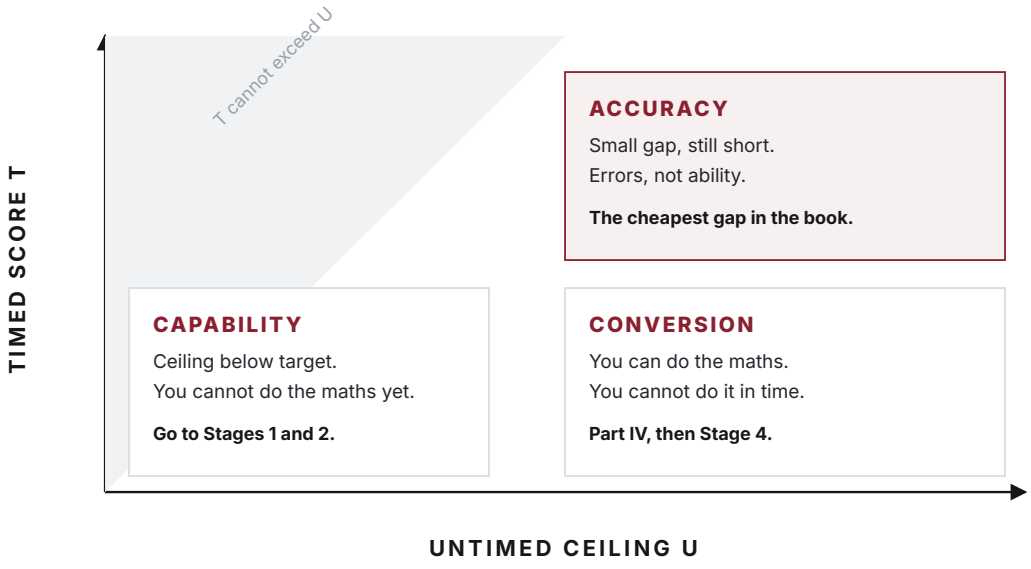
5 Find out where you actually are

Almost every argument about TMUA preparation dissolves the moment you realise that different advice is right for different people. So measure yourself before you decide anything. This takes one evening and it will save you weeks.

THE TWO-SCORE DIAGNOSTIC. DO THIS TODAY, THEN AGAIN EVERY THREE WEEKS.

1. Take an official past paper you have never seen. Any one, Paper 1 or Paper 2.
2. Sit it strictly timed. Seventy-five minutes, no calculator, no notes, working on a board or a single reusable sheet, and no writing on the question. Record your raw mark out of 20. Call it T.
3. Carry straight on, untimed, working on everything you did not get. No solutions, no internet, no notes. Keep going until you genuinely stop making progress, usually another forty to seventy minutes. Record the new raw mark out of 20. Call it U.
4. Write down three things: U, which is your ceiling; $U - T$, which is your conversion loss; and, for every question you missed even untimed, the error category from §15.

U is the number that matters and almost nobody measures it. T tells you what you would score tomorrow morning. U tells you what your mathematics is currently worth. The distance between them tells you which account to spend in.



The two-score diagnostic. Everything you do next is decided by which box you land in.

WHAT YOU FOUND	WHAT IT MEANS	WHAT TO ACTUALLY DO
U well below target	**Capability.** You cannot do this mathematics yet, at any speed.	Stop sitting timed papers completely for now. Foundations, then extraction of official papers. Sitting more papers against the clock will produce the same score for months.
U at or above target, $U - T \geq 4$	**Conversion.** The mathematics is there. The routing and the decisions are not.	Part IV in full, then timed community papers as decision training. Log every overrun as a decision failure, not a maths failure.
U at or above target, $U - T \leq 2$, still short	**Accuracy.** You are losing marks you already had.	Error taxonomy, distractor autopsy, and a verification habit. This is the cheapest gap in the book to close and it is often worth two marks in a fortnight.
$U - T$ large and U low	**Both.** Common, and not a disaster.	Capability first, always. Conversion work on a low ceiling only makes you fast at a low score.

THE MOST EXPENSIVE MISTAKE IN TMUA PREPARATION

Treating a conversion problem as a capability problem, or the other way round.

Somebody with $U = 18$ and $T = 11$ who responds by revising topics will not move, because there is nothing wrong with their topics. Somebody with $U = 11$ who responds by sitting three timed papers a week will not move either, because you cannot practise your way into an ability you do not have.

Both will eventually decide they have hit their limit. Neither of them has come anywhere near it.

5.1 The third number nobody records

Alongside U and T, write down R: how many of the twenty questions you *finished inside ninety seconds* on the timed run. That number is the best available proxy for how many trained circuits you own, it moves before your score does, and it is the reason §21 exists. Aim for ten.

6 Six things you probably believe that are not true

Jamie Baek opens his blueprint by attacking beliefs most candidates hold without examining them. What follows is in that spirit. Every one of these is mainstream advice, and every one of them is wrong or badly incomplete for somebody aiming at the top of the scale.

1 "Just do as much maths as possible."

Volume converts capability into speed. It does not create capability. If there is a class of problem you cannot solve, two hundred problems of other classes will not fix it, and two hundred of that class without extraction will just teach you two hundred separate facts. Volume comes after competence. Oxford and Cambridge both tell applicants to do a lot of maths and they are right, but that advice is aimed at a population of thousands, not at your particular bottleneck.

2 "Timed past papers are the best preparation."

A timed paper is a measuring instrument. Measuring yourself does not make you better, it tells you how good you are. Jamie Baek puts it perfectly: timing without analysis is a rehearsal of your current level. Running a broken program faster does not fix the program.

There is a second cost that is easy to miss. Official past papers are finite and non-renewable. There are about eighteen of them. Every one you burn as a timed measurement is one you can never use as a teaching document, because you will never meet it cold again.

3 "Don't look at the solution until you've solved it."

The most damaging piece of received wisdom in the whole field. The solution is your only source of information about the route you could not see. Refusing to look guarantees the routes you cannot generate stay ungenerated.

The rule is not *never*. The rule is: not until you have produced something worth comparing it against, and then only as much as unblocks you. See the hint ladder in §11. Two hours staring at one problem is not discipline. It is one extraction cycle you did not run.

4 "If it's harder than the TMUA it must help."

Difficulty is not the axis. Structural relevance is. A four-hour olympiad geometry problem is far harder than anything on the TMUA and trains almost nothing you will use, because what it develops is sustained construction over hours and you have three minutes. Meanwhile the first fifteen AMC 12 questions are individually easier than the TMUA's hard tail and are superb practice, because they train precisely what you need: fast idea generation on short unfamiliar problems.

Always ask what a resource trains, never how hard it feels.

5 "I'll write faster."

You won't. Writing faster produces more slips, a slip forces a restart, and a restart costs more than the writing ever saved. Every second of real speed is bought before the pen moves, at the point where you choose a route. Both of the people credited on the cover say this independently and both are right.

6 “My mock score predicts my real score.”

It does not, and believing it does real damage. Pre-2024 papers were scaled against their own cohorts with published tables. Current sittings are equated statistically with no public table. Community papers have no cohort at all, so any grade boundary printed on one was chosen by whoever wrote it. Chasing a number produced by three incompatible methods is an excellent way to feel terrible in a week when you actually improved.

> **AND ONE BELIEF WORTH KEEPING**
>
> Somebody once asked Viluxent whether it was realistically possible to go from 5.0 to 9.0. The answer is the right one, and it is not a motivational slogan.
>
> *If I told you it was impossible, would you stop revising? If I told you it was possible, would you revise more?*
>
> Letting somebody else’s estimate set your effort has never once improved an outcome. Trying your best always has.
>
> The related point is the only comparison worth making. Somebody improving one per cent a day will overtake somebody who started further ahead. That includes you, and the person you are overtaking is yourself three weeks ago, which is exactly why the diagnostic gets repeated.

02

PART II

THE ENGINE

What to do in the first four minutes in front of a problem you have never seen. This is the difference between a 7.5 and a 9.0, and it is the part almost nobody trains on purpose.

7 Before you write anything

8 When you are stuck, what to actually do

9 “I have no idea where to start”

10 Struggling well, and knowing when you are not

11 How long to spend on a hard question

12 Reading the examiner’s cues

13 Paper 2 is a different sport

Re-read §9 and §11 until you can run them without looking. §12 is the bridge into Book Two.

7 Before you write anything

The commonest failure on a hard question is not getting it wrong. It is starting to write eight seconds after you finish reading, because writing feels like progress and thinking feels like being stuck. That single reflex costs more marks than any topic gap you have.

Train these three questions to fire in that gap instead. They take about fifteen seconds and they save minutes.

THE OPENING THREE

1. **What kind of object am I actually holding?** A cubic and a line? A statement with two quantifiers? A geometric sequence dressed up as a diagram? A quadratic wearing an exponential costume? Naming it correctly pulls up everything you know about it. Naming it wrongly, calling something a trigonometry question when it is really a quadratic in $\sin x$, sends you straight down the expensive route.
2. **What is the cheapest thing that would settle this?** Not "how do I solve it". What is the smallest piece of information that lets me pick an option? You often need only the sign, the parity, the rough size, the number of solutions, or which of two things is bigger.
3. **What have they given me that I have not used?** Every condition in the question is there for a reason. An unused condition is not decoration, it is a signpost. If your method has not touched "where $p < 0$" or "for all positive integers n" or "x is real", your method is probably not the one they intended and is probably three times longer.

Install number three first. After the extraction protocol in Part III it is the highest-value habit in this book, and it is almost free. When you are stuck, the way in is very often sitting in the one clause you skimmed.

7.1 The conditions carry the information

A formula has three parts: the conditions under which it holds, the formula itself, and how it gets used. Most students learn only the middle one. TMUA9.'s material makes a point of this and it is worth taking seriously, because examiners build questions right at the edge of those conditions. That is where the interesting mathematics lives.

Take the area between a parabola and a line. If you know it depends only on the leading coefficient and the horizontal gap between the intersections, you can answer a question with deliberately monstrous numbers in about ten seconds. If you know the formula but not what it fails to depend on, you will start integrating. Same knowledge, four minutes apart. That result is §54.

7.2 Restating, on a screen, with no margin

On paper, "read the question properly" is advice. On a screen with a noteboard it is a procedure, because you cannot underline anything. Before the first line of working, write three short lines on the board:

— **Object.** Two circles. A cubic and a horizontal line. A nested quantifier. Four lines.

— **Asked.** The *quantity*, in your own words. Not "find x" but "the number of integer x", or "the value of c, not m".

— **Given.** Every condition, as a list. Then a dot next to each one as your method consumes it. Any condition still without a dot when you have an answer is either a check or a warning.

That is fifteen seconds and it removes most of an entire error category. §15 shows you what fraction of your losses are reading failures; for most candidates in the 5 to 7 range it is the largest single bucket and the one they never name.

---

8 When you are stuck, what to actually do

Every problem-solving guide gives you a list. Try small cases, draw a diagram, work backwards, look for symmetry. The lists are all correct and all useless, because in the moment you are not short of moves. You are short of a way to pick one. A checklist you read from the top burns the exact seconds you cannot spare.

So what follows is indexed by *why* you are stuck, not by technique. There are only about eight ways to be stuck. Work out which one you are in and the move comes with it.

8.1 Stuck because it is too general

You are staring at n, or an arbitrary function, or "for all real x", and there is nothing to grip.

Move: make it specific. Set $n = 1, 2, 3$. Set a parameter to 0 or 1. Take the simplest member of the family. Take the degenerate case.

Why it works. Making something specific turns an abstract statement into arithmetic, arithmetic gives you data, and data gives you a conjecture. On a multiple-choice paper this is often not the route to the answer, it *is* the answer, because one well-chosen value kills three options outright.

When it lets you down. When your special case is degenerate in a way that destroys the structure, like setting $a = 0$ in something that was only interesting because $a \neq 0$. Check your case still satisfies every stated condition.

8.2 Stuck because it is too specific

You have concrete horrible numbers, $252!$ or π^{74} or 2^{-33}, and no idea what to do with them.

Move: replace the horrible constant with a letter, then ask what the answer cannot possibly depend on.

Why it works. Grotesque numbers are a signal, not an obstacle. They are the examiner telling you that quantity is irrelevant and there is a structural shortcut waiting. A question that genuinely needs its numbers uses small ones. This is one of the most reliable tells on Paper 1, and once you have seen it three times you will never miss it again. §75 turns it into a systematic eliminator.

8.3 Stuck because you cannot see it

You understand every word and still have no picture.

Move: change the representation. Algebra to graph. Graph to algebra. Sequence to points on a line. Logical statement to a set inclusion. Inequality to a region. Geometry to coordinates, or coordinates back to geometry. A count to a listing.

Why it works. Difficulty is not a property of a problem, it is a property of a problem in a particular representation. The fact that takes three lines of algebra is sometimes a single glance at a picture. Sketching is the best twenty seconds you can spend on Paper 1.

When it lets you down. When you draw the picture accurately. Do not plot to scale. That is draughtsmanship, not mathematics. What you want is a sketch that is right about where things cross, which side is above, and how many turning points there are.

> **THE HIGHEST-VALUE SKETCH ON THE PAPER**
>
> "How many solutions does this equation have" is one of the most common question shapes in the whole test, and it is almost never an algebra question.
>
> Split the equation into two functions you can draw, put them on the same axes, and count crossings. $2^x = k$, $\tan x = c$, $|x^2 - 4| = x$, $x^3 - 3h^2x = 2h^3$, $a^x = \log_a x$. Every one of those becomes trivial the moment it is two curves rather than one equation.
>
> The rule underneath it: *solve* only when you need the values, *sketch* when you need the count. Reading which of the two the question wants is §7, step two.

8.4 Stuck because there are too many unknowns

You have set up five variables and the algebra is spreading across the page.

Move: count your constraints against your unknowns, then look at the relationship *between* the given equations instead of inside either one.

Why it works. When two equations arrive together, the intended route is nearly always a manipulation of the pair. One is the difference of squares of the other. One is the square of the other. Their sum or product collapses. Attacking either one alone is the long way round.

This is exactly the example TMUA9. uses to illustrate the whole method. Two simultaneous equations in 2^x and 2^y: substitute A and B, and $A^2 - B^2 = 6$ with $A + B = 3$ gives $A - B = 2$ in one line. Squaring the first equation, which is the reflex move, gives you a mess.

The lesson is not about exponentials. It is that before you manipulate anything, ask what the two given objects have to do with each other.

8.5 Stuck because forwards is not working

You know where you start and you cannot get anywhere.

Move: work backwards from the goal, or from the options. What would have to be true one step before the answer? If the answer is one of five expressions, what do all five share and what tells them apart?

Why it works. Searching backwards branches less than searching forwards on short problems, because the goal is more specific than the start. On multiple choice the goal is not just specific, it is listed for you.

8.6 Stuck because you do not know if it is true

Usually Paper 2. You are asked whether something holds, or which of several statements hold.

Move: try to break it before you try to prove it, and try to break it at the boundary. Equality cases, zero, one, negatives, non-integers, the empty case, the degenerate case.

Why it works, twice over. A failed attempt to break a statement usually shows you *why* it is true, which hands you the proof for free. And on multiple choice you often do not need a proof at all, because one counterexample settles it and building one is much faster than building an argument.

TMUA9. puts the instinct well: for a condition like $x > -2$, start at the forbidden value $x = -2$. Look at the boundary first. §69 gives the full search order for breaking cases.

8.7 Stuck because the algebra is exploding

Right method, but you are three lines in and it is getting worse. Moves, best first.

1. Substitute the repeated compound expression as a single letter. If $x + 1/x$ appears twice, it is one object, not two.
2. Factor before you expand. Group like powers before you evaluate.
3. Evaluate a bracket at both ends as a single object. $\left[ax^2 + bx\right]_{\beta}^{\alpha} = (\alpha - \beta)\big(a(\alpha + \beta) + b\big)$ instead of computing both ends and subtracting is a habit, not a trick, and it saves you time on about three questions a paper.
4. Ask whether you need the whole expression or just one coefficient of it.
5. Abandon. Explosion is usually evidence of a wrong route, not evidence of a hard problem.

8.8 Stuck because you have already tried this and it did not work

Move: change the *class* of approach, not the details of it. If algebra failed, sketch. If sketching failed, make it specific. Doing the same manipulation again more carefully is the single biggest source of wasted exam time there is. More on this in §10.

---

9 "I have no idea where to start"

This is the section that matters most in the whole book.

What follows is a ladder. You do not need inspiration to climb it. Every rung is a mechanical action that is available to you no matter how blank you feel, which is the entire point.

LEVEL 0	**Restate it in your own words.** Define every symbol. If you cannot restate it, you have a reading problem, not a maths problem.
LEVEL 1	**Knowns, unknowns, constraints, in three short columns.** Circle anything you have not used.
LEVEL 2 THE TRICK	**Produce one object that satisfies the conditions.** Any object. Always possible. This is the rung that matters.
LEVEL 3	**Produce a second, different one.** What changed and what stayed the same? The thing that stayed the same is usually the answer.
LEVEL 4	**Say a conjecture out loud, in one sentence.** Even if you are fairly sure it is wrong.
LEVEL 5	**Attack your own conjecture at the boundary and at the extreme.** Try hard to make it fail.
LEVEL 6	**Decide.** Either it survives and you answer, or it breaks and the break shows you the real structure.

The ladder. Every rung is available with no insight at all, which is the point.

You are never stuck. You are only ever short of examples. Producing an example needs no insight, is always available, and turns "I have no idea" into data. Nearly every experience of being stuck on a short problem is really the experience of refusing to compute something small.

9.1 Level 2 is the whole trick

Levels 0, 1, 3, 4, 5 and 6 come naturally once you are standing on level 2. The blockage is almost always between level 1 and level 2, and it is psychological rather than mathematical. Writing down a concrete example feels like giving up on the real solution. It is not. It is how the real solution gets found. Working mathematicians spend most of their lives on levels 2 and 3.

On the TMUA, "produce an example" means something concrete:

- a specific n, usually 1 then 2 then 3;
- a specific member of a family. If it says "a cubic", write $x^3 - x$;
- a specific point that satisfies the constraint;
- a small version of the configuration. Two circles instead of n. Three terms instead of a general series;
- for a logical statement, a specific small collection of objects you can check by hand.

> **CHOOSE THE FOURTH CASE, NOT THE THIRD**
>
> Three cases identify a linear or quadratic pattern and almost nothing else. If your first three cases were $n = 1, 2, 3$, the fourth one has to be chosen to be *awkward*: $n = 0$, or $n = 10$, or a negative, or a non-integer if the claim allows one. A convenient fourth case confirms a wrong guess, which is worse than having no fourth case at all.
>
> This matters on both papers. On Paper 1 it stops you generalising a coincidence. On Paper 2 it is the difference between a conjecture and a claim, and §71 makes it a procedure.

9.2 Write everything down, including the wrong turns

Not because you need it for the answer. Because you need it afterwards. The extraction protocol in Part III works by comparing your chain of reasoning against a better one, and you cannot compare against a chain you did not record.

TMUA9.'s guidance on this is the most useful single line in any of the source material, and it should be taken literally:

> *Your eraser is your enemy.*

Rubbing out the route that turned out to be wrong is destroying the raw material you were going to work with later. The wrong routes are the most informative thing on the page, because they are the evidence of what your instincts actually do under pressure. Do not be embarrassed by them. They are the thing you are trying to change, and you cannot change something you have deleted.

While you solve, keep a running honest commentary. Actual sentences, not just algebra.

> **WHAT HONEST JOURNALLING SOUNDS LIKE**
>
> *"What is this? There is no way this is how you are meant to do it. Why do I think that? Because it is far too much computation. Maybe I should pivot to a graph."*
>
> *"I don't get this bit. Okay, what exactly is causing the frustration. It is that I don't know the length of AB. So is it worth setting that as a variable? Probably, let's try."*
>
> It feels ridiculous for the first three questions and then becomes the most valuable thing you own. Those honest, ugly notes in your own handwriting will be worth more to you on the day than any PDF anybody sells you.

---

10 Struggling well, and knowing when you are not

Productive struggle and going round in circles feel identical from the inside and are worth completely different amounts. You need a test that does not depend on how you feel, because how you feel is the unreliable part.

> **THE NEW INFORMATION TEST**
>
> Look at what you have written in the last five minutes. **Did you write down one thing you did not know five minutes ago?** A value, a relationship, an option ruled out, a conjecture that failed?
>
> *Yes* means you are struggling productively. Keep going. A failed conjecture is new information. So is "it cannot be option C".
>
> *No* means you are spinning. Change the class of approach right now.

Spinning has three signatures. Learn to catch them in yourself.

SIGNATURE	WHAT IT LOOKS LIKE, AND WHAT TO DO INSTEAD
The redo loop	You do the same algebra a third time, slower, hoping the answer changes. It will not. If you think you have made an arithmetic error, do not redo the working. **Redo the question by a different route and compare the two answers.** That catches errors. Re-reading confirms them.
The sunk-cost grind	"I have spent four minutes, I cannot abandon it now." Those four minutes are gone whatever you do next. The only live question is what the *next* minute buys. Obvious on paper, very hard in the room, which is why it has to be rehearsed in practice.
The comfort detour	You start doing the bit you know how to do, expanding something, differentiating something, even though it leads nowhere, because it feels better than sitting in the blank. TMUA9. calls the general version of this *zombie maths*: reaching for the most primitive available method without asking whether it is the right one. It is extremely common in strong students, because strong students have a lot of methods to reach for.

> **THE THREE ATTEMPTS RULE**
>
> An attempt means a genuinely different class of approach: a different representation or a different direction. It does not mean the same algebra done more carefully.
>
> Attempt 1 fails, change representation. Attempt 2 fails, change direction, backwards or from the options. Attempt 3 fails: in the exam, guess with elimination, flag, move. In training, go to the hint ladder.

11 How long to spend on a hard question

Two completely different answers, because there are two completely different situations.

11.1 In a timed paper

ELAPSED	CHECKPOINT	WHAT YOU DO
0 to 20 s	Read, then the opening three	Name the object. Find the unused condition.
20 to 90 s	Have I got a route?	If not by ninety seconds, one move from §8.
90 s to 3 min	Am I producing new information?	Yes, continue. No, eliminate what you can, answer, flag, move.
Hard stop at 3 min		Answer with your best elimination. Flag it. Move. No exceptions on the first pass.
Second pass	5 to 6 min available	Now spend real time, because you know what is left and which questions have earned it.

The three-minute cap on the first pass is not about that question. It is about protecting the eleven questions you have not read yet. The most expensive thing that happens in a TMUA sitting is somebody spending eight minutes on question four and never reaching question seventeen, which they would have got in ninety seconds.

11.2 In training, on a genuinely hard problem

Here the goal is the opposite. You are deliberately buying the experience of generating an idea, and that experience is the product. But it still has a budget, because one extraction cycle is competing for the same evening and extraction is usually worth more.

THE SINGLE PROBLEM PROTOCOL

1. **0 to 10 min, cold.** No aids, no notes, no formula sheet. Write everything down, including the wrong turns.
2. **10 to 25 min, helped by your own system.** Now you may walk the moves in §8 deliberately, climb the ladder in §9, check your own circuit bank, and look up a definition you are unsure of. Not the solution.
3. **25 min: take the lowest useful rung of the hint ladder.** Then ten more minutes.
4. **35 min:** read the solution only as far as the point where it does something you did not. Close it. Carry on yourself from there.
5. **45 min: stop.** Read the whole solution, run the extraction protocol, then shelve the problem for at least forty-eight hours.

THE HINT LADDER. ALWAYS TAKE THE LOWEST RUNG THAT UNBLOCKS YOU.

1. Topic only. "This is a sequences question."
2. First move only. "Substitute." "Sketch it." "Try $n = 1, 2, 3$." "Look at the two equations together."
3. First line of the solution, then close it.
4. Solution as far as the point where it diverges from yours, then close it.
5. The whole solution.

Record which rung you needed. That number is one of your metrics. Across a set of problems it should be falling week by week. If it is not falling, your extraction is producing memories instead of rules, and §14 will tell you why.

11.3 The shelf

A problem you did not solve is not finished with you. Put it on the shelf and come back in forty-eight hours, cold. A surprising number of them fall over on the second attempt with no new input at all. Something happens in between, and that something is the thing you are actually training. If you read the solution straight away you never get to run it. That is a much better argument for waiting than "don't cheat".

---

12 Reading the examiner's cues

The vocabulary here is TMUA9.'s and it is the best description available of what a fast solver is really doing.

12.1 Command terms

The cues a question hands you. Particular phrasings, particular combinations of functions, particular configurations. The examiners are not trying to trick you. They are nudging you towards a route, and the nudge is in how the question is presented.

"A line" rather than $f(x) = mx + c$. "A circle" rather than an equation. A repeated root mentioned instead of a factor given. Ugly constants. A condition that looks unused. A question that asks *how many* rather than *which*.

12.2 Reaction circuits

Your trained response to a cue. Not a memorised solution, a first move that arrives without deliberation, because the cue and the move have been welded together by repetition.

A 9.0 sitting is mostly circuits firing, with three or four questions where you have to invent something in the room. That matches the time profile in §1.2 exactly, and it is why both halves of this book are necessary. Circuits buy you the time. Generation spends it.

> **THE CIRCUIT THAT MUST NEVER BECOME A MEMORY**
>
> A circuit is *cue to move*. It is not *question to answer*. That difference is everything, and it is the exact place where spaced repetition goes wrong (§18).
>
> If your bank is filed by technique, "the substitution method", it is unsearchable in the exam, because in the exam you are holding a question, not a technique. File everything by what you see.

12.3 The cue table

This is the bridge into Book Two, and it is worth learning before any of the individual results. Every row is something you can literally see on the screen, paired with the move it should trigger and the section that explains why.

WHAT YOU SEE	WHAT FIRES	WHERE
Constants deliberately grotesque: 2^{-33}, π^{74}, $x = -981$	The answer does not depend on them. Ask what it *can* depend on before computing anything.	§75
Area between a curve and a line, intersections given	Leading coefficient and the gap. Do not integrate.	§54
"How many solutions"	Sketch two curves and count crossings. Do not solve.	§8.3, §53
"Touches", "is tangent to", "exactly one solution"	Discriminant zero, or a repeated root, or equal derivatives.	§44
Roots named α, β but never asked for	Sum and product. Never solve the quadratic.	§45
The same compound expression twice: 2^x, $\log x$, $\sin x$, $x + 1/x$	Substitute it as a single letter, then impose its range.	§46
An absurd index: x_{2026}, the 100th term of a recursion	Compute terms until one repeats. Reduce the index modulo the period.	§63
Three terms said to be in arithmetic or geometric progression	Double the middle is the sum; square the middle is the product.	§61

WHAT YOU SEE	WHAT FIRES	WHERE
Nested similar shapes, total area or perimeter	Find the *length* ratio, then square it for areas.	§61
A functional equation: $f(3+x) = f(7-x)$	Add the arguments and halve. That is the mirror.	§50
$f(a-x)+f(x)=c$, or an integral over a symmetric interval	Symmetry inside the integral. Often the answer is half the width times c.	§51
A modulus sign in a graph question	Four cases, and only four. Inside is symmetry, outside is folding.	§52
A circle with a tangent, a chord, or a second circle	Draw to the centre, or subtract the two equations.	§57
"Is divisible by", "is prime", "for all integers n"	Parity and residues first. Two cases, ten seconds.	§64
"How many factors", "smallest k making kn a square"	Prime factorisation, then exponents.	§64
A statement offered with "which of these follows"	Translate into sets. Which condition is stricter?	§67
A stack of quantifiers and "which is the negation"	Flip each quantifier in place, left to right, then negate the claim.	§68
"Which of these is a counterexample"	The object must satisfy the hypothesis *first*.	§69
A proof printed out, "find the error"	Go straight to the flawed-step catalogue.	§70
"Exactly one of the following is true"	Rewrite all of them in one normal form and look for duplicates.	§72

Learn to read that table right to left as well. When you have finished a question, ask which row it was. If it was not on the table, you have found a new circuit, and it goes on a card.

13 Paper 2 is a different sport

Paper 1 rewards finding the cheapest valid route to a number. Paper 2 rewards deciding whether a claim is true and knowing exactly why. The habits are different enough that practising them in alternating blocks is a mistake. Give Paper 2 its own sessions.

13.1 Decide before you justify

Given a statement, your first act is a verdict. True or false. Not a proof, not a hunt for the argument, a verdict, formed by trying two or three cases. Then go looking for the reason, and now you know what kind of reason you want: a counterexample if you think it is false, a short argument if you think it is true.

Students who go hunting for a proof of a false statement lose four minutes and their composure at the same time.

13.2 Translate into sets

Nearly all the implication questions become mechanical the moment you stop trying to understand them in ordinary English. "If A then B", "A is sufficient for B", "B is necessary for A", "A only if B" and "$A \subseteq B$" are all the same sentence.

TMUA9. is blunt and correct about this: do not try to intuit the meaning, identify the structure and draw the diagram. There is one question to ask. *Which is the bigger set*, or equivalently which statement is the stricter one? The stricter condition implies the looser one, never the other way. §67 is the full table.

13.3 Negation is compartmentalising

Negating a nested quantified statement is a mechanical operation and at least one question per sitting rewards doing it mechanically. Split the sentence into its quantified pieces, flip each one in place, $\forall$ to $\exists$ and back, then negate the innermost claim. Do not paraphrase. Do not think about what it means. Perform the operation, then read what you produced. §68.

13.4 A counterexample has two jobs

To break "if p then q" you must show a case where p is true *and* q is not. An enormous number of Paper 2 marks are lost by offering something that fails the first job. It looks like a counterexample, but the hypothesis does not hold, so it proves nothing. Check both, out loud, every time. §69.

13.5 Why Paper 2 is the cheapest half of the exam

The content base is one page long. It is: true and false, *and*, *or*, *not*, the four if-then forms, converse, contrapositive, necessary, sufficient, for all, for some, there exists, negation, four proof methods, and identifying errors in an argument. That is the entire syllabus for half the exam and it can be learned properly in two days.

Most candidates never do, because it does not feel like maths. That is the single largest free-marks opportunity in TMUA preparation and Part XIV exists to close it.

PAPER 2 BLOCK, FORTY MINUTES, TWICE A WEEK

1. Work the official UAT-UK notes on logic and proof once, properly, with a pen. Then work a proof-and-logic worksheet that walks through the specification's technical vocabulary with examples. That is the entire content base, and it is small.
2. Take ten Paper 2 questions untimed. For each one write your **verdict first**, then the reason, then **name the reasoning idea being tested**: necessary or sufficient, converse, contrapositive, negation, quantifier order, counterexample, exhaustion, contradiction, vacuous truth, flawed step.
3. Wrong verdict goes on the shelf. Right verdict but you could not name the idea goes in the circuit deck as a cue card.

03

PART III

EXTRACTION

Turning one question into a rule that works on questions you will never see. This is where the score is actually made, and it is the work everybody skips because it produces no number.

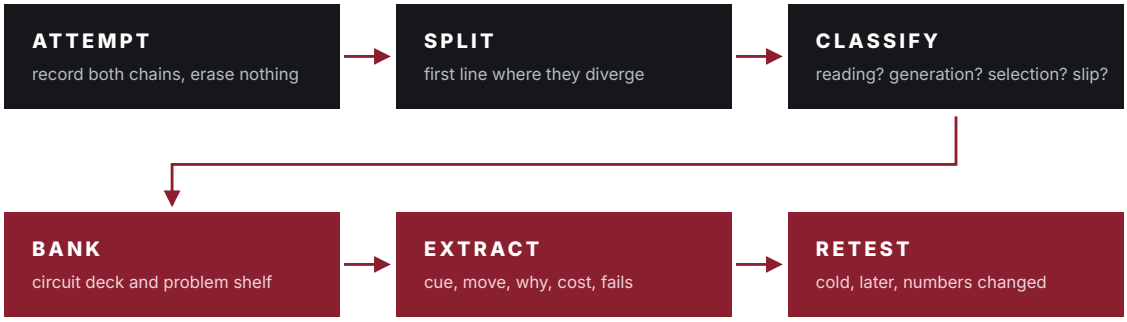
14 How to actually learn from a question
15 Why did you really get it wrong
16 Reading the wrong answers
17 The circuit card
18 Anki, done properly
19 When to redo a question
20 Lots of questions, or one question properly?

§14 and §15 are the operational heart of the book. If you read nothing else twice, read those.

14 How to actually learn from a question

The framework in this section is TMUA9.'s, presented as faithfully as a synthesis allows and then extended. It is the strongest single idea in any of the material this book is built on, and Jamie Baek attributes most of his own improvement to it.

Here is the problem it solves. You get a question wrong. You read the solution. You understand it. You think "ah, of course". You move on. That feeling of understanding is real and it is worth almost nothing, because understanding a route somebody else took does not install the ability to find it. The next unfamiliar question arrives and nothing has changed. Do that with forty questions and you have spent a fortnight to learn forty facts you will never see again.



The extraction cycle. One cycle is one unit of real work. Count these, not hours.

14.1 Layer 1: the WHAT

Put two chains of reasoning side by side. On the left, yours: the actual sequence of thoughts that took you to your answer, right or wrong or stuck. On the right, the model's. This only works if you wrote yours down while solving. There is no way round that, which is why §9 tells you to keep the honest commentary and never rub anything out.

TMUA9.'s recipe adds a refinement here that is worth adopting exactly. Record your thinking in two ways, not one:

1. the **mathematical chain**: the algebra, the substitutions, the sketches, in order;
2. the **verbal chain**: what was going through your head, in sentences, including the frustration and the false starts.

Then do the thing almost nobody does. Read your own thought process back as though you were marking somebody else's, looking for what they got wrong. In a different colour, next to each flawed step, write what it would have taken to remove it.

What you end up with is your own ideal chain of reasoning for that question, derived by you rather than handed to you. That document is worth more than any model answer, because you built it.

14.2 Layer 2: the HOW, and the exact point where you split

Find the first line where the two chains diverge. Everything before that, you got right. That single line is the entire lesson of the question and the rest of the solution is bookkeeping.

Then answer one question, and write the answer down word for word:

> *What is the critical thought I would have had to have, in the room, to take the other branch?*

For the quotient example TMUA9. uses: *"Hold on. I can see a differentiation method. But is there a cheaper one?"*

That sentence, generated by you, before the pen moved, is the whole difference. It feels impossible to imagine having thought it. That is exactly why it is worth extracting. Most candidates cannot, and you are training so that you can.

Most questions give you two or three of these, not one. Extract all of them, in whatever form they took: a sentence, a question you should have asked yourself, a reflex to write something down.

14.3 Layer 3: the WHY

Why is that the right thought? What definition, axiom or piece of structure makes it *natural* rather than clever? This is the layer that makes the lesson transfer.

"Reciprocal and expand beats the quotient rule here" is a fact about one question. "The quotient rule is derived from the product rule, so it is the same machinery at higher cost, and whenever the denominator is a simple power, rerouting is free" is a rule that fires on questions nobody has written yet. The first is a memory. The second is a circuit.

The habit generalises. When a circle question has exactly one intersection, the reflex is to draw the line joining the centres. Why? Because a circle is defined as the set of points equidistant from its centre, so the centre is where the information lives. Nothing esoteric there. Just a refusal to stop at the level of "the method".

> **THE TEST FOR A LAYER 3 ANSWER**
>
> A real WHY sentence generalises to a configuration nobody showed you. Try it: state your rule, then invent a question it should fire on. If you cannot invent one, you have written a layer 2 answer with a layer 3 label on it.
>
> Book Two is built on this. Every result there carries a one-line WHY, and the instruction is to rebuild the result from that line alone in thirty seconds (§42). If you cannot, you do not own it yet.

14.4 Layer 4: backtesting

Run this on every question you analyse. The name is TMUA9.'s and the trading analogy is right: you are testing your decision procedure against historical data.

> **THE BACKTEST**
>
> 1. **What was the first useful cue in the question?** Which word or expression should I have reacted to, and what should I have written down?

2. **What method did that cue suggest?** Did it suggest sketching? Substituting? Counting rather than solving?
3. **Where exactly did I hesitate**, and what thought would remove that hesitation permanently?
4. **What trap was the question setting?** Usually a computation the intended route avoids entirely.
5. **What would make me recognise this faster next time?** Extract the rule.
6. **What is the reusable decision rule here?**

Your accumulated answers to five and six are your circuit bank. Re-read it weekly and self-test with the moves covered up.

14.5 Say the general form out loud

One more step, and it takes thirty seconds. After you have the rule, articulate verbally what the question consists of, in general terms, and connect it to how you should have reacted.

Not "that was the 2019 area question". Something like: *"a curve and a line, intersections given, constants deliberately ugly, so the answer depends only on the leading coefficient and the gap."*

Speaking it out loud matters. Silent reading glides over the gaps. Your mouth does not.

> **ONE IS GREATER THAN ZERO**
>
> You will not do this perfectly at first. Your first ten analyses will be clumsy and you will suspect you are doing it wrong. Do them anyway. TMUA9. makes this point and it is the right one: a flawed analysis you conducted yourself is worth vastly more than reading somebody else's and nodding.
>
> Passive input feels efficient and changes nothing. Clumsy self-analysis feels inefficient and changes everything. And the notes you produce while doing it, in your own handwriting, will end up being the single most important resource you own. Not a PDF. Not a video. The messy notebook.

---

15 Why did you really get it wrong

This is the extension this book adds to the framework, and practically it is the most important thing in it.

The three-layer analysis quietly assumes one kind of failure: you never saw the route. But there are several structurally different places you can split from the model answer, and they need completely different responses. Applying the wrong response is why so much sincere, hard, exhausting revision produces nothing at all.

15.1 The four families

Before the detailed list, learn the four families, because in the room you can classify into these in about five seconds.

YOU SPLIT AT	WHAT HAPPENED	WHAT ACTUALLY FIXES IT
Reading	You solved a different problem, correctly. Misread a condition, missed a constraint, answered the wrong quantity.	A reading protocol: restate in one line, list the givens, tick each one as you use it, and re-read the last sentence of the question after you compute. Not more topic revision.
Generation	You never saw the route, even with time.	This is the one the three-layer analysis is built for. Extract the cue and the critical thought, write the card, then go and meet that structure again elsewhere.
Selection	You saw two routes and picked the expensive one.	A cost drill: re-solve five recent questions by a shorter method than you first used. You do not need new knowledge. You need the habit of pricing routes before committing to one.
Execution	Right route, arithmetic or algebra slip.	A slip log with the specific slip named. Grouping habits. A verification step. Never "revise the topic", because there is no topic gap.

NOT EVERY MISTAKE IS A KNOWLEDGE GAP

Revising the topic is the default response to every error, and it is the wrong response to three of the four families above.

Somebody who responds to a selection failure by rewatching a topic video has spent an hour reinforcing knowledge they already had, while the actual defect, not pricing routes, goes completely untouched. Then they conclude the material was the problem and go hunting for a better resource. That loop can eat a whole term.

Before any remedy, ask one question. **Did I not know this, or did I not do this?** Those are different illnesses and they do not share a cure.

15.2 The full taxonomy

Every wrong or guessed question gets exactly **one** primary label. Do not allow yourself two. Forcing a single choice is what makes the log tell you something.

CATEGORY	HOW YOU KNOW	REMEDY, AND HOW LONG IT TAKES
Knowledge gap	You did not know a definition, identity or standard result.	The relevant section of Book Two, plus ten targeted questions. **Days.**
Reading failure	Your answer is right for a question nobody asked.	The restate-and-tick protocol in §7.2. **Weeks,** and only with deliberate attention.
Structure blindness	You had the knowledge but did not see what kind of object you were holding.	A circuit card filed on the cue, not the topic. Work the cue table in §12.3. **Months of accumulation.**
Generation failure	With unlimited time and all your notes you still could not find it.	Extraction, then deliberate exposure to that family of structures. **Months.**
Selection failure	You found a route. A much shorter one existed and you could have seen it.	Two-route drill and cost drill (§21). **Weeks.**

CATEGORY	HOW YOU KNOW	REMEDY, AND HOW LONG IT TAKES
Assumption error	You assumed something the question never said: that a diagram is to scale, that a variable is positive, that a root is real, that a sequence starts at $n = 1$.	Write the assumption down as a line in your working, then challenge it. This is the single most common cause of a *confidently* wrong answer. **Weeks.**
Overcomplication	Your route was valid and four times longer than necessary, usually because you reached for the most powerful tool rather than the cheapest.	The zombie-maths check in §10: before line two, ask what the cheapest thing that would settle this is. **Weeks.**
Case loss	You solved it, correctly, and missed a branch. A negative root. The $a = 0$ case. The second solution in the interval. The other side of the modulus.	Case-checklist habit (§48 and §52). Ask "how many branches should this have" *before* solving. **Weeks.**
Execution error	Right route, wrong arithmetic or sign.	Named slip log, grouping, verification. **Weeks.**
Verification failure	The answer was implausible and you did not notice.	Estimate-first drill: sign, size, and which options are structurally impossible, *before* computing. **Fast to fix, high yield.**
Decision failure	Right route, but you overran and never came back, or gave up too late, or abandoned something you were ninety seconds from.	Hard-stop drill and the new-information test. **Weeks.**
Distractor capture	Your answer was an option built out of a specific common error.	Distractor autopsy (§16). **Weeks.**
Guessed correctly	You got the mark and did not earn it.	**Log it as a failure.** Unearned marks inflate every metric you have and will hide a real gap until the day it costs you.

READ THE MIX, NOT THE COUNT

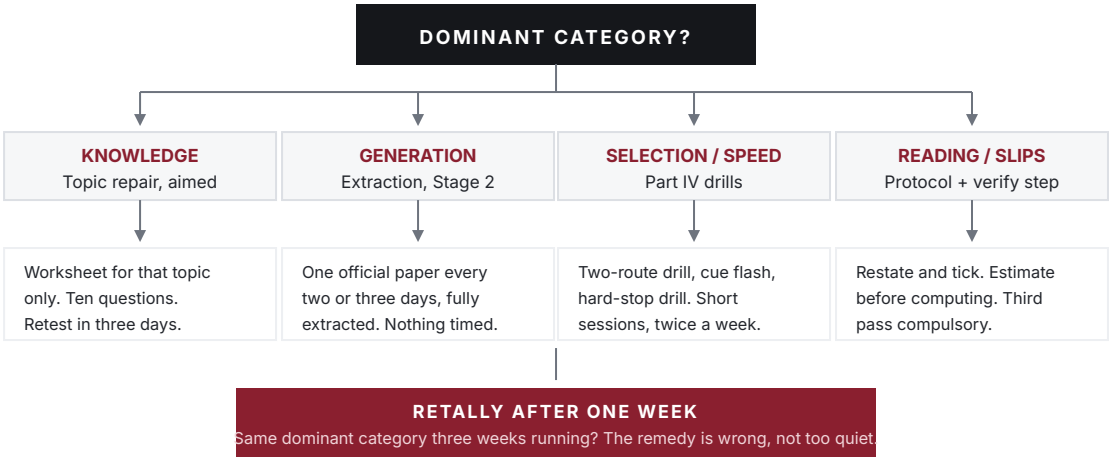
The *number* of errors tells you your score, which you already knew. The *mix* tells you what to do next.

Six execution errors and one generation failure is a completely different student from one execution error and six generation failures, even though both of them scored 13 out of 20. One needs two weeks of verification habits. The other needs two months of extraction.

Tally the mix every week, and when the dominant category changes, change what you are doing.

15.3 The remedy tree

When you have a week's tally in front of you, this is how to spend the next week.



The remedy tree. The whole point of a taxonomy is that it chooses your next week for you.

16 Reading the wrong answers

TMUA options are not random. The wrong ones are built out of specific anticipated errors: the sign you drop, the factor of two you forget, the case you do not consider, the root you should have rejected. The exam says as much about itself, and it is an enormous free gift, because it means the paper contains a model of the mistakes you are likely to make.

> **THE AUTOPSY. NINETY SECONDS PER ANALYSED QUESTION.**
>
> For each of the four wrong options, answer one question: **which precise error produces this option?**
>
> Some are obvious. Sign error. Forgot to halve. Some you will not be able to reverse-engineer, and that is fine, leave them. The ones you can identify go in your slip log, because that is the exam telling you which mistakes it expects from you.

Two payoffs, one in training and one in the room.

In training, after twenty questions of this you will have a personal list of six or seven error patterns that keep recurring. That list beats any generic “common mistakes” article, because it came from the real exam and was filtered through your actual behaviour.

In the room, when you finish a question and your answer is sitting there as an option, that is weak evidence you are right. But when you can see which error would have produced one of the other options, and your answer differs from that one by exactly that error, check again.

16.1 The distractor catalogue

After enough autopsies the same constructions keep appearing. Learning them in advance means you can often name a distractor before you have solved the question, which is worth real time.

THE WRONG OPTION IS BUILT FROM	WHICH LOOKS LIKE	GUARD
The root you should have rejected	$t = 2^x$ with a negative root, $u = \sqrt{x}$ with $u < 0$, a log with a non-positive argument	§46
The other branch of a square root	Keeping only $+\sqrt{}$, or writing $\sqrt{x^2} = x$	§70
A missing factor of 2, or an extra one	Sector against triangle, chord against half-chord, area against double area	§57
The converse	Proving $A \Rightarrow B$ and answering as though $B \Rightarrow A$	§67
Degrees where radians were needed, or the reverse	A sector or arc formula with a degree measure in it	§53
Transformations applied in the wrong order	$f(ax + b)$ read as stretch-then-shift	§50
An endpoint included or excluded wrongly	Two options differing only by $<$ against $\leq$, or by one	§64
The special case not separated	$a = 0$ in "$ax^2 + bx + c$", $n = 1$ in a sum formula, $r = 1$ in a geometric series	§44, §61
Counting the solutions in the substituted variable	Answering how many t, not how many x	§46
Sum of roots read with the wrong sign	b/a instead of $-b/a$	§45
The answer to the question before the last sentence	You found m; they asked for c	§7.2

DO NOT OVERCORRECT INTO PARANOIA

If your answer matches an option and you cannot name a reason to doubt it, take the mark and move on.

The autopsy is a training technique that leaves behind a light instinct in the exam. It is not permission to re-check everything, which is a decision failure wearing a diligence costume.

17 The circuit card

The atom of the whole system. Five lines. Written after extraction, filed by what you see.

CUE	What in the question fires this. The phrasing, the picture, the structure. Written as you would first meet it.
MOVE	The first thing to write down. One line. Not a solution.
WHY	The definition or fact that makes this the cheap route.
COST	What it saves against the obvious route. Be concrete: "40 s instead of 4 min", "avoids a quartic".
FAILS	When the cue is there but the move is wrong.

The FAILS line is not optional and it is the one people leave out. A rule you cannot falsify is not a rule, it is a memory with a confident voice. Forcing yourself to name a case where the cue appears but the move is wrong is what stops you misfiring on a question that only *looks* like one you know.

TWO CARDS IN THE FORMAT

TWO SIMULTANEOUS EQUATIONS, SAME TWO OBJECTS CORE

CUE	Two simultaneous equations in the same two compound expressions, one looking linear and one looking quadratic.
MOVE	Name them A and B, then look at the relationship *between* the equations before touching either.
WHY	A quadratic-looking pair over the same objects is nearly always a difference of squares, which factors through the linear one.
COST	One line instead of squaring and grinding.
FAILS	When the equations involve different objects, or only one substitution is available, in which case it is an ordinary substitution.

AREA WITH DELIBERATELY UGLY CONSTANTS CORE

CUE	Area between a curve and a line, intersection x-coordinates given, constants deliberately grotesque.
MOVE	Write down the leading coefficient and the gap between the x-coordinates. Do not integrate.
WHY	For the standard configurations the area depends only on those two things. The tilt of the line, the constant term and the roots themselves never enter.
COST	Ten seconds instead of a definite integral with absurd limits.
FAILS	When the configuration is not one of the standard ones, so check the shape and the multiplicity of the roots first.

Book Two writes every result in exactly this shape, with one extra line, DEFAULT, so you always have a slower method to fall back on. §42 explains the mapping.

18 Anki, done properly

Viluxent's Anki method is the best capture system in any of the source material, and the reason it works is that it has almost no friction. Any question you get wrong, or cannot finish in time, gets screenshotted into a deck. The deck gets reviewed every day. Cards are never deleted when you get them right, only graded. Everything gets unsuspended before the exam for a final sweep.

Keep all of that. It is right. But the system has four failure modes that show up around week three, and it is worth being precise about them so you can design around them rather than quietly giving up.

WHAT GOES WRONG WITH A SCREENSHOT DECK

1. **Recognition replaces generation.** By the third review you recognise the picture and recall the answer. The card now tests your memory of one specific problem, which is exactly the thing that will not transfer to an exam that never repeats a question.
2. **There is no cue index.** You are reviewing problems, not triggers. In the exam you retrieve from what you see, so a bank organised by problem is hard to search under pressure.
3. **The grading criterion is wrong.** "Did I get it right" is not the useful signal. The useful signal is *did the first move arrive without searching for it.*
4. **Volume creep.** Screenshot everything and you have four hundred cards by the sitting. At that point the deck is a chore that eats forty minutes a day and returns very little, and you will quietly stop doing it, usually in the exact fortnight you needed it most.

The fix is not to abandon it. It is to split it, because right now it is doing two jobs that need different mechanics.

18.1 Deck A: the circuit deck. Small, daily, fast.

Front is the cue only. Not the question. The stripped structural fragment. *"Statement of the form: for all N there is K such that, for all m. Asked for the negation." "Nested similar shapes, asked for a total area." "Two circles, one common tangent, asked for a length."*

Back is move, why, cost, fails.

Grade on arrival, not correctness. AGAIN if you had to think about what to do. GOOD if the move came but you had to reconstruct why. EASY only if it arrived instantly and felt boring.

Sixty to a hundred and fifty cards. It should take eight minutes. If it takes thirty you have put problems in it.

That last grading criterion is the right one. TMUA9. describes the signal of real internalisation as the material *feeling boring*, with almost no resistance in the chain of reasoning. Boredom is the target state, not a warning sign. When a card stops being interesting, it has started working.

18.2 Deck B: the problem shelf. Small, slow, cold.

The actual questions you failed. A screenshot is fine. Viluxent is right that friction kills capture.

— **Long intervals only:** 7, 21, 60 days. Never review a problem within forty-eight hours of first meeting it. Recognition is at its ceiling and you will learn nothing while feeling excellent.

— **Review means re-solving cold on paper**, not flipping a card. Flip it and you are testing recall of a solution.

— **Cap it at forty live problems.** Above that you are adding faster than you are learning. Stop adding and clear the backlog. This one rule is what keeps the system alive.

Retire a problem when you solve it cold, inside target time, by the efficient method, twice, at least two weeks apart. On retirement, pull it into Deck A if it produced a rule. Otherwise delete it. A retired problem that produced no rule was never worth keeping.

18.3 Three ways to stop it becoming memorisation

1. **The expiry rule.** If on review you recognise the answer before you have finished reading the question, that card has expired as a test. Convert it to a circuit card and delete the problem.
2. **Write your own version.** Change the numbers, alter one condition, swap what is being asked for. Then solve yours. This is the strongest anti-memorisation device there is and it doubles as the deepest possible check that you understood the structure. TMUA9.'s study loop includes exactly this step and it is the one everybody skips.
3. **Explain it out loud with the page turned over.** If your explanation contains "and then you do this bit", you have a memory rather than an understanding.

18.4 The final sweep

Viluxent's instruction to unsuspend everything before the exam is right, with one refinement. Sweep Deck A. The circuit deck can be swept in an hour and it reactivates every trigger you own. The problem shelf cannot be swept meaningfully, and re-reading forty solutions in the last week produces false confidence and no new capability.

In the final ten days you want to be firing cues, not consuming answers.

---

19 When to redo a question

WHAT HAPPENED FIRST TIME	FIRST RETRY	SECOND	RETIRE WHEN
Solved, but slowly or clumsily	3 days	14 days	Cold, inside target time, by the efficient method.
Failed, understood the solution the instant you read it	2 days, re-derive	10 days, re-solve	Cold, twice.
Failed, needed the whole solution and real effort to follow it	48 h re-derive, then 7 days	21 days	Cold, twice, at least a fortnight apart.
Failed twice	Rebuild the underlying topic first, requeue at 7 days	21 days	As above. Two failures means the gap is upstream of the question.
Right answer, wrong or absent reasoning	Treat exactly as a failure		As for a guessed question. The mark was luck.

> **RE-DERIVING IS NOT RE-SOLVING**
>
> Inside about forty-eight hours you cannot re-solve a problem, only re-derive it. You remember the route, and following a remembered route tests nothing except whether you were paying attention.
>
> Both are useful and they are useful for different things. Re-deriving soon consolidates the *why*. Re-solving after genuine forgetting, usually a week or more for a hard problem, tests whether the rule *transferred*. Schedule both on purpose and never confuse one for

the other.

20 Lots of questions, or one question properly?

Everybody asks this and the honest answer is that the question is underspecified. “Should I spend an hour on one question or do twenty?” depends on what that hour buys, which depends on where you are.

WHERE YOU ARE	LEAN TOWARDS	BECAUSE
Early, foundations shaky	**Volume**	You need to meet the standard structures before deep analysis has anything to bite on. Depth on a question you lack the vocabulary for is just slow confusion.
Foundations solid, ceiling below target	**Depth, heavily**	This is the extraction phase. One official paper pulled apart properly beats five sat superficially, and this is where scores move.
Ceiling high, timed score low	**Volume, timed**	You are training decisions, and decisions need variety under pressure.
The question exposed a real blind spot	**Depth**	Whatever stage you are at. A blind spot found is worth an hour of anybody’s evening.
You have already met four questions of this type	**Volume**	Diminishing returns. Extract the rule and move.
Final fortnight	**Volume**	There is no time left to turn new depth into reliability. Consolidate, sweep, simulate.

THE CADENCE THAT SETTLES IT IN PRACTICE

TMUA9. gives a specific and workable answer. One past paper taken through the full analysis every two to three days beats a paper a day at surface level, with analysis running roughly double to triple the sitting time. A seventy-five minute paper therefore earns two and a half to three and a half hours of extraction.

Two more rules from the same place, both cheap and both effective.

Never end a session on paper-sitting. End on analysis. The extraction is what consolidates overnight. The sitting is only raw material.

You do not need to sit every paper strictly timed before analysing it. Run a stopwatch to feel the flow of time and solve as much as you can. If you do not finish, that is better, not worse. You now have more raw material to compare against the model answer.

04

PART IV

SPEED

Speed is not a personality trait and it is not produced by hurrying. It is bought before the pen moves, and it breaks into six separate things, only one of which most people ever train.

21 Six kinds of slow

22 Using the answer choices

23 The noteboard nobody practises

§21 tells you which kind of slow you are. §23 is the one nobody rehearses and it is pure free marks.

21 Six kinds of slow

"Practise more and you will get faster" is true the way "eat less and you will lose weight" is true. Correct on average, useless as an instruction.

Being slow is a symptom and there are six different illnesses behind it. Work out which one you have, from your error log and from watching yourself on a paper, and train that one.

KIND	YOU HAVE THIS IF	THE DRILL
Reading	You read questions two or three times. You sometimes solve for the wrong quantity.	**Restate drill.** Twenty questions, twenty seconds each, no solving. Write only: what object, what is asked, what conditions. Mark the restatement, not the answer.
Recognition	There is a long silent gap between finishing reading and starting to write.	**Cue flash.** Twenty questions, thirty seconds each. Write only the first move. Do not solve. Mark the move against a worked solution.
Strategic	You start, abandon, restart. Two false starts a paper.	**Two-route drill.** On ten questions write both routes you can see and predict which is cheaper, *before* solving. Then do both and score your predictions.
Execution	Route was right, the arithmetic was slow or wrong.	**Grouping habits.** Squares to 25, cubes to 12, powers of 2 to 2^{12}, common surds and exact trig values, cold. Deliberately writing fewer lines.
Decision	You overrun on two or three questions and do not finish.	**Hard stop.** Sit a paper with an alarm every 3:45. You must move on when it goes off, whatever state you are in. Three of these and the instinct installs.
Verification	You submit answers with the wrong sign, wrong order of magnitude, or outside an obvious range.	**Estimate first.** Before computing, write the sign, a rough size, and which options are structurally impossible. Then compute.

WHERE THE TIME ACTUALLY IS

A question solved by the intended route takes forty seconds. The same question by brute force takes four or five minutes and is much more likely to be wrong, because five minutes of algebra has five times the surface area for a slip. So the brute-force route does not cost you four minutes. It costs you four minutes **plus** the expected value of the mark.

Both of the people credited on the cover reach this conclusion independently. Speed is a by-product of better routing, never of faster handwriting, and writing faster actively backfires because it raises the slip rate and a slip forces a restart.

21.1 The one number worth chasing

Do not chase your average time per question. Chase this:

> *How many questions on this paper did I finish in under ninety seconds?*

Aim for ten out of twenty. That number is the best available proxy for how many trained circuits you own, and it is what creates the room for the four questions that need thinking about.

Track it on every paper. It climbs before your score does, which makes it the early sign that the work is landing, and on a week when your score refuses to move it is the thing that will keep you going.

21.2 Speed you should not chase

There is a point where hunting for elegance costs more than it saves. If you have a valid route in front of you that takes ninety seconds and you suspect a twenty-second route exists, take the ninety-second one in the exam and find the elegant one in the analysis afterwards.

Hunting for beauty under time pressure is a decision failure, not sophistication. Elegance is built in training and spent in the exam, never the other way round.

21.3 The arithmetic you should own cold

Execution speed is the only one of the six that is genuinely mechanical, which makes it the fastest to fix. None of this is clever. All of it is worth about a mark.

Squares	To 25^2, and $30^2, 40^2, \ldots, 90^2$. Also 11^2 to 19^2 without thinking.
Cubes	To 12^3.
Powers of 2	To $2^{12} = 4096$, and $2^{10} \approx 10^3$ as an estimating anchor.
Surds	$\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$, and rationalising in one line.
Exact trig	The five standard angles, from the two standard triangles, rebuilt in five seconds rather than recalled.
Triples	$3, 4, 5$; $5, 12, 13$; $8, 15, 17$; $7, 24, 25$, and their multiples. Spotting one saves a Pythagoras.
Logs	$\log_{10} 2 \approx 0.30$, $\log_{10} 3 \approx 0.48$. Enough to place a number between two powers of ten.
Fractions	Halves, thirds, quarters, fifths, eighths and twelfths as decimals, both ways.

22 Using the answer choices

Multiple choice is not a weaker form of assessment. It is a different one, with structure you can exploit. Most candidates use the options purely as a place to put their answer, which is like being given the answers to half the question and ignoring them.

22.1 Five techniques, most profitable first

1. **Structural elimination.** Before computing anything, check sign, parity, degree, order of magnitude, and whether the option is even the right *kind* of object. This routinely kills two options in ten seconds and costs you almost nothing. §75 is the content-side version and it is worth more than this whole section.

2. **Substitute a specific value.** Pick a legal value of the parameter and evaluate all five options. Anything that fails is gone. Devastating on identity, transformation and sequence questions, and frequently faster than solving.
3. **Push it to the boundary.** Send a parameter to 0, to 1, or to its extreme legal value and see which options survive. Very strong on inequalities and on "for all n" claims.
4. **Work backwards from the options.** Substitute them into the condition. Use this **only when checking is cheaper than solving.** That is the whole criterion. If checking costs as much as solving, this is a trap rather than a technique.
5. **Read the options for the intended method.** They leak information. All of the form $a + b\sqrt{3}$? An exact trig value is involved. Differing only in a sign? The question is about a sign convention or a direction. All integers except one? That one is either the answer or the designed error. Three seconds here before you choose a method. §75 has the full table.

22.2 When not to use them

— When the options are numerically close. Elimination by estimate will not separate them and you will only fool yourself.

— When your likely error would produce one of the options, so checking reproduces the error.

— When there are eight options rather than five. Substitution still works, but working backwards gets expensive. Prefer structural elimination.

22.3 Guessing

Nothing is deducted, so this is pure arithmetic. Blind guess on five options: 0.2 marks. Kill two: 0.33. Kill three: 0.5. Across the five or six questions you will not fully solve, disciplined elimination is worth about one extra raw mark, and at the top of the scale raw marks are the entire difference.

> **THE RULE**
>
> Answer every question **before** you flag it, never after. A flag is a reminder to come back, not a placeholder for an empty box.
>
> If time runs out, a flagged question with an eliminated guess in it is worth a third of a mark. A flagged blank is worth nothing at all.

---

23 The noteboard nobody practises

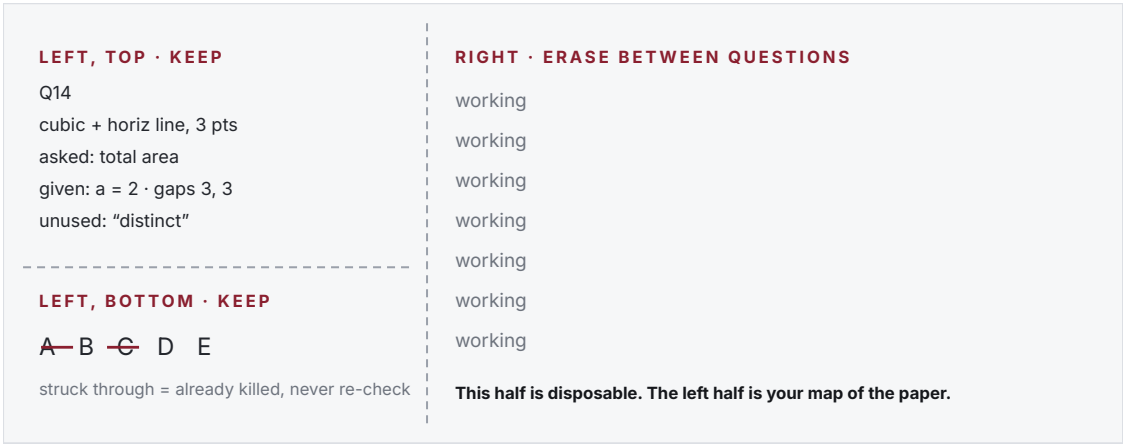
You will not be handed the paper. You will be handed a screen and an erasable board. Everybody prepares on printed papers, writing next to the question, crossing options out in the margin, circling the given conditions. On the day none of that exists. The consequences are concrete and they cost real marks.

— **You cannot cross out an option.** Elimination has to live on the board, which means you need a fixed rehearsed notation for it, or you will end up re-evaluating an option you already killed.

— **You cannot annotate the given conditions.** So the habit from §7, finding the unused condition, has to be done by writing the givens out rather than circling them.

— **Board space runs out.** Erase everything after each question and you cannot revisit your working on the third pass. Erase nothing and you have nowhere left to write.

— **Diagrams have to be redrawn**, because the figure on screen cannot be marked up. Take the wording as authoritative, not the picture: on-screen diagrams are not to scale.



The board layout. Set it up in the first thirty seconds, before you read question one.

THE RULE FOR THE BOARD

Between questions, erase **only the right column**. Keep the left column entries for anything you flagged, so that on the second pass you *resume* instead of starting again.

Restarting a flagged question from nothing is one of the largest hidden costs in the exam and it is entirely avoidable.

Rehearse this on at least three full papers. Print nothing. Work on a single reusable sheet in a plastic wallet, or a small whiteboard.

THE MOST AVOIDABLE LOSS IN THE WHOLE EXAM

Sitting a computer exam having practised only on paper.

Interface unfamiliarity does not make you worse at mathematics. It makes you slower for the first fifteen minutes and more anxious for the first five, which is precisely when the rhythm of your first pass is being set.

Two hours on the official practice and specimen tests removes this completely, and it is the best return per hour anywhere in this book.

ONE EXCEPTION TO “YOUR ERASER IS YOUR ENEMY”

In training, never erase. The wrong routes are the raw material for your analysis. In the exam, you have a finite board and no time to analyse anything, so erase the right column freely.

The two rules are not in conflict, they belong to different activities. Just make sure you know which one you are in.

05

PART V

MATERIAL

Not all practice is the same practice. Six kinds of resource, what each one builds, what each one cannot build, and the order to take them in.

24 What each resource is for
25 The order everybody agrees on
26 The six stages
27 The full order, resource by resource
28 Which community papers to bin
29 Things that feel productive and are not
30 Stop grading yourself

§27 is the full order. §30 is why you should stop grading yourself.

24 What each resource is for

The commonest preparation failure is not laziness. It is using a good resource for the wrong job: burning official past papers as timed measurements, or grinding community papers for exposure before you have anything to expose. So classify before you choose.

CATEGORY	WHAT IT BUILDS	WHAT IT CANNOT BUILD	WHEN IT IS WRONG
A. Official TMUA past papers ***2016 to 2023, plus specimens***	Everything, if analysed. The exam's own idea of what a question is: phrasing, cue design, how distractors are built, the difficulty gradient. The only fully authentic training data in existence.	Volume. There are about eighteen, nowhere near enough for exposure.	Any time you sit one as a timed measurement before extracting it. Irreplaceable material spent to produce a number.
B. Foundations and technique ***topic worksheets, topic MCQ, logic notes***	The content floor and the topic-level circuits. Turns "I sort of know this" into "this is automatic".	Tolerance for the unfamiliar. Everything is labelled by topic, so you never have to work out what you are looking at.	When you are already fluent. Redoing foundations you have is the most comfortable way to avoid the work that would move you.
C. Adjacent official banks ***MAT, ECAA, ENGAA, PAT, AMC, STEP***	Idea generation on unfamiliar problems, at professional question quality, in effectively unlimited quantity, free. The best substitute there is for more TMUA papers.	Format fidelity. The accent differs. MAT is more constructive, AMC more combinatorial, STEP longer form. Some of what they train is not tested.	Before the foundation exists, when it is just confusion at a higher difficulty. Or as a way of avoiding official papers because those feel higher stakes.
D. Community TMUA papers	Format fidelity in volume, and timed decision-making. The only place to get forty TMUA-shaped papers.	Calibration. No community paper has a cohort, so its difficulty and its boundaries are guesses. Quality varies enormously.	Early, before the official papers have been analysed. This is the single commonest sequencing error.
E. Topic repair	Fast closure of a diagnosed gap. Very high yield when aimed by the error log.	Anything you have not diagnosed.	When it is your default response to every error.
F. Retention ***circuit deck, problem shelf, error log***	Retention of everything above. Without it roughly half of what you extract evaporates.	New capability. The deck does not teach, it stops you forgetting.	When it has grown past the point where you will actually do it every day.

25 The order everybody agrees on

There is a lot of noise online about what to do first, and it dissolves once you notice that the people who have actually scored at the top all describe the same sequence.

THE AGREED SEQUENCE

1. **Build the foundation.** Topic worksheets and topic multiple choice until the content is automatic. Viluxent's phrasing: what is the Taj Mahal without its foundations?
2. **Analyse the official papers properly, untimed.** Not *sit* them. *Analyse* them. This is the step that everything else depends on, and it is the step people skip.
3. **Then everything else is fair game.** MAT, ECAA, ENGAA, AMC, STEP Foundation, STEP 1, community papers, mocks. TMUA9.'s own guidance says the same thing: adjacent resources are offered and they help, but if the earlier steps have not been done properly you are wasting your time on them.
4. **Simulate at the end.** The most recent and most realistic papers, timed, on screen, saved for last.

The condition attached to step three is the whole point, and it is worth saying plainly. Community papers and adjacent exams are genuinely useful. They are not a distraction and nobody serious thinks they are. What ruins them is doing them *instead of* the analysis, so that you accumulate hundreds of attempted questions and no extracted rules.

COLLECTING IS NOT PREPARING

There are roughly two hundred community papers in circulation. The marginal value of the two hundred and first is zero and the marginal cost of hunting for it is an evening.

Viluxent's instruction is exact: filter, and do not save the best for last. Material you can learn from should be used while you can still learn from it.

TEACHING MATERIAL AND MEASURING MATERIAL

That last instruction seems to argue with "save the recent papers for the end". It does not, and the distinction settles most sequencing questions you will ever have.

Teaching material should be used as early as you can benefit from it. Hoarding it wastes the time in which it could have compounded.

Measuring material has to be reserved. Once you have seen a paper it can never measure you again. Keep two or three genuinely unseen, recent, realistic papers for the final fortnight and do not touch them before.

The best community papers are teaching material. The most recent official papers are measuring material. Use each as what it is.

26 The six stages

STAGE	WHAT IT IS
0 Calibrate	The two-score diagnostic. One evening. Repeat every three weeks.
1 Floor	Topic worksheets, topic MCQ at 3 min a question, logic and proof.
2 Extract	Official papers 2016 to 2023, untimed, full three-layer analysis.
3 Stretch	MAT, ECAA, ENGAA, AMC, STEP Foundation, STEP 1.

STAGE	WHAT IT IS
4 Simulate	Community papers, timed, on screen. Decision training.
5 Peak	Recent papers and official interface practice, full conditions.

Running underneath all six, every single day: the circuit deck, the problem shelf, and the error log. Those are not a stage. They are the ground the stages stand on.

26.1 Move on when you pass the test, not when the week ends

STAGE	YOU MAY LEAVE WHEN	IF YOU CANNOT
0	You have U, T and an error mix written down.	You have not done it. It takes one evening and it will save you a month.
1	8 out of 10 on any topic multiple-choice set at three minutes a question, on a topic you have not just revised.	Do that topic's worksheet properly, with a pen rather than by reading, and retest in three days.
2	Untimed ceiling of 17 out of 20 on an unseen official paper, and your dominant error is generation, not knowledge gap.	Knowledge-gap dominance means you left Stage 1 too early. Go back. There is no shame in it and everybody who plateaus refused to.
3	On a genuinely unfamiliar problem you reliably make real progress inside ten minutes. Measured as: hint rung 0 or 1 on most attempts.	More exposure, and check you are extracting rather than accumulating.
4	Timed score within two marks of your untimed ceiling, across three papers.	The gap is a decision problem, not a maths problem. Hard-stop drill, then retest.
5	Two full paired sittings under real conditions at target, back to back, on screen.	You are where you are. Stop adding material, sweep the circuit deck, and rest.

27 The full order, resource by resource

This is the operational table. The order reflects increasing difficulty and development rather than chronology, and it is built on Viluxent's recommendations, cross-checked against what each resource actually is.

> **HONESTY ABOUT THIS TABLE**
>
> The comments on each resource follow its publisher's own stated intent and the recommendations in the source material. They are not the result of an independent review of every paper and should not be read as one. Where there is no evidence about a set, the table says so rather than inventing a verdict. Everything named here was free or freely findable at the time of writing.

#	RESOURCE	WHAT IT IS FOR	TIMED?	AFTER THE FIRST ATTEMPT
STAGE 0 · CALIBRATE				
0	Any one official TMUA paper, unseen	The two-score diagnostic (§5). Sit it to find your starting point, **not** to get a mark.	Timed, then untimed	Record U, T, error mix. Do not grade it. Repeat every 3 weeks on a new paper.

#	RESOURCE	WHAT IT IS FOR	TIMED?	AFTER THE FIRST ATTEMPT
STAGE 1 · FLOOR				
1	Official UAT-UK TMUA specification	Know exactly what is examinable. One hour, once.	No	Mark the topics you cannot confidently define.
2	Jacqueline Tyler Tutoring topic worksheets	Core knowledge plus TMUA-style technique, topic by topic, across the nine topic areas	No	Every error into the log. Redo that section, not the whole sheet.
3	Jacqueline Tyler Tutoring topic multiple choice	Turning knowledge into speed	Yes, 3 min	Wrong or overtime goes to the circuit deck or the shelf. Repeat until 8/10 at 3 min.
4	Official UAT-UK notes on logic and proof	The entire content base for Paper 2	No	Write your own example of every term. Once more before the exam.
5	JZ Maths logic notes	The same ground in a second voice, with fuller worked examples and exercises	No	Cue cards for every term. Read once properly, then use as reference.
6	Vantage TMUA Primer notes, one lesson per topic	A careful topic-by-topic second pass with worked examples	No	Use for any topic still shaky after 2 to 5. Book Two carries the distilled version.
STAGE 2 · EXTRACT. THIS IS WHERE THE SCORE IS MADE.				
7	Official TMUA 2016 to 2019, both papers	Learning the exam's own idea of a question	No. Stopwatch for awareness	Full three-layer analysis. Analysis 2 to 3 times the sitting time.
8	Official TMUA 2020 to 2023, both papers	As above. The later papers sit closest to current style.	No, until Stage 4	As above, plus a distractor autopsy on every question.
9	Official specimen and practice papers	More authentic material	No	As above.
10	Somebody else's full worked analysis of a paper	A second chain of reasoning to juxtapose against yours, per question	No	Write your own analysis first, then compare. Never the other way round.
STAGE 3 · STRETCH. UNFAMILIAR PROBLEMS, PROFESSIONAL QUALITY, MOSTLY FREE.				
11	MAT Section A, multiple choice	The single best substitute for more TMUA Paper 1	Loosely, about 4 min a part	Target 8/10 or better. Anything below that, extract it.
12	ECAA Advanced Maths, Section 1 Part B only	Close in style. Ignore the rest of the paper.	Loosely	Extract, shelf the misses.
13	Hodder *STEP, TMUA and MAT* book	Structured technique across all three exams, in one place	No	Skip the STEP-only material if it is not serving you.
14	JT question sets	Key techniques drilled in TMUA shape	Loosely	Extract.
15	STEP Foundation modules	Real problem-solving on A-level content only	No	Skip modules with no TMUA relevance rather than grinding them.

#	RESOURCE	WHAT IT IS FOR	TIMED?	AFTER THE FIRST ATTEMPT
16	AMC 12, questions 1 to 15	Fast idea generation on short unfamiliar problems. Outstanding value.	Yes, 3 min each	Extract. Generation failures surface here fastest.
17	ENGAA Section 1 Part B, maths only	More multiple choice under pressure	Yes	Extract.
18	STEP 1	The bridge between TMUA and real STEP. A-level syllabus only.	No	Anything missed or overtime goes to the deck. Do not be put off by the word STEP.
19	PAT maths, ignore the physics	Volume of quick multiple choice, speed	Yes	Light extraction only.
STAGE 4 · SIMULATE. COMMUNITY PAPERS, TIMED, ON SCREEN. SEE §27.1 FOR THE LIST.				
20	tmua.io	A realistic Pearson-style interface for sitting community and official papers, with a timer, flagging and a review screen. Your default way of sitting anything timed from here on.	Yes	Analyse as a paper, never as a score.
21	The community mock series in §27.1	Format fidelity in volume, and timed decision-making	Yes, strict	Every overrun logged as a decision failure. Full analysis the next day.
STAGE 5 · PEAK. FINAL TWO TO THREE WEEKS.				
22	Official Pearson VUE practice tests	Real past questions in the real layout	Yes	Rehearse the noteboard (§23).
23	Official Pearson VUE specimen tests	Original questions, interface familiarity	Yes	As above.
24	The most recent reconstructed sittings, saved	The closest available simulation of the current standard	Yes, full conditions, both papers back to back	Full analysis the next day. Keep the last one for the final week.

ONE VARIANT OF THIS ORDER THAT ALSO WORKS

The table puts the official papers at Stage 2, before the adjacent banks, because analysing them is what teaches you the exam's own idea of a question.

A second ordering, recommended by strong candidates and equally defensible, runs the adjacent banks **first** — MAT, ECAA, the Hodder book, JT sets, STEP Foundation, AMC 12, ENGAA, STEP 1 — and comes to the official TMUA papers once the mathematics feels comfortable. It trades a little cue familiarity for a lot of raw problem-solving before you spend irreplaceable material.

Either is fine. What is not fine is the third option, which is doing neither properly and going straight to timed mocks. And in both orderings the same two rules hold: **untimed with full analysis first, timed only once you are comfortable**, and **keep the post-2023 papers back** for the very end.

27.1 The community mock series, by name

Once you are comfortable enough that timed work is measuring decisions rather than mathematics, this is where the volume comes from. Not in any order of merit, and every one of them worth triaging first with §28.

SERIES	WHAT IS ACTUALLY KNOWN ABOUT IT
tmuamockz	Reported as the most realistic of the lot. If you sit only one series under strict conditions, sit this one, and save two of its papers for the final fortnight as measuring material (§25).
tmuaguru	A large set. Good for volume once your decisions, rather than your mathematics, are what you are training.
tmua9. mocks	Built around the same circuit philosophy that Part III of this book runs on, so the distractors tend to be designed rather than random, which makes them unusually good for the autopsy in §16.
viluxent mocks	Free, and written by one of the two names on the cover of this book.
dpid mocks	Recommended in the source material this edition was built from.
tmua euclid	Recommended in the source material this edition was built from.
jz maths papers	Stretch under time. Deliberately hard by their author's own description, so expect a low mark. Extract it and do not take the score personally: a hard paper sat badly is still four or five circuit cards.
hercules	In circulation and worth triaging. No independent verdict beyond that.
tmua.co.uk	In circulation and worth triaging. No independent verdict beyond that.
james chen papers	In circulation and worth triaging. No independent verdict beyond that.

Where the note says only that a set is in circulation, that is the honest limit of what this book can tell you, and §28 is how you decide for yourself in three minutes. Sit all of them on tmua.io rather than on paper (§23), and analyse every one of them the following day rather than the same evening (§35).

WHEN YOU HAVE FINISHED THOSE

If you still have time after working through the series above, there are only two sensible next moves and both are better than hunting for a twelfth publisher.

— **Move on to other community papers.** There are roughly two hundred sets in circulation. Triage with §28, take whatever passes, and treat them exactly as above: timed, on screen, analysed the next day.

— **Repeat official papers of your choice.** A paper you analysed four months ago is no longer a measurement, but it is still excellent training: sit it timed, and the questions you get wrong the second time are the ones whose rules never actually installed. That is worth more than a fresh mock of unknown quality.

What you should **not** do is keep collecting. Past the point where you have fifteen analysed papers and a circuit deck you review daily, the marginal paper is worth close to nothing and the marginal hour of extraction is worth a great deal (§29).

ABOUT THE PAPERS AFTER 2023

The public UAT-UK archive covers 2016 to 2023, plus specimen and practice material. Papers from the 2024 sitting onwards are not in that archive, and the versions circulating are reconstructions of varying fidelity.

Treat them as what they are. They are the best available simulation of the current standard and therefore the right thing to sit last. But a mark from a reconstruction is not a mark from an official paper and should not be read as one.

28 Which community papers to bin

There are around two hundred in circulation. You will complete maybe fifteen. The filtering matters more than the sitting.

THREE-MINUTE TRIAGE, BEFORE YOU COMMIT SEVENTY-FIVE MINUTES

Open it and check four things.

1. **Does it stay on specification?** The commonest failure, and one flagged directly by tutors who review community mocks. Questions needing content outside the specification are training something the exam will not ask.
2. **Twenty questions, five options, a plausible difficulty gradient?** A paper that is uniformly hard is a question set, not a mock, and sitting it against the clock will teach you a pace that does not exist.
3. **Are there worked solutions, not just an answer key?** A paper without solutions cannot be extracted, so it can only measure you. You do not need more measurement.
4. **Do the wrong options look designed?** Real distractors are built from specific errors. If the wrong options are random numbers, the paper cannot train elimination or the autopsy.

Fails two of these, bin it. There are one hundred and ninety-nine others.

28.1 Discard without hesitation

- Papers substantially recycled from official questions you have already analysed. You will score well, learn nothing, and believe the score.
- Papers with no solutions.
- Papers whose stated aim is to be as hard as possible, when used as timed mocks. They are fine as question sets. Against the clock they teach a false pace.
- Anything that takes more than ten minutes to find. The search cost is real and the marginal paper is worth almost nothing.

29 Things that feel productive and are not

Every one of these is common, every one feels like work, and every one has a specific way of failing.

THE HABIT	HOW IT FAILS
Sitting official past papers timed before extracting them	There are eighteen and you cannot unsee one. A timed sitting converts an irreplaceable teaching document into a number you could have estimated anyway. Sit them untimed, extract them fully, and do your timed work on community papers, which are effectively unlimited.

THE HABIT	HOW IT FAILS
Grading every paper	See §30. The number is unreliable, the emotional cost is real, and it crowds out tracking that would actually tell you something.
Responding to every error by revising the topic	The right remedy for one of thirteen error categories, applied to all thirteen. This single habit accounts for an enormous amount of sincere, hard, wasted work.
Reading the solution the moment you feel uncomfortable	And equally, refusing to read it for two hours. Both are the same failure: substituting a rule for a judgement. Use the hint ladder.
Memorising solutions	It feels like learning, produces short-term confidence, and transfers to nothing, because the exam does not repeat questions. The test: can you write your own version of the question and solve that? If not, you have a memory.
Practising what you are already good at	The strongest gravitational pull in all revision. The topics you enjoy get five hours and the topic that makes you feel stupid gets none. Your error log exists specifically to overrule that instinct. Follow the log, not the mood.
Collecting formulae	A shortcut has a price: recall, condition-checking, and the cost of misfiring. §74 is the arithmetic. A list of ninety results you half-know is worth less than twelve you own.
Chasing every new resource	If you have three unused resources already, a fourth is not the problem. The right number of question banks to have open at once is one or two.
Spending the last month entirely on timed papers	Peak volume belongs two or three weeks out, not in the final week. Fatigue costs marks, and there is no capability to be gained in the last ten days. There is only capability to be lost through exhaustion.
Comparing your timeline to somebody else's	The person you are measuring yourself against might be lying, or might simply have started earlier. Neither fact changes what your next hour should contain. The only comparison carrying information is with yourself three weeks ago, which is why the diagnostic gets repeated.
Confusing being busy with being stretched	Three hours of comfortable questions is worth less than forty minutes of genuine struggle followed by extraction. Hours are a terrible unit because they measure attendance. Count extraction cycles.

30 Stop grading yourself

30.1 What is actually true about the scale

Raw marks from both papers are combined and converted to one score from 1.0 to 9.0. Since 2024 the conversion is done by equating the test forms statistically, adjusting for how hard the particular questions turned out to be, and no raw-mark table is published for current sittings.

Before that, tables were published, per paper and overall, and they differed every year precisely because the papers differed. In 2023, an overall raw score around 38 out of 40 converted to 9.0.

30.2 Three things that follow, all of which people get wrong

1. **A score from a pre-2024 paper is not comparable to a target on the current scale.** Published analyses put mid-range scores roughly a point to a point and a half lower on the current scale than the old tables suggest. Apply a 2019 conversion table to your raw mark, get 7.5, and you have established nothing.
2. **A community paper has no scale at all.** No cohort, so any boundary printed on it was chosen by whoever wrote it. Applying it produces a number that looks like a measurement and is not one.
3. **You cannot reproduce the current scale.** Equating needs the whole cohort's responses. You do not have them. There is no arithmetic you can do at your desk that turns 31 out of 40 into a scaled score.

Your job between now and the exam is not to predict your final score from every practice paper. It is to become able to solve harder problems, accurately, faster. Only one of those two things is under your control, and it happens to be the one that matters.

30.3 What to record instead

Raw marks out of 20, per paper. That number is real, comparable to itself over time, and free of invented conversions. Alongside it, the metrics in §34.

> **THE ONE LEGITIMATE USE OF A SCALED SCORE**
>
> As a coarse trend over months, on official papers only, understood as plus or minus one.
>
> "I was around 5 in June and around 7 in September" is a real and useful statement. "I got 7.3 on Tuesday and 6.9 on Thursday so I have gone backwards" is not a statement about you at all. It is noise from two differently calibrated instruments, and treating it as signal will cost you a week of morale you cannot spare.

06

PART VI

YOUR PLAN

The same programme is wrong for seven different students. Find yourself here, then build the week around what your diagnostic actually said rather than around what feels productive.

31 Find yourself here

32 How long have you got?

33 Building your week

34 How to tell you are improving

§32 has the plan for somebody with roughly a month left. §34 is what to track instead of a score.

31 Find yourself here

Each of these is a real pattern, and in several cases the instinctive response is exactly the wrong one.

31.1 Year 12, more than a year out

Your risk is not lack of time. It is wasting it. Fourteen months of past papers is fourteen months of diminishing returns, and you will exhaust the official set within a term and have nothing to show for it.

Spend the first six months on mathematical development rather than TMUA preparation. Foundations until the floor is genuinely solid, then category C heavily: MAT Section A, the first fifteen AMC 12 questions, STEP Foundation modules, STEP 1. Prove the standard results instead of using them, which also pays at interview. Start the circuit deck now, at a low rate. Book Two's BEYOND material belongs in exactly this window and nowhere else.

Come to the official papers late, around six months out, by which point you should be so far past their content difficulty that the only things left to learn from them are cue design and pace. Which is exactly what they are best at teaching.

What not to do: start sitting TMUA papers a year out. You will burn the entire authentic supply on a version of yourself who could not have learned from it.

31.2 Six months out

The cleanest case and the one the six stages are written for.

Stage 1 for four to six weeks. Stage 2 for eight to ten. Stage 3 running alongside from the midpoint. Stage 4 in the final six weeks. Stage 5 in the final fortnight. Diagnostic every three weeks throughout.

31.3 Three months out

Compress Stage 1 to two weeks and be ruthless about it. Worksheets only for the topics where the diagnostic shows knowledge-gap errors, not all nine. Stage 2 gets the largest share and that is not negotiable, because it is where the marks are. Stage 3 becomes two sessions a week rather than a phase. Stage 4 the last three weeks. Stage 5 the last ten days.

31.4 Already scoring well and stuck at 7.5

Diagnose first, always. A 7.5 that is $U = 18, T = 15$ is a completely different animal from a 7.5 that is $U = 15, T = 15$.

High ceiling means you have a conversion problem, so Part IV is your entire programme plus enough timed volume for the decisions to go automatic.

Ceiling also 15 means more papers will not help: the remaining marks live in the hard tail, and the hard tail is not on the papers you have already seen. That is what category C and Part XV of Book Two are for.

The specific trap at this level: you are good enough that every paper feels productive, because you always learn a little. But a little from each of many loses to everything from a few. Halve your paper volume and triple your extraction time.

31.5 Stuck around 5 or 6

The commonest profile and the most fixable one. The ceiling is almost never the problem. The extraction is.

Jamie Baek's own account is exactly this. Stuck at around 5 for a long time, and what broke it was not more questions but the three-layer analysis.

Your programme: stop all timed work for three weeks. One official paper every three days, worked untimed, extracted for two to three hours each. Build the circuit deck from every question, not only the ones you got wrong. Tally the error mix weekly.

Expect your timed score to sit still for two or three weeks and then move sharply. That lag is normal and it is the single most common point at which people abandon the method about ten days before it would have worked. Write that sentence somewhere you will see it in week two.

31.6 Fast but careless

You finish with fifteen minutes spare and drop four marks to slips. This is the best problem to have and the cheapest to fix. The spare fifteen minutes are already yours. The issue is that you are not spending them.

Make a third pass compulsory. Run the distractor autopsy until you can name your own six recurring slips. Add a verification step to every question: sign, size, and does the answer respond correctly to the conditions. Then slow your first pass down by about ten per cent, which sounds wrong and usually gains two marks.

31.7 Accurate but too slow

You get nearly everything right untimed and run out of road at question fifteen. This is a routing problem, not a hurrying problem, and trying to write faster will make it worse.

Work out which of the six speeds is your bottleneck. For this profile it is almost always recognition or strategic speed. Run the cue flash and the two-route drill twice a week. Track your sub-90-second count rather than your total time. Then the hard-stop drill, so the questions you cannot do quickly get abandoned quickly.

> **THE TRAP FOR THIS PROFILE**
>
> Responding to slowness by sitting more full timed papers. You already know you are slow. Another paper confirms it.
>
> Timed papers *measure* speed, they do not *train* it. Speed is trained in short drills on single skills, then verified on papers.

31.8 Strong on Paper 1, weak on Paper 2

More common than anybody admits, and the cheapest imbalance in the book to correct, because the Paper 2 content base is one page long (§13.5).

Two forty-minute Paper 2 blocks a week for three weeks, run exactly as §13 describes, will usually move that half of your score more than a month of general practice. Work Part XIV of Book Two until the vocabulary table is automatic, then do nothing but verdict-first practice.

32 How long have you got?

> **ABOUT DATES AND BOOKING**
>
> Deadlines, test windows and access arrangements are published by UAT-UK and change every cycle. **Check them yourself**, and note that the access arrangements deadline falls *before* the booking deadline, which is the one people miss. You register and book your own centre; your school will not do it for you.
>
> Nothing else about the calendar belongs in this book. Anything printed here is wrong within a year, and a stale deadline is worse than no deadline at all. §79 explains why this box replaced a longer one.

What this section is actually for is the time you have left, which is the part under your control. Roughly a month, and §32.1 below is the plan. Longer than that, and §31 finds your profile while §33 builds the week around it.

32.1 Roughly a month left

If you are reading this in the last few weeks, the arithmetic changes. You cannot build capability from nothing in a month. But a month is comfortably enough to convert capability you already have, close accuracy leaks, and stop giving away marks you had already earned.

Jamie Baek's own transformation ran on a runway of about a month, and his account of it is specific: the compression works, as long as the time goes into extraction rather than volume. Four weeks of the right work beats four months of the wrong work, and you are about to find out which one you have been doing.

> **FOUR WEEKS, AND WHAT ACTUALLY GOES IN THEM**
>
> **Day 0.** The two-score diagnostic. Write down U, T and the error mix. Everything below depends on what it says.
>
> **Week 1, extract.** Four official papers, untimed, fully analysed. Two to three hours of analysis each. Build the circuit deck from every question, not only the wrong ones. Topic repair only where the log shows knowledge gaps, and only in forty-minute blocks.
>
> **Week 2, extract and stretch.** Three more official papers, same treatment. Add two sessions of AMC 12 questions 1 to 15 at three minutes each, purely to keep meeting unfamiliar structures. Repeat the diagnostic at the end of the week and compare.
>
> **Week 3, convert.** Now timed. Three or four community papers under strict conditions, on screen. Every overrun logged as a decision failure. Sit the official practice and specimen tests so the interface holds no surprises. Rehearse the noteboard layout every single time.
>
> **Week 4, peak then taper.** Two full paired sittings under real conditions using the most recent papers, back to back, no break. Analyse each one the following day. Then stop adding material. Sweep the circuit deck, re-read the error log, and rest. Two days out,

stop hard study entirely, confirm your centre and your route, and sleep.

IF YOU HAVE A MONTH AND YOU ARE A LONG WAY BELOW TARGET

Be strategic rather than heroic. Find the six or seven question archetypes that turn up on almost every paper and make those automatic. The cue table in §12.3 is the shortlist. In four weeks that is worth far more than a broad sweep.

Then train elimination hard, because a disciplined guess on every remaining question is worth about a mark.

This will not turn a 4.0 into a 9.0 in four weeks and any document promising you that is lying. It will reliably produce a substantial improvement, and the same work is the first phase of a later attempt if you need one. Nothing you do in these four weeks is wasted either way.

33 Building your week

33.1 A week in the middle of preparation

Not a timetable to obey. A set of proportions: change the hours, keep the ratios.

BLOCK	WHAT	SHARE	NOTES
Extraction	Official paper untimed, then full analysis	40%	The biggest block, always. One paper every two or three days; analysis two to three times the sitting time.
Stretch	Unfamiliar problems from category C	20%	Short sessions: three or four problems, under the single problem protocol.
Retention	Circuit deck daily, problem shelf on schedule	10%	Eight to twelve minutes a day. Never skipped, especially on a bad day.
Repair	Topic worksheets aimed by the error log	10%	Only where the log says knowledge gap. Zero if it does not.
Speed	One targeted drill from §21	10%	Pick the drill for your diagnosed bottleneck. Twice a week.
Simulation	Timed community paper	10%	Rises to 30 or 40% in the last three weeks. Near zero early on.

33.2 A session, in ninety minutes

0 to 10	Circuit deck. Cues only, fast. Warms up retrieval.
10 to 45	The hard block. New problems or a paper, under the single problem protocol. Your attention is at its best here, so do not spend it on admin.
45 to 50	Break, properly away from the desk.
50 to 85	Extraction of what you just did. Split points, circuit cards, error labels.
85 to 90	Two sentences in the log: what you learned, and what to do next session.

NEVER END ON THE SITTING

End every session on analysis, never on paper-sitting. The extraction is what consolidates while you sleep. The attempt is only raw material.

This is TMUA9.'s rule and it is worth following literally: if you run out of time, cut the *attempt* short rather than cutting the analysis.

33.3 Quality over hours

Two hours of real struggle and extraction beats six hours of comfortable questions, and it is not close. Hours measure attendance, which is why they are a terrible unit. Count **extraction cycles** instead: one problem taken from attempt, through the split point, to a written circuit card. Four good cycles is a good day; twelve rushed ones is a wasted one. On a day when you can manage one, do one. One is greater than zero, and a great deal greater than a scrolled evening.

---

34 How to tell you are improving

Score is a lagging, noisy, badly calibrated indicator. Record these instead. All of it fits on one sheet a week.

METRIC	WEIGHT	WHY IT MATTERS
Untimed ceiling U	Highest	What your mathematics is worth: the only number measuring capability rather than performance. It moves slowly, and when it moves everything follows.
Independent solve rate ***unfamiliar problems solved cold, first attempt***	Highest	A direct measurement of the thing a 9.0 needs. Track weekly on category C. This is what predicts the hard tail.
Hint rung needed ***0 to 5, per hard problem***	High	Should fall over weeks. If it is not falling, your extraction is producing memories rather than rules.
Sub-90-second count ***per 20 questions***	High	How many trained circuits you own. It rises *before* your score does. Aim for ten.
Conversion gap $U - T$	High	Separates decision and speed problems from capability problems. Should shrink through Stage 4.
Error mix ***proportions by category***	High	Tells you what to do next week. When the dominant category changes, change the programme.
Retry survival ***shelf problems solved cold at 21 days***	Medium	The cleanest test of whether anything actually stuck.
Raw mark out of 20	Medium	Real and comparable to itself. Use the trend, never the point.
Scaled score estimate	Lowest	Plus or minus one at best. Monthly trend, official papers only. Never week to week.

If your untimed ceiling, your independent solve rate and your sub-90-second count have all gone up, your score has already improved. It just has not been measured yet.

34.1 Ten minutes every Sunday

Two questions, written down. **What was my dominant error category this week, and what is the remedy?** If the answer is the same as the last two weeks, the remedy is not working; change it rather than repeating it louder. And: **what is the single most likely reason I will not hit my target?** Then make next week's biggest block about that.

THE WEEKLY SHEET. ONE SIDE OF PAPER, NINE LINES.

Copy this out once and fill it in every Sunday. The value is not in any single week's numbers; it is that eight of these side by side show you a trend that no individual paper ever will.

LINE	THIS WEEK	WHAT IT TELLS YOU
Untimed ceiling U		Capability. If it has not moved in a month, you are not extracting.
Timed score T		Performance. Meaningless on its own.
Conversion gap $U - T$		Which account to spend in next week.
Sub-90-second count		How many circuits you own. Should climb first.
Unfamiliar problems solved cold		The number that predicts the hard tail.
Hint rung, average		Falling means rules; flat means memories.
Dominant error category		Chooses next week's biggest block.
Extraction cycles completed		The only honest measure of effort.
One sentence: what changes next week		If you cannot write it, you have not reviewed.

THE TEST FOR WHETHER THIS IS WORKING

Put four consecutive sheets side by side. If the untimed ceiling, the sub-90-second count and the independent solve rate are all higher than they were four weeks ago, the programme is working and the score will follow, whatever this week's paper said.

If all three are flat, the programme is not working, and no amount of additional volume will change that. Go back to §5, re-run the diagnostic, and change what the biggest block contains.

07

PART VII

THE EXAM

Two and a half hours, back to back, on a screen you have hopefully seen before. Everything here sits downstream of the training. None of it is a gimmick.

35 The last two weeks

36 Exam day

37 When it goes wrong in the room

§36 is the protocol. Read it twice in the final week and not before.

35 The last two weeks

The final fortnight cannot add capability. It can very easily subtract it, through fatigue, panic volume, and the particular misery of a bad mock four days out. Treat this period as protecting something you have already built.

WINDOW	WHAT GOES IN IT
Days 14 to 8	Peak volume now, not later. Two full paired sittings under real conditions, both papers back to back, on screen, no break, board only. Analyse each the *next* day rather than the same day. One or two community papers besides. Circuit deck daily.
Days 7 to 3	Volume halves. One last paired sitting early in this window, then stop sitting papers. Read the error log end to end and sweep the circuit deck. Redo five or six hard shelf problems cold, not to learn anything new but to confirm the rules still fire.
Days 2 to 1	Stop hard study. Confirm your centre, your route, and that the name on your booking matches your photographic ID exactly. Look at the circuit deck once, lightly. Sleep properly for two nights, not one: it is the night before the night before that matters most.
After day 7	**No new material.** A technique learned in the final week is a technique you will not trust under pressure, and mistrust costs more than the technique gains.

> **THE FOUR-DAYS-OUT DISASTER MOCK**
>
> It happens to nearly everybody. A paper sat late, badly, on a tired evening, producing a score that sends you into the exam convinced you have gone backwards. This is exactly why peak volume belongs at day 14 and the last paper you sit belongs early in the final week.
>
> If it happens anyway, understand what you are looking at: one noisy sample from an uncalibrated instrument on a bad night. Skim it for anything mechanical, then put it down. It is not information about the exam.

36 Exam day

36.1 Before you start

Arrive early. Eat. A rushed arrival costs you the first three questions, usually among the cheapest on the paper, and that is the most expensive place to be flustered.

The two papers run one after the other with no guaranteed break, so prepare for a hundred and fifty continuous minutes of concentration rather than two seventy-five minute sprints. If you have not trained that in one sitting at least three times, the back half of Paper 2 will be worse than your mathematics deserves.

36.2 The three passes

This structure is TMUA9.'s and it is the right one. The on-screen flag and review navigator make it cheaper to run than it ever was on paper.

PASS	TARGET	WHAT YOU ARE DOING
One	35 to 40 min	Answer everything a circuit fires on straight away. Anything needing real deliberation: eliminate what you can, **put an answer in anyway**, flag it, move. Hard cap of three minutes, no exceptions. You are harvesting the cheap marks and building a map of the paper.
Two	about 25 min	Return to the flagged questions, **easiest first and hardest last.** This matters and it is counter-intuitive. You now know how many there are and how much each can have. Spend real time here. This is what pass one was protecting.
Three	whatever is left	**Re-solve** from question one. Do not re-read, re-solve, and compare outputs. Jamie Baek caught a calculation error in an early question on his third pass through Paper 1 in the real sitting, and fixed it in the final minutes.

WHY RE-SOLVING BEATS RE-CHECKING

When you re-read your own working you reconstruct the reasoning that produced it, including the error, which will look correct to you for exactly the same reason it did the first time. Re-solving generates an *independent* answer: agree and you have real evidence, disagree and you have found something. This is the highest-value ten minutes available in the exam, and most candidates spend it re-reading.

36.3 The rules that never change

- **Never leave a blank.** Nothing is deducted. Answer before you flag, so a flag can never become a zero.
- **Eliminate before you guess.** Sign, parity, size, kind of object. Ten seconds, and it roughly doubles what the guess is worth.
- **Read the whole question before touching the options.** Traps live in the wording and the unused condition is usually in the clause you skimmed.
- **On Paper 2, decide whether it is true before hunting for why.**
- **Flag aggressively on pass one.** A flag costs nothing. Grinding costs you the questions you never reach.
- **Take the wording as authoritative, not the diagram.** On-screen figures are not to scale and cannot be marked up.
- **Use the last five minutes to fill everything in**, even blindly.

36.4 The board on the day

Set up your two-column layout in the first thirty seconds, before you read question one. Keep the left column entries for anything flagged, so on pass two you resume instead of starting again. Restarting a flagged question from nothing is one of the largest avoidable costs in the exam.

37 When it goes wrong in the room

37.1 Question 1 is hard

It happens and it means nothing. Question order is not strictly difficulty order and one question is a sample of one. Flag it, answer with whatever elimination you have, move. Candidates who read a hard opener as a forecast lose the next four questions to adrenaline rather than to mathematics.

37.2 The whole paper feels harder than expected

Then it probably is, and here is the thing to hold onto: **the scale is equated.** Scores are adjusted for how hard the particular questions turn out to be. If the paper is hard it is hard for everybody and the conversion reflects that.

The feeling of "this is harder than the past papers" is not evidence you are doing badly. It is frequently evidence the paper is difficult, which is neutral. The only thing that changes your score in that moment is whether you keep collecting marks. Lower your threshold for flagging, raise your rate of educated guessing, and keep moving. A hard paper rewards discipline more, not less.

37.3 Paper 1 went badly

The two papers combine into one score. Every question on Paper 2 is worth exactly what it was worth an hour earlier. Writing off the second half because of the first is the most expensive single decision available to you in that room, and it gets made on emotion every year by people who would have been fine.

EMOTIONS IN THE ARENA

TMUA9.'s formulation is the most useful one available: **your emotions during the exam are independent variables from your outcome.** They are not data. Confidence and panic are equally uninformative about whether your last answer was right. Execute in the present and do not consult your feelings, because in the exam they are noise.

Two related bottlenecks from the same place, both real.

Insecurity: second-guessing a calculation, reflexively going back to check, hesitating. It destroys momentum and costs far more than the errors it catches.

Pride: refusing to abandon a question you feel you ought to be able to do. Sunk cost in a flattering costume.

Both are trainable, and practice is the only place you can train them.

SOMETHING WORTH KNOWING BEFORE YOU WALK IN

Watch anybody who scores a 9.0 solve a paper live and you will see something reassuring. They make arithmetic errors. They have ideas that go nowhere. They start down routes they abandon.

Errors never get eradicated, not at any level. What changes is how often they show up and how fast they get caught.

So do not walk in expecting to feel flawless, because nobody does, and treating a mistake as evidence you are not good enough is a mistake that costs more than the original one.

37.4 You run out of time

Then pass one was too slow, and you fix it next time with the hard-stop drill. Right now, the last two minutes go on filling every blank with an eliminated guess. Nothing else in those two minutes is worth as much.

37.5 You realise afterwards that you got one wrong

Between Paper 1 and Paper 2, that thought will arrive. It is worth nothing and it will cost you if you let it sit. You cannot go back, the mark is already decided, and the only variable still under your control is the twenty questions in front of you. Note it on the board you are about to erase, then let it go until you are outside.

37.6 If you are going to sit it again

There are two sittings a cycle, and a second attempt is a normal thing rather than an admission of anything. If you are already fairly sure you will take one, the useful work starts on the way home, while the paper is still in your head.

THE FORTY-EIGHT HOURS AFTER

1. **Write down every question you remember, the same evening.** Not the answers, the *structures*: what kind of object each one was, and which of them you had no route into. This list decays within a day and it is the most accurate diagnostic you will ever get, because it is the real paper rather than a reconstruction of it.
2. **Label each remembered failure with a category from §15.** Reading, generation, selection, execution. The mix from the real sitting is worth more than the mix from any practice paper, because pressure changes it.
3. **Do nothing else for a week.** Rest is not a delay in the programme; it is the part of the programme that makes the next eight weeks possible.
4. **Then re-run the two-score diagnostic (§5) on an unseen paper**, and start again from whichever of the three problems it says you have. It will very often not be the one you assumed while walking out.

The trap for a resit is treating the second run as a continuation of the first. It is not. Almost everybody who improves substantially between sittings does so because they changed *what kind of work* they were doing, not because they did more of the same for another two months.

08

PART VIII

THE ROADMAP

Seven phases, and one page to keep where you work. The last two pages of Book One are the two you will actually re-read.

38 Seven phases

39 One page to keep

Pin §39 above your desk. Everything else in Book One is the argument for it.

38 Seven phases

They overlap at the edges. The order matters more than the boundaries.

~	PHASE	WHAT YOU ARE BUILDING	YOU ARE DONE WHEN
1	**Calibrate**	An honest picture: ceiling, conversion gap, error mix.	You can say in one sentence which of the three problems you have.
2	**Build**	The content floor. Topic worksheets, topic MCQ, logic and proof.	8 out of 10 on any topic set at three minutes a question, cold.
3	**Extract**	The machinery. Three-layer analysis on official papers, untimed. The circuit bank starts here.	Untimed ceiling 17 out of 20, and your dominant error is generation rather than knowledge.
4	**Stretch**	Tolerance for the unfamiliar. Category C, deliberately stranger than the exam.	You can make real progress on an unseen problem inside ten minutes, reliably.
5	**Sharpen**	Speed and strategy. The six speeds, elimination, route pricing, the board.	Ten questions a paper under ninety seconds.
6	**Convert**	Performance. Timed papers, on screen, decisions under pressure.	Timed score within two marks of your ceiling, three papers running.
7	**Peak**	Simulation and composure, then taper.	Two paired sittings at target. Then stop, and rest.

THE THROUGH-LINE

Phases 2, 3 and 4 spend in the **capability** account. Phases 5, 6 and 7 spend in the **performance** account. Phase 1 tells you which one you are short of.

The commonest failure in TMUA preparation is spending four months in phases 5 and 6 trying to fix a problem that lives in phase 3. The diagnostic exists to stop you doing that, and it takes an evening.

39 One page to keep

The method edition

IN FRONT OF A PROBLEM

What kind of object is this? What is the cheapest thing that would settle it? What have they given me that I have not used?

Restate on the board in three lines: object, asked, given. Tick each given as your method uses it. An unticked given at the end is a warning.

STUCK? NAME THE OBSTRUCTION, THEN MOVE

Too general, make it specific. Too specific, generalise. Cannot see it, change representation. Too many unknowns, look at the relationship *between* the equations. Forwards fails, work backwards or from the options. Not sure it is true, try to break it at the boundary. Algebra exploding, substitute, group, or abandon the route. Already tried it, change the *class* of approach, not the details.

NO IDEA AT ALL

Produce one example. Then a second, different one. What stayed the same is usually the answer. Then a conjecture. Then attack it at the boundary. Then decide.

Five minutes with nothing new written down means you are spinning.

AFTER A PROBLEM

Erase nothing. The wrong routes are the raw material.

Put your chain beside the model's. Find the first line where they split. What thought would have taken the other branch? Write it word for word. Why is that thought natural? Which definition makes it the cheap route?

Classify into one category, not two. Different failures need different remedies.

Which error produced each wrong option? Write the card: CUE, MOVE, WHY, COST, FAILS.

EVERY DAY

Circuit deck, cues only, eight to twelve minutes. Grade on *did the move arrive*, not on *was I right*.

Problem shelf on schedule: 7, 21, 60 days. Re-solved cold on paper. Capped at forty.

Never end a session on paper-sitting. End on analysis.

TRACK THESE, NOT YOUR SCORE

Untimed ceiling. Independent solve rate. Hint rung needed. Sub-90-second count. Conversion gap. Error mix. Raw marks out of 20 only, and no invented conversions.

IN THE ROOM

Set the board up before question one. Left column: givens, and the A B C D E elimination row. Right column: working, erased between questions.

Pass 1, three-minute cap: answer what fires, eliminate, answer anyway, flag, move. Pass 2: flagged questions, easiest first. Pass 3: re-solve from question one, do not re-read.

Never a blank. Eliminate before guessing. Paper 2: verdict first, reason second.

A hard paper is hard for everybody and the scale is equated. Keep collecting marks.

CLOSING BOOK ONE

The work that counts

Everything in Book One comes down to one uncomfortable idea. The work that moves your score is the work that produces no number.

Sitting a paper produces a number. Reviewing a topic you already know produces a warm sense of coverage. Sitting with a problem you cannot do for twenty-five minutes and then writing down the one sentence you failed to think produces nothing you can show anybody.

That last one is the trade. Everyone at the top of this scale has made it. Most people who plateau never do, and it is not because they are lazy. It is because the alternative looks so much more like studying.

The second idea is smaller and matters just as much. You are not becoming somebody who can do TMUA questions, because TMUA questions will not exist for you the day after the sitting. You are becoming somebody who can be handed an unfamiliar problem and find a way in. That is what maths degrees are actually selecting for, what an interview will actually test, and what you will actually be doing for the next three years. The exam is a proxy. Train the thing itself and the proxy looks after itself.

And then the part nobody can do for you. Trust this for longer than feels reasonable. The ceiling moves before the number does. Your score will sit still for two or three weeks and then jump, and almost everyone who abandons this method abandons it inside that window, about ten days early.

You are going to have evenings where nothing works, where you get a question wrong that you got right last week, where somebody online posts a score that makes you feel sick. None of that is information about you. The only information about you is whether you sat down again the next day.

Do that enough times and the exam takes care of itself.

Book Two begins overleaf.

Book One was what you do. Book Two is what you know.

09

PART IX

THE LIBRARY

Why the content half was rebuilt, how to tell an examinable result from an ornament, and the six lines every result in here has to earn before it gets printed.

40 Why Book Two was rebuilt

41 Tiers, and how they were decided

42 The frame, and the card it becomes

Read §40 once. Keep §41 and §42 open while you use Parts X to XIII.

40 Why Book Two was rebuilt

The first printing of the content volume was a list of results. It was correct and it was not usable, and the reason it was not usable is worth stating properly, because the same mistake is in almost every formula sheet ever sold to a TMUA candidate.

A strong student read it and wrote down six objections. They were sharp, mostly correct, and worth more than any amount of praise. Here they are, each with an honest verdict, because the verdicts are what shaped this rebuild.

1 "We do not need proofs."

Partly right, for the wrong reason.

A proof is not there to satisfy anybody's standard of rigour. It is there so that you can rebuild the result when your memory of it is half gone, and so that you recognise it in a configuration nobody showed you. A student who knows *why* the area between a line and a parabola depends only on the leading coefficient and the gap will see the same idea in a cubic. A student who memorised $|a|D^3/6$ will not.

What was actually wrong was the length. A full display proof, worked line by line, is the wrong instrument for that job. It reads like something to be studied rather than something to be reconstructed, so it gets studied, which trains recognition of a page rather than the ability to rederive.

In this edition every result has a one-line WHY, and the instruction is to reproduce it from that line in about thirty seconds. If you cannot, you do not own the result yet.

2 "Writing out long proofs is a waste of practice time."

Right. And the first printing did tell you to write them out. That instruction is withdrawn. Reconstruction, out loud, from the one-line cue, at speed, is the drill. Transcription is not.

3 "Some of the formulae are far too niche."

Right, and now provable. The published specification says what can be asked, and read against the first printing a pattern appears.

Exact values are required for $0°, 30°, 45°, 60°$ and $90°$ only, and compound angle formulae appear nowhere, so the $15°$ and $75°$ triangle cannot be needed. Points of inflexion are explicitly not examined beyond a qualitative understanding, which strands any result whose statement begins with the gradient at the inflexion point. Binomial expansion is specified for $(1+x)^n$ and $(a+f(x))^n$ with n a positive integer, so the full multinomial general term is decoration. Nothing in the specification asks for Ceva, Menelaus, Stewart or Apollonius.

None of that material is bad mathematics. It is misfiled. It now lives in §77, labelled BEYOND, with the honest note that it pays at interview and on STEP and not on this paper.

4 "Applying the formula takes longer than the original method."

Right, and this is the deepest of the six.

A shortcut is not free. You pay to recall it, you pay to check that its conditions hold, and you pay again, much more, on the occasions when you fire it in a configuration where it does not apply. When the frequency is low and the conditions are fiddly, that bill is larger than the four minutes of ordinary integration you were trying to avoid.

So every result in this book states its price. COST says what it saves against the ordinary route. BREAKS says when the cue appears and the result is wrong anyway. §74 turns that into a rule you can apply to anything, including everything in here: **if you cannot confirm the conditions in five seconds, use the default method and be right slowly.**

The deeper fix was structural. Eight separate area formulae, indexed by configuration, is exactly the situation where retrieval costs more than derivation. §54 replaces all eight with one identity and a four-line table.

5 "The later sections drop in difficulty. We know about circle theorems."

Right about the level, wrong about the remedy.

The circle theorems are on the specification twice and cannot be cut. But *listing* them is worthless to anybody reading this document, because the examinable skill is not stating a theorem. It is looking at a configuration and knowing, in two seconds, which of seven facts the diagram is built around.

So the content stays and the task changes. The drills are configuration recognition against the clock, then exam-standard questions, then composites that need two chapters at once. **Every chapter with drills opens at A**, recognition at ten to twenty seconds, where you name the move and do not solve. **B** follows, exam standard at sixty to ninety seconds. **C**, the 9.0 layer at two ideas and three minutes, appears wherever the section has one. If you found the first printing easy, start at B and drop to C.

6 "Written obtusely. Complexity for complexity's sake."

Right. Some of it was ornament and some of it was actively costly, because notation you have to decode is notation you cannot fire under pressure. Three examples of what has gone.

- *Prologue, formula, epilogue* is a good idea buried in a bad label. The condition, the result and the use are now three lines with those names on them.
- "If the product of the powers of each term is odd" is true and unreadable. It now says: *if every term has odd degree.*
- The line symmetry test read $f(\square) = f(\triangle)$ with $\square + \triangle = 2a$. It now says: *if* $f(p + t) = f(p - t)$ *for every* t*, the mirror is at* $x = p$. Same result, no symbols to decode, and you can see the sentence is just the definition of a mirror.

All six objections were design faults rather than mathematical ones. The mathematics in the first printing was correct, give or take about ten arithmetic slips. The packaging was costing more than the content was worth.

And the one place the critic was wrong

The claim underneath objections 1 and 2 is that the test is not rigorous, so reasoning about *why* things are true is wasted effort.

Half the exam is Paper 2, which is a paper about why things are true: necessary and sufficient conditions, negation, counterexamples, and finding the false step in a proof somebody else wrote. It is the half most candidates prepare least, and it is the half where the marks are cheapest. A student who has trained himself out of caring why a result holds has trained himself out of twenty questions.

> **WHY THE SAME DOCUMENT CANNOT SERVE BOTH READERS**
>
> There is a real phenomenon behind this disagreement, and it has a name.
>
> Guidance that helps a novice reliably hurts an expert: worked examples and step-by-step support are the best available instruction for somebody without the schema, and become redundant load for somebody who has one, at which point open problem solving overtakes them. The research calls it the **expertise reversal effect**, and the practical consequence is that no single presentation is right for both ends of the range.
>
> The first printing chose one presentation and served the lower end. The critic was at the upper end and felt every line of support as friction. Both experiences were accurate.
>
> This edition is built to be read two ways. The WHY line and the level A drills carry the reader who needs the scaffold. The tier labels, the cost lines and the level C drills carry the reader who wants the result, the price, and a hard question. Same page, two depths, and an explicit instruction to skip.

What is new here that was in neither printing

This edition also absorbs a second body of teaching material: a full ten-lesson TMUA primer covering polynomials, inequalities, functions, coordinate geometry, plane geometry, sequences and series, combinatorics, number theory, logic and proof, and a miscellany of results that do not fit elsewhere. It is careful, it is well explained, and it predates the 2024 move on screen, so it has been re-tiered against the current specification rather than reprinted.

Seven areas the exam leans on and neither previous printing covered at all are now written from scratch: polynomial structure and the factor theorem, symmetry inside a definite integral, the floor and ceiling functions, series and the method of differences, recursions, counting and selections, and curvature with the trapezium rule. Paper 2 has grown from three pages to a part of its own, which is closer to the half of the exam it actually is.

Everything it draws on is named where you would actually use it, in §27.

---

41 Tiers, and how they were decided

Every result carries one of three labels. The label is a claim about examinability, and it was decided by reading the specification rather than by taste.

TIER	WHAT IT MEANS	WHAT YOU DO WITH IT
CORE	On the specification, and it recurs.	Automatic. Card in Deck A, reviewed daily, until the move arrives before you have finished reading.

TIER	WHAT IT MEANS	WHAT YOU DO WITH IT
EDGE	On the specification but infrequent, or a shortcut that is genuinely faster when the cue is unmistakable.	Know it. Do not force it. Check the conditions in five seconds or take the default route.
BEYOND	Not on the specification.	Read once for the pleasure and the transfer. Do not drill it in the last two months. It pays at interview, on the MAT and on STEP.

Four constraints in the specification do most of the tiering work, and they are worth knowing directly because they let you price any new resource anybody sells you.

1. **No calculator, no formula booklet.** Every number in a question was chosen to be workable by hand. An answer that requires real arithmetic is evidence you are on the wrong route, not evidence you should work harder.
2. **Exact values only for the five standard angles.** No compound angle formulae anywhere in the specification. The only identities named are $\tan\theta = \sin\theta / \cos\theta$ and $\sin^2\theta + \cos^2\theta = 1$.
3. **Points of inflexion are not examined**, beyond knowing qualitatively that a cubic has one. Stationary points are maxima and minima only.
4. **No penalty for a wrong answer**, so every question has a value even when you cannot do it, and elimination is worth real marks. That is Book One §22, and §75 adds the content-side version.

There is a fifth constraint that is not in the specification but is in the papers. Since the test moved on screen the questions have been reported as longer and more often set inside an applied context, which means each one carries a translation step before any of this content becomes relevant. That does not change what is examinable. It changes the first fifteen seconds, which is Book One §7.

> **ONE HONEST CAVEAT ABOUT TIERS**
>
> A tier is a judgement about what the paper rewards, made by reading the published content specification and a decade of past papers. It is not a guarantee, and the specification is revised.
>
> Two consequences. Read the current specification yourself once, for an hour, at the start of Stage 1: it is the only authoritative document and it is short. And treat BEYOND as "cannot be *required*", not as "will never be *useful*" — a BEYOND result that shortens a question you could have done anyway has still earned its keep, it just must never be the thing you were relying on.

42 The frame, and the card it becomes

Six lines, in the same order, for every result in this book.

TRIGGER	What you see in the question. Written as you would first meet it, not as a topic name. This is the only line you should be able to recall cold.
CLAIM	The result, with its conditions inside the sentence rather than in a footnote.

WHY	One line. Enough to rebuild the result in thirty seconds. Not a proof.
COST	What it saves against the ordinary route, in seconds, and the frequency you can expect.
BREAKS	A case where the trigger is present and the claim is false. If a result has no such case, it is not a rule, it is a memory with a confident voice.
DEFAULT	The ordinary method to fall back on. Always available, always slower, never wrong.

Book One §17 asks for five lines: cue, move, why, cost, fails. The mapping is exact. TRIGGER becomes the cue and goes on the front of the card. CLAIM becomes the move. WHY, COST and BREAKS go on the back, unchanged. DEFAULT is the line the first printing left out, and it is the one that stops a half-remembered shortcut from turning a question you could have done slowly into a question you got wrong quickly.

HOW TO ACTUALLY USE PARTS X TO XIII

Once per result, in this order, and it takes about four minutes.

1. Read TRIGGER and CLAIM. Cover the page. Say the claim out loud.
2. Read WHY. Cover the page. Rebuild the result on paper from that line alone. Thirty seconds. If it takes three minutes, the result is not yours yet and no amount of rereading will change that, so do the A drill and come back tomorrow.
3. Read BREAKS. **Construct a second breaking case yourself.** This is the step that stops misfiring, and it is the step everybody skips.
4. Do drill A. If you are at or above 7.0, skip to B, and if B is boring, go to C.
5. Write the card **only if** COST beat the §74 threshold for you. Not every result in this book deserves a place in your deck, and a deck of ninety cards you review is worth more than a deck of two hundred you abandon.

The signal that a result has landed is the one from the first printing, and it was right. **It stops being interesting.** There is no resistance in the chain of reasoning, and the verbal step falls away. Boredom is the target state.

THE ONE THING TO READ BEFORE ANY OF IT

The cue table in Book One §12.3 is the index to everything that follows. It maps what you literally see on the screen to the section that handles it. Learn that table first and the rest of this book becomes a reference rather than a syllabus.

10

PART X

ALGEBRA

Polynomials, quadratics, roots, substitutions, logarithms, inequalities and ranges. Roughly half of Paper 1 is somewhere in this part, and almost all of it is one of six or seven recurring shapes.

43 Polynomials: degree, roots and shape

44 Quadratics: the discriminant machine

45 Roots and coefficients, and related equations

46 Hidden polynomials, and the constraint the substitution carries

47 Exponentials and logarithms

48 Inequalities and casework

49 Ranges: what a function can and cannot reach

Everything labelled CORE here belongs in Deck A. Work §46 until the substitution and its constraint arrive together, because that single habit is worth about two marks.

43 Polynomials: degree, roots and shape

Almost every Paper 1 question that is not obviously geometry or obviously logic is a polynomial question wearing something else's clothes. This section is the vocabulary the rest of Part X assumes.

43.1 Degree, and why it is the first thing to write down

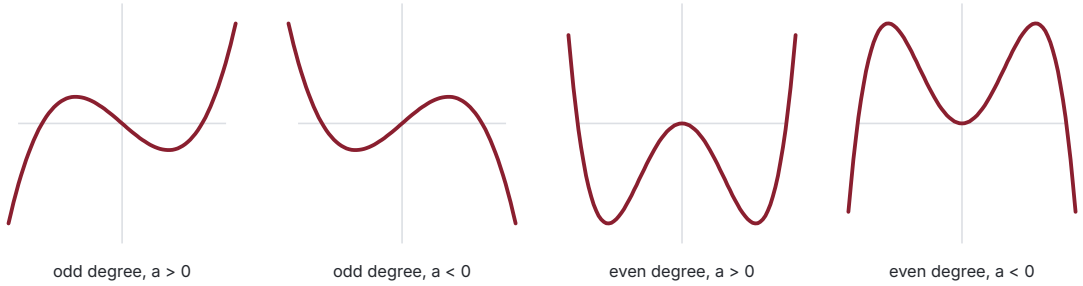
43.1 DEGREE UNDER THE THREE OPERATIONS — CORE

TRIGGER	A question about the degree, the number of roots, the shape, or the end behaviour of a combination of polynomials, with no numbers you would want to expand.
CLAIM	Multiplication adds degrees, $\deg(pq) = \deg p + \deg q$. Composition multiplies them, $\deg\big(p(q(x))\big) = \deg p \times \deg q$, in either order. Addition and subtraction give **at most** the larger of the two, and less exactly when the leading coefficients cancel.
WHY	Only the leading terms matter. $x^n \cdot x^m = x^{n+m}$; $(x^m)^n = x^{mn}$; and in a sum the two leading terms may destroy each other.
COST	Answers a whole class of question with no expansion at all. Ten seconds against a page of brackets.
BREAKS	The "at most" in addition. If two polynomials of equal degree are added, check the leading coefficients, then the next ones down, until one survives. This is the designed trap.
DEFAULT	Expand, which is always available and almost always the wrong use of ninety seconds.

Two consequences worth holding directly, because they answer questions on their own.

— **A polynomial of degree n has at most n real roots**, and the graph crosses or touches the x-axis at most n times.

— **End behaviour is decided by two things only:** the parity of the degree and the sign of the leading coefficient. Odd degree means the two ends go opposite ways, so *every odd-degree polynomial has at least one real root*. Even degree means both ends go the same way.



End behaviour is a two-bit question: parity of the degree, sign of the leading coefficient. Everything between the ends is turning points.

43.2 The factor theorem, used the way the exam intends

43.2 GUESS A ROOT, THEN DIVIDE CORE

TRIGGER	A cubic or quartic you are asked to solve, factorise, or count the roots of, with small integer coefficients.
CLAIM	If $p(a) = 0$ then $(x - a)$ is a factor. On this paper the root you are meant to guess is small: try $0, \pm 1, \pm 2, \pm 3$ first, then the divisors of the constant term over the divisors of the leading coefficient.
WHY	A polynomial with integer coefficients and a rational root p/q in lowest terms has p dividing the constant term and q dividing the leading one. You do not need that statement, only its consequence: the list of candidates is short.
COST	Turns an unsolvable-looking cubic into a quadratic in about twenty seconds. The alternative is the cubic formula, which is grotesque and is not the intended route.
BREAKS	When there is no nice root, which happens, and is the exam telling you the question is not about *finding* the roots. Then it is about *counting* them, and the tool is a sketch (§43.4).
DEFAULT	Sketch and count, or use the discriminant of a factor you already have.

Do not use long division. Once you have the factor, write down the quotient directly by matching the leading and constant coefficients and then checking one middle term. For $x^3 - 12x^2 + 41x - 42$ with root $x = 2$:

$$x^3 - 12x^2 + 41x - 42 = (x - 2)(x^2 + bx + 21),$$

because the leading term must be x^2 and the constant must multiply with -2 to give -42. Matching the x^2 coefficient gives $b - 2 = -12$, so $b = -10$, and the quadratic $x^2 - 10x + 21 = (x - 3)(x - 7)$ finishes it. Roots $2, 3, 7$. That is fifteen seconds and no division.

43.3 Touching against crossing: multiplicity

43.3 EVEN MULTIPLICITY BOUNCES, ODD MULTIPLICITY CROSSES CORE

TRIGGER	A factorised polynomial, a sketch to identify, or the word “touches”, “tangent”, or “repeated root”.
CLAIM	At a factor $(x - a)^m$, the curve **touches and turns back** if m is even, and **crosses** if m is odd, flattening against the axis as m grows. Two curves $y = f(x)$ and $y = g(x)$ touch at $x = a$ exactly when $f - g$ has a repeated root there, which is exactly when $f(a) = g(a)$ **and** $f'(a) = g'(a)$.
WHY	Near $x = a$ the factor $(x - a)^m$ controls the sign, and an even power cannot change sign.
COST	Identifies a graph from an equation, or an equation from a graph, in five seconds. At least one question a paper.
BREAKS	Reading multiplicity off a picture. On screen a tangency and a very shallow crossing look identical. Count the multiplicity algebraically.
DEFAULT	Evaluate the polynomial just either side of a and compare signs.

> **THE TWO TESTS FOR TANGENCY, AND WHEN TO USE WHICH**
>
> Two curves touch at a point. There are two equivalent conditions and they are convenient in different situations.

Repeated root. Set $f(x) = g(x)$, bring everything to one side, and impose a repeated root. If what you get is a quadratic, that is the discriminant equal to zero, which is one line. Use this when the difference is a quadratic.

Equal values and equal gradients. $f(a) = g(a)$ and $f'(a) = g'(a)$. Use this when the difference is not a quadratic, or when the point of contact is what you actually want.

Both are on the specification, both are quick, and choosing between them in three seconds is a genuine selection skill. Practise it: on any tangency question, name which one you are about to use and why, before you write.

43.4 Counting roots without finding them

This is the highest-yield idea in the section, and it turns an entire family of questions into a sketch.

43.4 THE SLIDING CONSTANT — CORE

TRIGGER	"For how many values of k", or "for which k does $p(x) = k$ have exactly three solutions", or a polynomial with an unknown constant term and a question about the number of roots.
CLAIM	Write the equation as $f(x) = k$, where f carries everything except the constant. Sketch $y = f(x)$ once, marking only the **stationary values**. Then slide the horizontal line $y = k$ and read off how many crossings you get in each band. The answer changes exactly as the line passes a stationary value.
WHY	$p(x) = f(x) - k$ and subtracting k translates the graph vertically, so the number of x-intercepts of p is the number of intersections of $y = f(x)$ with $y = k$.
COST	Enormous. It converts a case analysis you cannot do into a picture you can, and it is the intended route for a whole family of questions. Once or twice a paper.
BREAKS	You need the stationary *values*, not just the stationary points, and you need all of them. Miss one turning point and every band is wrong.
DEFAULT	Discriminant arguments, which work for quadratics and get ugly above that.

WORKED: THE QUARTIC WITH AN UNKNOWN CONSTANT

For how many solutions does $x^4 - 8x^3 + 22x^2 - 24x + k = 0$ have, in terms of k?

Write it as $f(x) = -k$ with $f(x) = x^4 - 8x^3 + 22x^2 - 24x$. Differentiating, $f'(x) = 4x^3 - 24x^2 + 44x - 24 = 4(x-1)(x-2)(x-3)$, so the stationary points are at $x = 1, 2, 3$, and

$$f(1) = -9, \qquad f(2) = -8, \qquad f(3) = -9.$$

A positive leading coefficient means the curve comes down from $+\infty$, bottoms at -9, rises to a local maximum of -8, falls to -9 again, and returns to $+\infty$. Now slide the line $y = -k$:

- above -8: two crossings, so $k < 8$ gives **two** solutions;
- exactly -8: the line meets the local maximum, so $k = 8$ gives **three**;
- between -9 and -8: **four**;
- exactly -9: the line meets both minima, so $k = 9$ gives **two**;
- below -9: **none**.

No case algebra, no quartic formula, and the whole thing is one sketch and three substitutions. Notice what made it cheap: the derivative factorised, which it was designed to.

43.5 Two variables, one of them held still

43.5 FREEZE ONE VARIABLE

EDGE

TRIGGER	A polynomial in two variables, or a question of the form "for which values of y does this equation have two solutions".
CLAIM	If the question asks about solutions in x, treat y as a constant and read the expression as a polynomial in x alone. Every tool you have then applies: discriminant, factor theorem, sum and product of roots.
WHY	"How many solutions" always means solutions in one named variable. Naming it is the entire step.
COST	Turns something that looks unfamiliar into a quadratic. It is also the standard way a hard-looking question gets built.
BREAKS	The coefficient of x^2 may itself vanish for some y, and that case is separate. This is the same trap as §44.1 and it is where the missing option hides.
DEFAULT	There is not really one. Naming the variable is the question.

Two related habits from the same idea.

Symmetry as a factorisation hint. If an expression is unchanged when you swap x and y, it can be written in terms of $x+y$ and xy, and a factor of $(x-y)$ appears whenever setting $x=y$ makes it zero. Similarly, if setting $x=-y$ gives zero, then $(x+y)$ is a factor. Checking two substitutions is much faster than trying to factorise blind.

A compound variable is a variable. x^2+1/x^2 is a polynomial in $v=x+1/x$; $\sin^2\theta$ and $\cos^2\theta$ are polynomials in one of them via the Pythagorean identity; 4^x is a polynomial in 2^x. That is §46 and it is the single most productive idea in Part X.

DRILL A RECOGNITION ONLY. NAME THE DEGREE OR THE SHAPE, TEN SECONDS EACH.

1. p has degree 3 and q has degree 4. Degrees of pq, of $p(q(x))$, of $q(p(x))$, and the largest possible degree of $p+q$?
2. $y=-x(x-1)^2(x+3)^3$. How many times does the curve cross the x-axis, and how many times does it touch?
3. Does $x^5-4x^2+1=0$ definitely have a real root? Does $x^4-4x^2+7=0$?
4. $p(x)=2x^3+ax^2+bx+c$ and $q(x)=-2x^3+dx^2+ex+f$. What can you say about $\deg(p+q)$?

ANSWERS. 1. 7; 12; 12; 4. 2. Crosses at $x=0$ (multiplicity 1) and $x=-3$ (multiplicity 3), so twice; touches at $x=1$, once. 3. Yes, odd degree always has a real root. Not necessarily: $x^4-4x^2+7=(x^2-2)^2+3>0$ always, so no real root. 4. At most 2, and it is exactly 2 unless $a+d=0$ as well, and so on downwards.

DRILL B EXAM STANDARD, NINETY SECONDS EACH.

1. Solve $x^3-6x^2+11x-6=0$.
2. The line $y=c$ meets $y=x^3-3x$ at exactly two points. Find both values of c.
3. For which k does $x^4-2x^2+k=0$ have exactly four real solutions?
4. $y=x^3+ax^2+bx+c$ touches the x-axis at $x=2$ and passes through the origin. Find a, b and c.

ANSWERS. 1. $x=1$ is a root; $(x-1)(x^2-5x+6)=(x-1)(x-2)(x-3)$, so $1,2,3$. 2. Stationary values of x^3-3x are $f(-1)=2$ and $f(1)=-2$; a horizontal line meets the curve twice exactly when it passes through a turning point, so $c=\pm 2$. 3. Put $t=x^2\geq 0$: $t^2-2t+k=0$ needs two distinct **positive** roots, so $\Delta=4-4k>0$ and product

$k > 0$ and sum $2 > 0$; hence $0 < k < 1$. Each positive t gives two values of x, so four in all. 4. Touching at 2 means a repeated factor: $y = x(x-2)^2 = x^3 - 4x^2 + 4x$, so $a = -4$, $b = 4$, $c = 0$.

DRILL C THE 9.0 LAYER, THREE MINUTES EACH.

1. $f(x) = x^3 + px + q$ has three distinct real roots. Show that $p < 0$, and find the condition on p and q in terms of the stationary values.
2. For how many integers k with $1 \le k \le 20$ does $x^4 - 4x^3 + k = 0$ have exactly two real solutions?
3. $p(x)$ is a polynomial of degree 4 with $p(x) = p(6-x)$ for all x, and p has a root at $x = 1$. Name three more roots or explain why you cannot.

ANSWERS. 1. $f'(x) = 3x^2 + p$, which needs two real zeros for three roots, so $p < 0$ and the stationary points are at $x = \pm\sqrt{-p/3}$. Three distinct real roots requires the two stationary values to have opposite signs, i.e. $f(\sqrt{-p/3})\, f(-\sqrt{-p/3}) < 0$, which reduces to $4p^3 + 27q^2 < 0$. 2. $f(x) = x^4 - 4x^3$ has $f'(x) = 4x^2(x-3)$. At $x = 0$ the derivative does not change sign, so that is a stationary point of inflexion and creates no extra crossings; the only turning point is the minimum at $x = 3$, where $f(3) = -27$. So f falls from $+\infty$ to -27 and rises again, and the line $y = -k$ meets it exactly twice whenever $-k > -27$. Every k from 1 to 20 satisfies that, so **20**. 3. The symmetry $p(x) = p(6-x)$ has mirror $x = 3$ (average the arguments, §50), so $x = 5$ is also a root. A quartic has at most four roots and nothing forces the other two to be real, so you can name one more and no more.

44 Quadratics: the discriminant machine

The discriminant is on the specification by name, it is the standard way the examiners ask about tangency, about how many solutions something has, and about the range of a function, and it is the engine behind at least three questions a sitting.

44.1 The three questions the discriminant answers

44.1 DISCRIMINANT, AND THE CASE THAT IS NOT A QUADRATIC

CORE

TRIGGER	Anything about how many solutions, whether a line touches a curve, or whether an expression can be zero or is always positive.
CLAIM	For $ax^2 + bx + c$ with $a \neq 0$ and $\Delta = b^2 - 4ac$: two distinct real roots if $\Delta > 0$, one repeated root if $\Delta = 0$, none if $\Delta < 0$. Tangency of a line and a curve is the case $\Delta = 0$ of their difference. And $ax^2 + bx + c > 0$ for every real x exactly when $a > 0$ **and** $\Delta < 0$.
WHY	A repeated root is a double intersection, which is what touching means. Completing the square gives the same three cases.
COST	It is not a shortcut, it is the intended route. Doing these questions by sketching or by trial costs two minutes each.
BREAKS	The $a \neq 0$ condition. When a contains the parameter you are solving for, the case $a = 0$ is a **separate case** and it is where the missing option usually hides. If a question says "for all x" and a is a parameter, write down the $a = 0$ case before you touch the discriminant.
DEFAULT	Complete the square. Slower, and it answers the same three questions.

THE $A = 0$ TRAP, WRITTEN OUT ONCE SO YOU NEVER LOSE IT AGAIN

"$kx^2 + (k+2)x + 3 = 0$ has exactly one real solution. Find every possible k."

Almost everybody writes $\Delta = 0$ and gets $(k+2)^2 - 12k = k^2 - 8k + 4 = 0$, so $k = 4 \pm 2\sqrt{3}$. Two answers, and the option list contains "two values" precisely because that is the answer you were expected to give.

But $k = 0$ makes the equation $2x + 3 = 0$, which is linear and has **exactly one real solution**. Three values in total.

The general rule: whenever the leading coefficient contains a parameter, the sentence "this is a quadratic" is an assumption, and every assumption you make without writing it down is a case you will lose (§15.2).

44.2 Always positive, never zero, and the range of a quadratic

Completing the square is the on-specification route to every inequality question that looks as though it wants a named inequality.

$$x^2 + px + q = \left(x + \tfrac{p}{2}\right)^2 + q - \tfrac{p^2}{4}$$

A square is at least zero, so the minimum value is $q - p^2/4$, attained at $x = -p/2$, and the expression is always positive exactly when that minimum is. That single line answers "find the least value", "show it is always positive", "find the range", and "for which k does it never equal zero", and it does all four without a formula.

WHY THIS IS WORTH MORE THAN THE DISCRIMINANT

The discriminant tells you *whether* there is a root. Completing the square tells you *where the minimum is and what it equals*, which is strictly more information for the same effort.

If a question asks anything about the *value* of a quadratic rather than about its roots, complete the square. If it asks only about the number of roots, use Δ. Choosing correctly saves a line and, more importantly, stops you computing a discriminant when what you needed was a vertex.

DRILL A RECOGNITION ONLY. NAME THE CONDITION, TEN SECONDS EACH.

1. $x^2 + kx + 9 = 0$ has a repeated root. Write the condition.
2. A line meets a parabola at two distinct points. Write the condition.
3. "Find the least value of $3x^2 - 6x + 1$." Discriminant, or completed square?
4. "$px^2 + 2x + p > 0$ for every real x." What must you write down before you touch Δ?

ANSWERS. 1. $\Delta = 0$, so $k^2 = 36$. 2. $\Delta > 0$ for the difference of the two. 3. Completed square: it asks for a **value**, not a root count. 4. The case $p = 0$, which is a separate case and is where the missing option hides.

DRILL B EXAM STANDARD.

1. For which values of k does $x^2 + kx + k + 3 = 0$ have two distinct real roots?
2. The line $y = mx + 2$ is tangent to $y = x^2 + 3x + 6$. Find both values of m.
3. Find the least value of $2x^2 - 12x + 7$ and the value of x at which it occurs.
4. For which k is $kx^2 + 4x + k > 0$ for every real x?

ANSWERS. 1. $k^2 - 4(k+3) > 0$, so $k^2 - 4k - 12 > 0$, giving $k < -2$ or $k > 6$. 2. $x^2 + (3-m)x + 4 = 0$ with $\Delta = 0$ gives $(3-m)^2 = 16$, so $m = -1$ or $m = 7$. 3. $2(x-3)^2 - 11$, least value -11 at $x = 3$. 4. Need $k > 0$ and $16 - 4k^2 < 0$, so $k > 2$. The $k = 0$ case gives $4x + 2$, which is not always positive, so it is correctly excluded.

---

45 Roots and coefficients, and related equations

A question that names the roots α and β and then never asks you for them is telling you not to find them.

45.1 Sum and product before roots

45.1 VIETA, AND EVERYTHING SYMMETRIC

CORE

TRIGGER A question about the roots of a polynomial that does not ask you for the roots: their sum, their squares, their reciprocals, their difference, or a new polynomial with related roots.

CLAIM For $ax^2+bx+c=0$, $\alpha+\beta=-b/a$ and $\alpha\beta=c/a$. Everything symmetric follows: $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$, $\dfrac{1}{\alpha}+\dfrac{1}{\beta}=\dfrac{\alpha+\beta}{\alpha\beta}$, $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=\Delta/a^2$, and $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$.

WHY Expand $a(x-\alpha)(x-\beta)$ and compare coefficients. That is the whole of it, and it generalises: for any polynomial the sum of the roots is $-a_{n-1}/a_n$ and the product is $(-1)^n a_0/a_n$.

COST Thirty seconds against solving with the quadratic formula and then manipulating two surds, which is four lines of arithmetic you will get wrong once a paper.

BREAKS Nothing, as an identity. But it gives you **symmetric** functions only. If the question distinguishes α from β you still need $(\alpha-\beta)^2$ and a sign decision.

DEFAULT Solve for the roots.

For a cubic $ax^3+bx^2+cx+d=0$ with roots α,β,γ:

$$\alpha+\beta+\gamma=-\frac{b}{a},\qquad \alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a},\qquad \alpha\beta\gamma=-\frac{d}{a}.$$

You do not need to memorise these. You need to be able to expand $a(x-\alpha)(x-\beta)(x-\gamma)$ and compare, which takes fifteen seconds and never goes wrong.

45.2 Building the equation whose roots are related

Two methods, and knowing which to reach for is worth more than either.

45.2 SUBSTITUTE, OR REBUILD FROM SUM AND PRODUCT

CORE

TRIGGER "The equation … has roots α and β. Find an equation with roots …".

CLAIM If the new roots are a **rational transformation** of the old ones, say $y=2x$ or $y=1/(x-1)$ or $y=x+3$, substitute: rearrange to make x the subject, put it into the original equation, and clear denominators. If the new roots involve **powers or roots** of the old ones, say $\sqrt{\alpha}$ or α^2, compute the new sum and the new product from Vieta instead.

WHY A substitution maps each old root to exactly one new root, so the transformed equation has exactly the roots you want and the same degree. A power is not one-to-one, so substituting doubles the degree and you get spurious roots.

COST The substitution route is often three lines against a page. The Vieta route is the only one available for powers.

BREAKS Substituting $y = \sqrt{x}$ into a quadratic gives a quartic, which is the wrong degree and contains the negative square roots you did not want. That is the tell: if the substitution changes the degree, you chose the wrong method.

DEFAULT Solve the original, transform the roots, rebuild. Always works, usually involves surds you do not want.

THE TWO ROUTES, ON THE SAME KIND OF QUESTION

Substitution. $x^2 - 2x + \frac{1}{2} = 0$ has roots α, β; find the equation with roots $\alpha/2, \beta/2$.

Put $y = x/2$, so $x = 2y$. Then $(2y)^2 - 2(2y) + \frac{1}{2} = 0$, that is $4y^2 - 4y + \frac{1}{2} = 0$, and dividing by 4, $y^2 - y + \frac{1}{8} = 0$. Three lines.

Vieta. $x^2 - 8x + 4 = 0$ has roots α, β; find the equation with roots $\sqrt{\alpha}, \sqrt{\beta}$.

Substitution would give a quartic, so use sum and product. $\sqrt{\alpha}\sqrt{\beta} = \sqrt{\alpha\beta} = \sqrt{4} = 2$. And $\left(\sqrt{\alpha} + \sqrt{\beta}\right)^2 = \alpha + \beta + 2\sqrt{\alpha\beta} = 8 + 4 = 12$, so $\sqrt{\alpha} + \sqrt{\beta} = 2\sqrt{3}$, taking the positive root because both terms are non-negative. Hence

$$x^2 - 2\sqrt{3}\,x + 2 = 0.$$

Notice the one place judgement was needed: choosing the sign of the square root. That is the step the options will punish.

DRILL A **RECOGNITION ONLY. NAME THE FIRST LINE, TEN SECONDS EACH.**

1. α, β are the roots of $2x^2 - 5x + 1 = 0$. Asked for $\alpha^2 + \beta^2$. First line?
2. The same roots, asked for the equation with roots $\alpha + 3$ and $\beta + 3$. Substitution or Vieta?
3. The same roots, asked for the equation with roots α^2 and β^2. Substitution or Vieta?
4. What is the sum of the roots of $3x^3 - 4x^2 + x - 7 = 0$?

ANSWERS. 1. $\alpha + \beta = \frac{5}{2}$ and $\alpha\beta = \frac{1}{2}$, then $(\alpha + \beta)^2 - 2\alpha\beta$. 2. Substitution, $y = x + 3$. 3. Vieta: a power changes the degree, so substitution would give a quartic. 4. $\frac{4}{3}$.

DRILL B **EXAM STANDARD.**

1. α, β are the roots of $2x^2 - 6x + 1 = 0$. Find $\alpha^2 + \beta^2$ and $\dfrac{1}{\alpha} + \dfrac{1}{\beta}$.
2. $x^2 + 5x + 2 = 0$ has roots α, β. Find the equation with roots $\alpha + 2$ and $\beta + 2$.
3. $x^3 + x^2 - 1 = 0$ has roots α, β, γ. Find the equation with roots $\dfrac{1}{\alpha - 1}, \dfrac{1}{\beta - 1}, \dfrac{1}{\gamma - 1}$.
4. $x^2 - px + q = 0$ has roots whose difference is 4. Express p and q in one relation.

ANSWERS. 1. $\alpha + \beta = 3$, $\alpha\beta = \frac{1}{2}$, so $\alpha^2 + \beta^2 = 9 - 1 = 8$ and $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = 3/(1/2) = 6$. 2. Put $y = x + 2$, so $x = y - 2$: $(y-2)^2 + 5(y-2) + 2 = y^2 + y - 4$, giving $x^2 + x - 4 = 0$. 3. Put $y = 1/(x-1)$, so $x = (y+1)/y$. Then $\left(\frac{y+1}{y}\right)^3 + \left(\frac{y+1}{y}\right)^2 - 1 = 0$; multiply by y^3: $(y+1)^3 + (y+1)^2 y - y^3 = 0$, which expands to $y^3 + 5y^2 + 4y + 1 = 0$. 4. $(\alpha - \beta)^2 = 16$ and $(\alpha - \beta)^2 = p^2 - 4q$, so $p^2 - 4q = 16$.

---

46 Hidden polynomials, and the constraint the substitution carries

Half the quadratics on the paper are not written as quadratics. The substitution is the easy part. **The constraint the substitution carries is where the marks are**, and it is almost never stated in the question.

46.1 The move, and the rule underneath it

46.1 SUBSTITUTE THE REPEATED OBJECT, THEN IMPOSE ITS RANGE

CORE

TRIGGER The same compound expression appears twice, once squared and once not. 4^x with 2^x. $(\log x)^2$ with $\log x$. $\sin^2\theta$ with $\sin\theta$. x with $\sqrt{x}$. $x^2 + 1/x^2$ with $x + 1/x$.

CLAIM Name the repeated object t and solve the resulting polynomial. Then **map every root of t back**, rejecting any that is outside the range of the substitution and counting how many values of the original variable each surviving root produces.

WHY A substitution is a map, and a map has a range. Solving in the new variable is only half the question; the other half is asking which of those answers were reachable.

COST It is the intended route. Without it there is no route. The rejection step is worth about a mark a paper on its own.

BREAKS Forgetting that one value of t can give **two** values of x, or none. That is the difference between "how many solutions in t" and "how many solutions in x", and the option list always contains both numbers.

DEFAULT None. This is the method.

WRITTEN AS	SUBSTITUTE	THE CONSTRAINT YOU MUST IMPOSE
$4^x - 3\cdot 2^x + 2 = 0$	$t = 2^x$	$t > 0$ strictly. A negative or zero root is not a solution. Each valid t gives exactly one x.
$25^x - 3\cdot 5^x + 2 = 0$	$t = 5^x$	As above. Note $25^x = (5^x)^2$ and $2^{x+1} = 2\cdot 2^x$ and $8^x = (2^x)^3$: rewrite before substituting.
$2\cos^2\theta = 1 + \sin\theta$	$s = \sin\theta$	$-1 \le s \le 1$, and then each valid s gives a different **number** of θ in the interval. Use $\cos^2 = 1 - \sin^2$ to eliminate the squared one.
$2\sin^2\theta + 4\sin\theta\cos\theta - 4\cos^2\theta = 0$	$u = \tan\theta$	Divide through by $\cos^2\theta$ first. Legal because $\cos\theta = 0$ would force $\sin\theta = \pm 1$, which does not satisfy the equation — check that, do not assume it.
$x - 5\sqrt{x} + 6 = 0$	$u = \sqrt{x}$	$u \ge 0$, and $x = u^2$, so two values of u give two values of x.
$x^2 + 1/x^2 = k$	$v = x + 1/x$	$x^2 + 1/x^2 = v^2 - 2$, and $\lvert v\rvert \ge 2$ for real x. The options are built from the roots you should have rejected.
$(\log x)^2 - 3\log x + 2 = 0$	$L = \log x$	The argument of every logarithm must stay positive, here $x > 0$, and the same check applies to every log that appears during the working.

WORKED, WITH THE COUNTING DONE PROPERLY

"How many solutions does $3\cdot 4^x - 2^{x+1} - 1 = 0$ have?"

Rewrite: $4^x = (2^x)^2$ and $2^{x+1} = 2\cdot 2^x$. With $t = 2^x$,

$$3t^2 - 2t - 1 = 0 \iff (3t+1)(t-1) = 0 \iff t = -\tfrac{1}{3} \text{ or } t = 1.$$

Two roots in t. Now map back. $t = -\frac{1}{3}$ is rejected because $2^x > 0$ always. $t = 1$ gives $x = 0$, one value. So the answer is **one**, and "two" is sitting in the option list waiting for you.

The same question with $s = \sin\theta$ instead would go the other way: a single valid s can give two values of θ in $[0, 2\pi)$, so the count in the original variable can be *larger* than the count in the substituted one. Always ask which direction the mapping goes.

46.2 Why $|v| \geq 2$

Worth deriving once, because it is the least obvious constraint in the table and it recurs.

For real $x \neq 0$, $\left(x - \frac{1}{x}\right)^2 \geq 0$ expands to $x^2 + \frac{1}{x^2} \geq 2$, and $\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \geq 4$. So $|v| \geq 2$, with equality exactly at $x = \pm 1$. That is a completed square and nothing else, which is the general theme of §48.

DRILL A RECOGNITION ONLY. NAME THE SUBSTITUTION AND ITS CONSTRAINT, TEN SECONDS EACH.

1. $9^x - 4 \cdot 3^x + 3 = 0$
2. $2\sin^2\theta + \cos\theta = 1$
3. $x - 7\sqrt{x} + 10 = 0$
4. $(\log_2 x)^2 = \log_2 x + 2$

ANSWERS. 1. $t = 3^x$ with $t > 0$; each valid t gives one x. 2. Use $\sin^2 = 1 - \cos^2$, then $c = \cos\theta$ with $|c| \leq 1$; each valid c can give **two** values of θ. 3. $u = \sqrt{x}$ with $u \geq 0$, and $x = u^2$. 4. $L = \log_2 x$, and the domain $x > 0$ throughout.

DRILL B EXAM STANDARD.

1. Solve $25^x - 3 \cdot 5^x + 2 = 0$.
2. How many solutions does $2\cos^2 x = 3\sin x$ have for $0 \leq x < 2\pi$?
3. Solve $x - 5\sqrt{x} + 6 = 0$.
4. How many real x satisfy $x^2 + \frac{1}{x^2} = 3$?

ANSWERS. 1. $t = 5^x$ gives $t = 1$ or $t = 2$, both positive, so $x = 0$ or $x = \log_5 2$. 2. $2(1 - s^2) = 3s$ gives $2s^2 + 3s - 2 = 0$, so $s = \frac{1}{2}$ or $s = -2$; reject $s = -2$; $\sin x = \frac{1}{2}$ has **two** solutions in the interval. 3. $u = \sqrt{x}$ gives $u = 2$ or $u = 3$, both ≥ 0, so $x = 4$ or $x = 9$. 4. $v^2 - 2 = 3$ gives $v = \pm\sqrt{5}$. Now *check* the constraint rather than reciting it: $\sqrt{5} \approx 2.24 > 2$, so **both** values are legal. Each gives $x^2 - vx + 1 = 0$ with discriminant $v^2 - 4 = 1 > 0$, so two real x each: **four** in total.

> **THE CONSTRAINT IS A TEST YOU APPLY, NOT A RESULT YOU RECALL**
>
> Question 4 is built to catch one specific habit. Somebody who has learned "$|v| \geq 2$" as a slogan sees $\sqrt{5}$, feels the rule fire, and rejects both roots without doing the comparison. "No real solutions" is on the option list for exactly that reader.
>
> Every time you impose a constraint, do the arithmetic out loud: is $\sqrt{5}$ bigger than 2, or not? Book One §15.2 calls the failure an assumption error, and it is the most common cause of a confidently wrong answer.

---

47 Exponentials and logarithms

A reliable question a paper, and the one topic where the specification's own limits are a usable reading cue.

47.1 The laws, and the three that get forgotten

$$\log_a(xy) = \log_a x + \log_a y, \qquad \log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y, \qquad \log_a(x^k) = k\log_a x.$$

Then $\log_a a = 1$, $\log_a 1 = 0$, and the one people lose under pressure, $\log_a\left(\frac{1}{x}\right) = -\log_a x$, which is just the power rule with $k = -1$.

Three things worth reading as cues rather than as facts.

— **Change of base is not on the specification's list of named laws.** So a question that appears to need it usually has another route, and finding that route *is* the question. Know the formula anyway, $\log_a b = \dfrac{\log_c b}{\log_c a}$, because it costs one line and occasionally saves three.

— $y = a^x$ **and** $y = \log_a x$ **are reflections in** $y = x$**.** A question about the number of solutions of $a^x = \log_a x$ is a sketching question, not an algebra question.

— **The domain is a condition, not a formality.** Every logarithm you write down carries "the argument is positive". Solutions that violate it are exactly the distractors.

47.2 Comparing sizes without a calculator

47.2 RAISE TO A COMMON POWER — EDGE

TRIGGER	"Which is largest" with powers, roots or logarithms, and no calculator.
CLAIM	Raise everything to a common power, or take logarithms and compare multiples. To compare 2^{100} and 3^{60}, take twentieth roots: $2^5 = 32$ against $3^3 = 27$, so 2^{100} wins.
WHY	Both operations are strictly increasing, so they preserve order. Choosing the common power is the whole skill: take the highest common factor of the exponents.
COST	Fifteen seconds against an impossible calculation.
BREAKS	Strictly increasing needs base greater than one. With base between 0 and 1 the inequality **reverses**, and that is the designed trap in logarithmic inequalities.
DEFAULT	Estimate with known anchors: $2^{10} \approx 10^3$, $\log_{10} 2 \approx 0.30$, $\log_{10} 3 \approx 0.48$.

47.3 How many digits, and how many leading zeros

47.3 DIGIT COUNTING BY LOGARITHM — EDGE

TRIGGER	"How many digits does N have", or "how many zeros immediately after the decimal point", with N a large power you cannot compute.
CLAIM	A positive integer N has $\lfloor \log_{10} N \rfloor + 1$ digits. A number $0 < x < 1$ has $-\lceil \log_{10} x \rceil$ zeros between the decimal point and the first non-zero digit, which is easiest to get by writing x between two powers of ten.
WHY	N has d digits exactly when $10^{d-1} \le N < 10^d$, and taking $\log_{10}$ turns that into $d - 1 \le \log_{10} N < d$.
COST	Ten seconds against something you cannot do at all. Uncommon but unmistakable when it appears.
BREAKS	Getting the direction of the floor wrong at exact powers of ten. Check with $N = 1000$: $\log_{10} = 3$, so four digits. Correct.
DEFAULT	Bracket the number between powers of ten by hand using $2^{10} \approx 10^3$.

DRILL A RECOGNITION ONLY. NAME THE MOVE OR THE CHECK, TEN SECONDS EACH.

1. $\log_3 x + \log_3(x - 6) = 3$. What must you check at the end?

2. Which is larger, 2^{30} or 3^{20}? What common power do you take?
3. $\log_{0.5}(x-1) > 3$. What happens to the inequality sign?
4. "How many digits does 3^{50} have?" What do you compute?

ANSWERS. 1. That every logarithm's argument came out positive. 2. Tenth roots: $2^3 = 8$ against $3^2 = 9$, so 3^{20} wins. 3. It **reverses**, because the base is below 1. 4. $\lfloor 50 \log_{10} 3 \rfloor + 1$.

DRILL B EXAM STANDARD.

1. Solve $\log_2 x + \log_2(x-2) = 3$.
2. Which is larger, 3^{40} or 5^{25}?
3. Find every x with $\log_{1/2}(x-1) > 2$.
4. How many digits does 2^{40} have, given $\log_{10} 2 \approx 0.30103$?

ANSWERS. 1. $x(x-2) = 8$ gives $x = 4$ or $x = -2$; reject $x = -2$ because both $\log_2 x$ and $\log_2(x-2)$ need positive arguments, so $x = 4$. 2. Fifth roots: $3^8 = 6561$ against $5^5 = 3125$, so 3^{40}. 3. Base below one reverses: $0 < x - 1 < \frac{1}{4}$, so $1 < x < \frac{5}{4}$. 4. $40 \times 0.30103 = 12.04$, floor 12, so **13** digits.

---

48 Inequalities and casework

The specification asks for linear and quadratic inequalities. That sounds tame and it is where a surprising number of marks die, because every step you take with an inequality has a condition attached and the conditions are invisible.

48.1 The four moves that need a sign

48.1 NEVER MULTIPLY BY SOMETHING WHOSE SIGN YOU DO NOT KNOW CORE

TRIGGER	An inequality with a variable in a denominator, under a square root, or on both sides.
CLAIM	Four rules, and they are one rule. **Multiplying or dividing by an expression of unknown sign is not allowed** — bring everything to one side, factor, and read the sign off a sketch or a sign table. **Squaring is valid only when both sides are known non-negative**, otherwise it invents solutions. **Taking reciprocals reverses the inequality only when both sides have the same sign. Cancelling a common factor throws away the case where that factor is zero.**
WHY	An inequality is a statement about order, and multiplication by a negative number reverses order. So any step whose sign you have not established is a step you have not justified.
COST	These are not shortcuts. They are the difference between right and wrong, and Paper 2 sets whole questions on exactly this as flawed-step questions (§70).
BREAKS	Nothing. This is the safe method.
DEFAULT	There is nothing safer than this.

THE SIGN TABLE, IN FIFTEEN SECONDS

Get everything on one side as a single quotient or product. Mark on a number line the zeros of every factor **and** every point where the denominator vanishes. Then walk along the line and track the sign of each factor, multiplying as you go.

The answer is a union of intervals. Points where a denominator vanishes are **always excluded**, even when the inequality is weak ($\leq$ rather than $<$).

That last clause is the trap, and it is the one that generates the two options differing only by a bracket.

48.2 Integers change the answer

A quietly examinable subtlety, and one that almost nobody trains.

If n is an integer, "$n > 3$" and "$n \geq 4$" are the same statement, and "$n > 3.2$" also means "$n \geq 4$". So an inequality over the integers should always be finished by *rounding to the nearest legal integer*, in the correct direction, and the direction is where people slip.

— $n > c$ with n an integer means $n \geq \lfloor c \rfloor + 1$.

— $n < c$ with n an integer means $n \leq \lceil c \rceil - 1$.

— The number of integers in $[a, b]$ is $\lfloor b \rfloor - \lceil a \rceil + 1$. **Forgetting the $+1$ is the most common single slip in this area.**

Related, and worth a card of its own: a strict inequality between integers can often be strengthened for free. If a and b are integers and $a < b$, then $a \leq b - 1$. That step converts an unbounded-looking problem into a finite check surprisingly often.

48.3 Proving an inequality on this paper

Everything the test can set is reachable by completing the square, by sketching, or by considering a square that must be non-negative. That is not a coincidence: it is the toolkit the specification names.

— (something)$^2 \geq 0$ is the workhorse. To show $x^2 + xy + y^2 \geq 0$, write it as $\left(x + \frac{y}{2}\right)^2 + \frac{3}{4}y^2$.

— **Difference of two sides.** To compare A and B, look at $A - B$ and establish its sign. Do not manipulate the inequality itself.

— **A sum of squares is zero only if every square is zero.** This turns an equation into a system and it is a favourite construction.

> **ABOUT AM–GM AND CAUCHY–SCHWARZ**
>
> Both are correct, both are lovely, and both are BEYOND. Neither appears anywhere in the specification, and the reason matters more than the fact.
>
> The two-variable AM–GM inequality is literally $\left(\sqrt{a} - \sqrt{b}\right)^2 \geq 0$ rearranged, and Cauchy–Schwarz in two variables is $(ay - bx)^2 \geq 0$ rearranged. So on this paper they are not extra power. They are the same power with a longer name and a precondition, non-negativity, that you have to remember to check.
>
> Where they genuinely pay: interviews, olympiad-style problems, and STEP, where you are writing an argument rather than choosing an option and naming the standard inequality is the fastest way to communicate it. They are in §77 with the rest.

DRILL A RECOGNITION ONLY. NAME THE FIRST MOVE, TEN SECONDS EACH.

1. $\dfrac{x+2}{x-1} < 4$. First move?
2. $\sqrt{x+6} > x$. When are you allowed to square?
3. $x^2 > 4x$. May you divide by x?
4. n is an integer with $n > 7.3$. Smallest n?

ANSWERS. 1. Everything onto one side over a common denominator, then a sign table. Never multiply by $x-1$. 2. Only where both sides are known non-negative; the branch where the right side is negative is true automatically and is handled separately. 3. No. That loses $x=0$ and reverses the inequality for $x<0$. 4. 8.

DRILL B EXAM STANDARD.

1. Solve $\dfrac{x+1}{x-2} \geq 3$.
2. Find every x with $\sqrt{x+4} > x+2$.
3. Show that $x^2+xy+y^2 \geq 0$ for all real x and y.
4. How many integers n satisfy $-3.4 < n \leq 12.7$?

ANSWERS. 1. Move over: $\dfrac{x+1-3(x-2)}{x-2} \geq 0$, that is $\dfrac{7-2x}{x-2} \geq 0$, giving $2 < x \leq \frac{7}{2}$. Multiplying up would have lost the lower end. 2. Domain $x \geq -4$. For $-4 \leq x < -2$ the right side is negative and every such x works. For $x \geq -2$ both sides are non-negative, so square: $x+4 > x^2+4x+4$, giving $x^2+3x<0$, so $-3<x<0$; intersecting with $x \geq -2$ leaves $-2 \leq x < 0$. Answer $-4 \leq x < 0$. 3. $\left(x+\frac{y}{2}\right)^2 + \frac{3}{4}y^2 \geq 0$. 4. From $n \geq -3$ to $n \leq 12$: $12-(-3)+1 = 16$.

49 Ranges: what a function can and cannot reach

An enormous number of TMUA questions are settled not by solving anything but by noticing that one side of an equation cannot reach the other. This section is short and it is worth more than its length.

49.1 The bounding argument

49.1 IF THE RANGES DO NOT MEET, THERE ARE NO SOLUTIONS CORE

TRIGGER	An equation whose two sides are visibly different kinds of object: a bounded trigonometric expression against something growing, a sum of squares against a negative, a square root against a negative.
CLAIM	Find the range of each side separately. If the ranges are disjoint there are no solutions. If they touch at exactly one value, the only solutions are where **both** sides equal that value, which usually pins everything down at once.
WHY	An equation asserts that two values coincide, so anything neither side can reach is irrelevant.
COST	It answers "how many solutions" instantly on questions that look as though they need work, and it is the intended route more often than it looks.
BREAKS	You need the *exact* range, not a bound you half-remember. $\sin x + \cos x$ ranges over $[-\sqrt{2}, \sqrt{2}]$, not $[-1,1]$, and getting that wrong turns a right answer into a wrong one.
DEFAULT	Solve, or sketch both sides and count intersections.

THE THREE STANDARD RANGES YOU SHOULD NEVER RECOMPUTE

$$\sin x,\ \cos x \in [-1,1]; \qquad x^2 \geq 0; \qquad a^x > 0 \text{ for } a > 0.$$

And three that are one line of work each: $x + \dfrac{1}{x} \geq 2$ for $x>0$ (and ≤ -2 for $x<0$); $\sqrt{\cdot} \geq 0$ by definition, so $\sqrt{x^2} = |x|$ and not x; and a quadratic has range $[q - p^2/4,\ \infty)$ when it opens upwards, read straight off the completed square.

49.2 Ranges of nested functions

For $g(f(x))$, work outwards. Find the range of f over the given domain; that range is the *domain* available to g; then find what g does on it. The only subtlety is that g may not be monotonic, in which case the extremes of the output need not come from the extremes of the input — check the stationary point of g inside the interval as well as the two ends.

The same discipline gives you the maximum of a function on a restricted domain: **candidates are the stationary points inside the interval and the two endpoints, and nothing else.** Evaluate all of them and take the largest. Forgetting the endpoints is a standard loss.

DRILL A RECOGNITION ONLY. NAME THE RANGE, TEN SECONDS EACH.

1. What values can $\sin x + 2$ take?
2. Can $x^2 + 1$ ever be zero for real x?
3. What values can a^x take, for $a > 0$?
4. One side of an equation is at most 3; the other is at least 3. What follows?

ANSWERS. 1. $[1, 3]$. 2. No: it is at least 1. 3. Strictly positive values only. 4. Both sides must equal 3, which usually pins every variable down at once.

DRILL B EXAM STANDARD.

1. How many real solutions does $\sin x = x^2 + 1$ have?
2. $x^2 + y^2 - 4x + 6y + 13 = 0$. Find all real pairs (x, y).
3. Find the greatest value of $f(x) = x^3 - 3x$ on $-1 \le x \le 3$.
4. $f(x) = \dfrac{1}{x^2 + 2}$. Find the range of f, and then the range of $f(f(x))$.

ANSWERS. 1. The left side is at most 1 and the right side is at least 1, so both must equal 1: $x^2 + 1 = 1$ forces $x = 0$, and $\sin 0 = 0 \neq 1$. **No solutions.** 2. Complete both squares: $(x-2)^2 + (y+3)^2 = 0$, a sum of squares equal to zero, so $x = 2$ and $y = -3$. 3. Candidates: stationary points $x = \pm 1$ inside the interval, and the endpoints $x = -1, 3$. Values $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Greatest is 18. 4. $x^2 + 2 \ge 2$ so $f(x) \in (0, \tfrac{1}{2}]$. Then $f(f(x))$ needs $u \in (0, \tfrac{1}{2}]$ in $1/(u^2 + 2)$, and $u^2 + 2 \in (2, \tfrac{9}{4}]$, so $f(f(x)) \in [\tfrac{4}{9}, \tfrac{1}{2})$.

11

PART XI

FUNCTIONS, GRAPHS AND CALCULUS

Where a picture beats an equation, where symmetry does the work an integral would have done, and the small number of calculus facts the paper actually leans on.

50 Graphs, symmetry and transformations

51 Symmetry inside a definite integral

52 Modulus, floor and ceiling

53 Trigonometry: counting, not computing

54 Areas: one identity, not eight formulae

55 Curvature, the trapezium rule, and honest estimates

56 Calculus checkpoints

§51 is the section least likely to be in any other document you own, and it is worth a mark a paper.

50 Graphs, symmetry and transformations

The strongest chapter in the first printing and the one that needed least work. Two additions and one simplification.

50.1 Where the mirror is, and where the centre is

50.1 AVERAGE THE ARGUMENTS — CORE

TRIGGER	A functional equation rather than a formula: the function evaluated at two different expressions, set equal to each other, or added to give a constant.
CLAIM	If $f(p+t) = f(p-t)$ for every t, the graph has a **mirror at** $x = p$. If $f(p+t) + f(p-t) = 2q$ for every t, the graph has a **centre of rotational symmetry at** (p, q). To use either, take the two arguments you are given, add them, and halve: that average is p. If the two sides also add to a constant, halve that for q.
WHY	That is what a mirror and a centre *mean*, written down.
COST	Turns "$f(77 - x) = f(-79 + x)$, where is the line of symmetry" into one addition. Five seconds against a page of substitution.
BREAKS	**For every** t, not for one t. A single coincidence is not a symmetry. And read the signs: the arguments in that example sum to -2, so the mirror is at $x = -1$, not $x = 1$.
DEFAULT	Substitute two or three values of t and look at the pairs of points you get.

Two special cases are worth naming, because they are the ones the paper states in words rather than in symbols.

— **Even function:** $f(-x) = f(x)$. Mirror in the y-axis. This is $p = 0$.

— **Odd function:** $f(-x) = -f(x)$. Centre of symmetry at the origin. This is $p = q = 0$.

50.2 The three reflections

REFLECT THE GRAPH IN	NEW EQUATION	HOW TO SEE IT
the vertical line $x = \alpha$	$y = f(2\alpha - x)$	The point at distance t to the right must now show what used to be t to the left, so $x \mapsto 2\alpha - x$.
the horizontal line $y = \alpha$	$y = 2\alpha - f(x)$	Heights measured from $y = \alpha$ get negated: $f \mapsto \alpha - (f - \alpha)$.
the line $y = x$	$x = f(y)$, i.e. the inverse	Swapping the roles of the two coordinates is exactly what reflecting in $y = x$ does.

The third row is the one worth internalising, because it explains a family of questions in one sentence: $y = a^x$ **and** $y = \log_a x$ **are reflections of each other in** $y = x$, and any question about where they meet or how many times is a sketching question.

50.3 Composite transformations, in the safe order

50.3 FACTOR BEFORE YOU INTERPRET CORE

TRIGGER	$y = f(ax + b)$, or a described sequence of stretches and shifts, or $f(g(x))$.
CLAIM	Factor first. $f(ax + b) = f\big(a(x + b/a)\big)$, which is a **shift left by b/a and then a stretch by $1/a$ parallel to the x-axis.** Inside the bracket everything is backwards; outside it, everything reads the way it is written.
WHY	Replacing x by $x + k$ asks for the value the function had k further along, which is why the graph moves the *other* way.
COST	Removes the single most common transformation error, which is applying the shift and the stretch in the wrong order and landing on the option that was built for exactly that mistake.
BREAKS	Composition is not commutative, and the specification asks for $f(g(x))$ by name. Work from the inside out and check the domain at each stage.
DEFAULT	Track two or three specific points through the transformation. Slower, and it never lies.

> **THE DEFAULT IS BETTER THAN YOU THINK**
>
> Tracking points is not a fallback for weak candidates. It is a two-second check that catches the order error every time.
>
> Take the vertex, or an intercept, or any point you can name. Push it through the transformation as a coordinate pair. If the answer you were about to give sends it somewhere else, your ordering is wrong. On a question worth one mark, this is the cheapest verification available anywhere in Book Two.

50.4 The modulus family

50.4 INSIDE IS SYMMETRY, OUTSIDE IS FOLDING CORE

TRIGGER	A modulus sign anywhere in a graph question.
CLAIM	Four cases, and only four. $y = f(\lvert x\rvert)$: keep $x \geq 0$ and mirror it into $x < 0$, so the result is symmetric about the y-axis. $y = \lvert f(x)\rvert$: fold everything below the x-axis upwards. $\lvert y\rvert = f(x)$: delete where $f < 0$ and reflect what is left in the x-axis, giving $y = \pm f(x)$. $\lvert y\rvert = f(\lvert x\rvert)$: keep the first quadrant and copy it into the other three.
WHY	Modulus on the input is about symmetry of the *domain*; modulus on the output is about the sign of the *range*. Those are two different operations and the whole table follows from that one distinction.
COST	These appear as "how many solutions" questions, where the sketch answers instantly and algebra takes four minutes and produces spurious roots.
BREAKS	Order of operations with combinations. $\lvert f(\lvert x\rvert)\rvert$ is not the same as $f(\lvert x\rvert)$, and you have to do them in the order written.
DEFAULT	Case-split on the sign inside each modulus and sketch each branch on its own domain.

Two things that follow and are worth having as separate reflexes.

— $|x-a|$ **is a distance.** "$|x-3|<2$" reads as "x is within 2 of 3", which is $1<x<5$, with no algebra at all. On Paper 2 this is often the fastest way to see whether one condition implies another.

— $\sqrt{x^2}=|x|$**, not** x**.** This is item three of the flawed-step catalogue (§70) and it is examined in both directions.

DRILL A **RECOGNITION ONLY. NAME THE ANSWER, TEN SECONDS EACH.**

1. $f(5+t)=f(11-t)$ for every t. Where is the mirror?
2. Which of $y=|f(x)|$ and $y=f(|x|)$ folds the graph upwards?
3. $y=f(2x-6)$. What happens first, and by how much?
4. $g(x)+g(4-x)=10$ for every x. Where is the centre of symmetry?

ANSWERS. 1. Average the arguments: $x=8$. 2. $|f(x)|$ folds; $f(|x|)$ mirrors the right half into the left. 3. Factor to $f(2(x-3))$: shift right 3, then stretch by $\frac{1}{2}$ horizontally. 4. $(2,5)$.

DRILL B **EXAM STANDARD.**

1. $f(3+x)=f(7-x)$ for all x, and f has exactly three real roots, one at $x=9$. Find the other two.
2. How many solutions does $|x^2-4|=3$ have? And $|x^2-4|=x$?
3. Describe $y=2f(3x-6)+1$ as a sequence of transformations, in order.
4. $f(x)+f(2-x)=6$ for all x, and $f(5)=1$. Find $f(-3)$.

ANSWERS. 1. Arguments sum to 10, so the mirror is at $x=5$; the root at 9 pairs with 1, and the third must be its own partner, so $x=5$. Roots $1, 5, 9$. 2. First: $x^2=7$ or $x^2=1$, so four. Second: sketch $y=|x^2-4|$ against $y=x$ — two. 3. Shift right 2, stretch by $\frac{1}{3}$ horizontally, stretch by 2 vertically, shift up 1. 4. Centre at $(1,3)$, so $f(-3)=6-f(5)=5$.

---

51 Symmetry inside a definite integral

An area of the specification that most preparation material ignores completely, and one where a single identity replaces an integration you cannot do.

51.1 The reflection substitution

51.1 REPLACE X BY $A+B-X$ CORE

TRIGGER A definite integral you cannot evaluate directly, whose limits are symmetric about something, or whose integrand contains a function evaluated at two arguments that add to a constant.

CLAIM For any integrable f, $\displaystyle\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$. Adding the two forms often collapses the integrand to a constant.

WHY The substitution $u=a+b-x$ maps the interval to itself, reverses direction, and $du=-dx$, so the two minus signs cancel.

COST Turns an impossible integral into one line. On the TMUA it shows up disguised as a symmetry question rather than as an integral, which is why it is worth recognising.

BREAKS The limits have to be the ones you reflected about. Reflecting about the wrong midpoint gives a true but useless statement.

DEFAULT Integrate directly, when you can.

Three consequences, in increasing order of usefulness.

— **Odd function, symmetric limits.** If $f(-x) = -f(x)$ then $\displaystyle\int_{-a}^{a} f = 0$. Spot the odd function and stop: no integration at all. x^3, $x\cos x$, $\sin x$, and any sum of odd powers.

— **Even function, symmetric limits.** If $f(-x) = f(x)$ then $\displaystyle\int_{-a}^{a} f = 2\int_{0}^{a} f$. Halves the work and often makes the arithmetic clean.

— **Centre of symmetry.** If $f(x) + f(2p - x) = c$ for every x, then over any interval centred on p, $\displaystyle\int_{p-h}^{p+h} f(x)\,dx = c\,h$. Because the graph has a centre at $(p, c/2)$, the region above the centre-height exactly compensates the region below.

WORKED: WHEN THE INTEGRAL LOOKS IMPOSSIBLE

"Evaluate $\displaystyle I = \int_0^1 \frac{x^5}{x^5 + (1-x)^5}\,dx$*."*

Nothing you know integrates that. But $a + b = 1$, so replace x by $1 - x$:

$$I = \int_0^1 \frac{(1-x)^5}{(1-x)^5 + x^5}\,dx.$$

Add the two expressions for I. The integrands sum to 1 at every point, so $2I = \int_0^1 1\,dx = 1$, giving $I = \frac{1}{2}$.

The whole question is one substitution and one addition. Notice what the cue was: **an integrand built from x and $1 - x$ in the same shape**, with limits 0 and 1. That pairing is the trigger, and once you have seen it you will never miss it.

WHERE THIS ACTUALLY TURNS UP ON THIS PAPER

Rarely in that pure a form. Much more often as one of these.

— An area under a curve between limits symmetric about the vertex of a parabola, where the even-function halving makes the arithmetic trivial.

— A polynomial integrand containing odd powers over $[-a, a]$, where half the terms vanish and the question is testing whether you noticed.

— A sketch question about a curve with a stated centre of symmetry, asking for a signed area.

The identity earns its place because it converts all three into the same reflex: **before integrating anything over a symmetric interval, ask what the symmetry kills.**

DRILL A RECOGNITION ONLY. NAME WHAT THE SYMMETRY KILLS, TEN SECONDS EACH.

1. $\displaystyle\int_{-3}^{3} \left(x^5 - 2x^3 + x\right) dx$.
2. $\displaystyle\int_{-2}^{2} \left(x^4 + 3\right) dx$. Rewrite it more cheaply.
3. An integrand built from x and $5 - x$ in the same shape, with limits 0 and 5. Which move?
4. $h(x) + h(6 - x) = 4$ for every x. What is $\displaystyle\int_1^5 h$?

ANSWERS. 1. Zero: every term is odd and the limits are symmetric. 2. $2\int_0^2 (x^4+3)\,dx$. 3. Replace x by $a+b-x$ and add the two forms. 4. Centre at $p=3$ with $c=4$ and half-width 2, so 8.

DRILL B EXAM STANDARD.

1. Evaluate $\displaystyle\int_{-2}^{2} (x^5 - 3x^3 + 4x)\,dx$.
2. Evaluate $\displaystyle\int_{-1}^{1} (3x^4 + 2x^3 - x + 5)\,dx$.
3. $f(x) + f(4-x) = 10$ for all x. Find $\displaystyle\int_1^3 f(x)\,dx$.
4. Without integrating, say which of $\displaystyle\int_{-1}^{1} x^2 \sin x\,dx$ and $\displaystyle\int_{-1}^{1} x^3 \cos x\,dx$ is larger.

ANSWERS. 1. Every term is odd, so the integral is 0. 2. Odd terms vanish; $2\int_0^1 (3x^4+5)\,dx = 2\left(\frac{3}{5}+5\right) = \frac{56}{5}$. 3. Centre at $p=2$ with $c=10$, and the interval $[1,3]$ is centred on 2 with $h=1$, so the integral is $10\times 1 = 10$. 4. Both integrands are odd ($x^2\sin x$ is even times odd; $x^3\cos x$ is odd times even), so both integrals are 0 and neither is larger.

---

52 Modulus, floor and ceiling

The modulus graphs are in §50.4. This section is the arithmetic side, plus the two step functions the specification uses for counting.

52.1 Equations and inequalities with a modulus

Three routes, and picking the right one is the skill.

1. **Sketch.** Best for "how many solutions". Draw both sides, count crossings, done.
2. **Case split.** Best when you need the actual values. Split at each point where something inside a modulus changes sign, solve on each piece, and **check each solution lies in the piece you assumed**. That last check is where the marks are: a solution that falls outside its own case is not a solution.
3. **Square both sides.** Legal only for $|A| = |B|$ or $|A| < |B|$, where both sides are non-negative by construction. It gives $A^2 = B^2$, hence $(A-B)(A+B) = 0$, which is often the fastest route of the three. Not legal for $|A| = B$ unless you already know $B \geq 0$.

For an integral containing a modulus, split the range at every point where the inside changes sign and integrate each piece with the sign it actually has. Integrating $|f|$ as though it were f is a guaranteed loss.

52.2 Floor and ceiling

52.2 THE STEP FUNCTIONS, AND THE THREE FACTS THAT MATTER EDGE

TRIGGER $\lfloor\ \rfloor$ or $\lceil\ \rceil$ notation, or a question about counting integers in a range, or a sum whose terms change at integer boundaries.

CLAIM $\lfloor x \rfloor$ is the greatest integer not exceeding x, so $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$. $\lceil x \rceil$ is the least integer not below x. The number of integers in $[a, b]$ is $\lfloor b \rfloor - \lceil a \rceil + 1$. An integral of a floor function over a range is a sum of rectangles, one per integer step.

WHY	Both are defined by an inequality, not by a formula, so every question about them is answered by writing that inequality down.
COST	Small but reliable. The counting formula alone is worth a mark whenever it appears.
BREAKS	**Negatives.** $\lfloor -1.5 \rfloor = -2$, not -1. Floor always moves left on the number line, ceiling always moves right, regardless of sign. This is the designed error.
DEFAULT	Write out the first few values by hand and read the pattern.

Two habits that make these questions mechanical. Write $x = n + \theta$ with $n = \lfloor x \rfloor$ an integer and $0 \le \theta < 1$; almost every floor identity falls out immediately in that form. And when a floor function appears inside an integral or a sum, **draw the staircase**: the picture converts the problem into a finite list of rectangles you can add up.

DRILL A RECOGNITION ONLY. TEN SECONDS EACH.

1. $\lfloor -2.3 \rfloor$ and $\lceil -2.3 \rceil$?
2. How many integers lie in $[-4.2,\ 9]$?
3. $|2x - 1| = |x + 5|$. Fastest route?
4. $\displaystyle\int_0^4 \lfloor x \rfloor \, dx$. What do you draw?

ANSWERS. 1. -3 and -2. Floor always moves left, ceiling always right. 2. $9 - (-4) + 1 = 14$. 3. Square both sides: both are non-negative by construction. 4. A staircase of four unit rectangles of heights $0, 1, 2, 3$, total 6.

DRILL B EXAM STANDARD.

1. Solve $|2x - 1| = |x + 4|$.
2. How many solutions does $|x - 1| + |x + 2| = 5$ have?
3. Evaluate $\displaystyle\int_0^3 \lfloor x \rfloor \, dx$.
4. How many integers n with $1 \le n \le 1000$ have $\lfloor \log_{10} n \rfloor$ even?

ANSWERS. 1. Square both sides: $(2x - 1)^2 = (x + 4)^2$, so $(2x - 1 - x - 4)(2x - 1 + x + 4) = 0$, giving $x = 5$ or $x = -1$. 2. Split at -2 and 1. For $x < -2$: $-(x - 1) - (x + 2) = 5$ gives $x = -3$, which is in range. For $-2 \le x \le 1$: $(1 - x) + (x + 2) = 3 \ne 5$, no solutions. For $x > 1$: $2x + 1 = 5$ gives $x = 2$, in range. **Two.** 3. Rectangles of height 0, 1 and 2 on unit intervals: $0 + 1 + 2 = 3$. 4. Floor is 0 for 1 to 9 (nine values) and 2 for 100 to 999 (nine hundred), so 909; $n = 1000$ has floor 3.

53 Trigonometry: counting, not computing

The examinable thing here is not exact values. It is how many solutions an equation has in an interval, and what a radian measure lets you compute.

53.1 Counting solutions in an interval

53.1 TRANSFORM THE INTERVAL, NOT THE EQUATION CORE

TRIGGER	"How many solutions", or "the number of values of x in the given range".
CLAIM	Transform the **interval**. If x runs over $[0, 2\pi]$ and the argument is $3x$, then the argument runs over $[0, 6\pi]$, which is three full periods; $\sin\theta = k$ has two solutions per period for $\lvert k \rvert < 1$, one per period at $\lvert k \rvert = 1$, and none beyond. Count in the argument, then stop.

WHY	A stretch parallel to the x-axis multiplies the number of periods in view and changes nothing else.
COST	Twenty seconds against solving, listing, and then worrying about whether you missed a branch. The missed second branch is the classic loss here.
BREAKS	**Endpoints.** A closed interval can contain one extra solution that a half-open one does not, and that is usually the difference between two adjacent options.
DEFAULT	Sketch $y = \sin\theta$ across the transformed interval and draw the horizontal line. Twenty seconds, and more reliable than counting in your head.

The periods worth knowing without thinking: $\sin(bx + c)$ and $\cos(bx + c)$ have period $2\pi/|b|$; $\tan(bx + c)$ has period $\pi/|b|$, and one solution per period for every value.

53.2 The symmetries, as a picture rather than a list

Everything you need about where the second solution lives comes from four facts, and all four are visible on a sketch.

$\sin(\pi - \theta) = \sin\theta$	The second solution in a period is the reflection about the peak.
$\cos(2\pi - \theta) = \cos\theta$	Cosine is even, so solutions pair up about the vertical axis.
$\tan(\theta + \pi) = \tan\theta$	Period π, so solutions are spaced by π, and there is only ever one per period.
$\sin^2\theta + \cos^2\theta = 1$	The identity that converts a mixed equation into a hidden quadratic (§46).

53.3 Radian measure, including the segment

53.3 ARC, SECTOR, SEGMENT CORE

TRIGGER	An angle in radians with an arc, a sector, or a shaded region between a chord and an arc.
CLAIM	Arc $= r\theta$. Sector $= \frac{1}{2}r^2\theta$. Segment $= \frac{1}{2}r^2(\theta - \sin\theta)$.
WHY	The segment is the sector minus the triangle, and the triangle is $\frac{1}{2}r^2\sin\theta$ by the area formula. Do not memorise the third one, derive it.
COST	Free if you know the subtraction. The segment is on the specification and appeared in neither previous printing.
BREAKS	**Radians only.** A degree measure in a sector formula is a wrong answer that looks plausible, and it is a distractor on every question of this type.
DEFAULT	Sector minus triangle, written out.

53.4 What to keep and what to drop

— **Keep.** The unit circle read as *coordinates* rather than lengths. The gradient of a line as the tangent of its angle with the x-axis. The quarter-by-quarter view of the graphs, which is the reliable way to get the symmetries right. The exact values for the five standard angles, rebuilt from the two standard triangles rather than recalled. The Pythagorean triples $3, 4, 5$ and $5, 12, 13$ and $8, 15, 17$ and $7, 24, 25$: spotting one saves you a Pythagoras.

— **Add.** The ambiguous case of the sine rule, which is on the specification and was missing. Given two sides and a non-included angle there can be **two** triangles, and the question is usually about which. See §59.

— **Drop to BEYOND.** The $15°$ and $75°$ triangle. It is elegant, it comes from compound angle formulae, and those are not on the specification, so no question can require it.

53.5 A QUARTER OF A SINE WAVE

EDGE

TRIGGER	An area under $y = a\sin bx$ or $y = a\cos bx$ over a quarter period.
CLAIM	The area of one quarter is $a/\lvert b\rvert$.
WHY	$\int_0^{\pi/2} \sin x\,dx = 1$; the vertical stretch multiplies by a and the horizontal stretch by $1/b$. It is the same substitution argument as §54.
COST	Ten seconds against an integration. Uncommon but genuinely quick.
BREAKS	Quarter periods measured from the right place. From a maximum to a zero is a quarter; from a zero to a zero is a half.
DEFAULT	Integrate.

DRILL A RECOGNITION ONLY. TEN SECONDS EACH.

1. $\sin 4x = k$ with $0 \le x < 2\pi$ and $\lvert k\rvert < 1$. How many solutions?
2. What is the period of $\tan(3x + 1)$?
3. A sector of radius r and angle θ measured in degrees. What is wrong with $\frac{1}{2}r^2\theta$?
4. Which identity turns $3\cos^2 x = 2 + \sin x$ into a quadratic?

ANSWERS. 1. The argument runs over $[0, 8\pi)$, four periods, two solutions each: **eight**. 2. $\pi/3$. 3. The formula is radians only. 4. $\cos^2 x = 1 - \sin^2 x$.

DRILL B EXAM STANDARD.

1. How many solutions does $\sin 3x = \frac{1}{2}$ have for $0 \le x \le 2\pi$?
2. How many solutions does $\tan 2x = 1$ have for $0 \le x < \pi$?
3. A sector of radius 6 has angle $\pi/3$. Find the area between the chord and the arc.
4. How many solutions does $2\sin^2 x = 3\cos x$ have for $0 \le x < 2\pi$?

ANSWERS. 1. The argument runs over $[0, 6\pi]$, three periods, two each, so **6**. 2. The argument runs over $[0, 2\pi)$ and tan has period π, so **2**. 3. $\frac{1}{2} \cdot 36\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right) = 6\pi - 9\sqrt{3}$. 4. $2 - 2c^2 = 3c$ gives $2c^2 + 3c - 2 = 0$, so $c = \frac{1}{2}$ or $c = -2$; the second is impossible, and $\cos x = \frac{1}{2}$ gives **2**.

54 Areas: one identity, not eight formulae

The first printing gave eight area results. They are all correct and they are all the same result. Here is the one they come from.

54.1 The two-point area identity

54.1 AREA DEPENDS ON THE LEADING COEFFICIENT AND THE GAP

CORE

TRIGGER	An area enclosed between a polynomial and a straight line, with the x-coordinates of the intersections given, and constants chosen to look unusable.

CLAIM If the difference between curve and line factorises as $a(x-\alpha)^m(x-\beta)^n$, then with $D=|\beta-\alpha|$ the enclosed area is

$$S=|a|\,D^{\,m+n+1}\,\frac{m!\,n!}{(m+n+1)!}.$$

WHY Substitute $x=\alpha+Dt$. Every factor becomes D times something in t, so $S=|a|D^{m+n+1}\int_0^1 t^m(1-t)^n\,dt$, and that integral is $m!\,n!/(m+n+1)!$ by parts. The whole content of the identity is that the shape does not care where it sits or how the line is tilted, only how wide it is.

COST Ten to fifteen seconds against two to five minutes of definite integration with deliberately horrible limits. Expect one question a paper, sometimes two.

BREAKS The multiplicities have to be the **real** ones. A cubic that crosses the line three times is not $m=2,n=1$, and reading a tangency off a small on-screen diagram is exactly how this gets misfired. Count intersections algebraically, not visually.

DEFAULT Expand the difference and integrate between the two intersections. Always works.

You need four values of the constant. Learn the table, not the formulae.

(M,N)	CONFIGURATION	CONSTANT	RESULT
$(1,1)$	Line cuts a quadratic twice	$\frac{1}{6}$	$S=\lvert a\rvert D^3/6$
$(2,1)$	Line is tangent to a cubic and cuts it once	$\frac{1}{12}$	$S=\lvert a\rvert D^4/12$
$(3,1)$	Line meets a quartic at a triple root and a simple root	$\frac{1}{20}$	$S=\lvert a\rvert D^5/20$
$(2,2)$	Line is tangent to a quartic at both touch points	$\frac{1}{30}$	$S=\lvert a\rvert D^5/30$

The first two are CORE. The second two are EDGE, because a quartic with prescribed multiplicities is rare.

> **WHY THE TILT OF THE LINE NEVER MATTERS**
>
> This is the sentence to keep. Subtracting $mx+c$ from a polynomial changes the coefficient of x and the constant term. It **cannot touch the leading coefficient**, and the intersections are by definition the roots of the difference. So every result above is a statement about the difference function, and the line has already been absorbed into it before the calculation starts.
>
> Horizontal, tilted, or passing through nothing you can identify: same answer.
>
> That also tells you what the answer *cannot* depend on: the value of c, the roots themselves as opposed to their separation, and any monstrous constant in the question. Which is worth marks on its own — see §75.

54.2 Three intersections, and the splitting trick

54.2 WRITE THE FAR FACTOR IN TERMS OF THE NEAR ONE — CORE

TRIGGER A cubic cut by a line at three points, and you are asked for one lobe or for both.